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DENVER EMU EAGLE P3 DOOR SYSTEM

Project # 7230

Door Control Unit Application

Software Integration Test Plan (SITP)

Contract Number		CDRL Number	
		Customer Ref. Number	

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0.1	---	2011-10-25	José Trejo & Anthony C. Boulais	Implemented comments from inspection I-1883.
0	---	2011-09-30	José Trejo	Initial Issue.

Preface

This Software Integration Test Plan complies with the requirements of the IEEE Std. 829-1998 and IEEE Std. 1558-2004.

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1 TEST PLAN IDENTIFIER

201111453 Rev. 1, 2011-11-08

2 INTRODUCTION

This document is the TC1 Door Control Unit (DCU) Application Software Integration Test Plan (SITP) for the DENVER EMU EAGLE P3 DOOR SYSTEM project.

This document lists the test cases required to validate the Door Control Unit (DCU) Application.
The Software Integration Test Plan (SITP) is created for each of the DCU SCIs and includes the major sections:

- **Sections 1 to 3: Software Integration Test Planning** identifies the tests required to verify and validate each DCU. Its content is based and comply with the requirements of IEEE Std 829, standard for Software Test Documentation. The SITP shall also comply with the documentation requirements of VSRS's Test Plan Template S17-08.
- **Sections 4 to 16: Software Integration Test Design Specification** describes the design of the tests to be performed in order to validate the requirements of each DCU. Its content is based on the IEEE Std. 829. This section can be included inside the "Software Integration Test Planning", and inside the "Software Integration Test Procedure Specification".
- **Sections 17: Software Test Procedures Specification** describes the procedures that need to be executed in order to validate the requirements of each DCU.

2.1 Objectives

The purpose of this Software Integration Test Plan (SITP) is to specify the tests required to validate the Door Control Unit Application that is part of the Door Control Unit (DCU).

The intended audience is the Vapor Stone Rail Systems software development and maintenance team members, Hyundai Rotem and Denver RTD.

2.2 Scope

The scope of this SITP is to cover the requirements of the DCU Application. The DCU Application controls and monitors two door panels. It also reports the status and diagnostic information to the car network through the Neuron Gateway Application (NGA).

2.3 References, Definitions and Acronyms

2.3.1 References

This section lists the number, title, revision, and source of all documents referenced in this SITP.

2.3.1.1 Standards

CMU/SEI-93-TR-25	Key Practices of the Capability Maturity Model, Version 1.1, February 1993, Software Engineering Institute, Carnegie Mellon University, www.sei.cmu.edu
IEEE Std 1558-2004	IEEE Standard for Software Documentation for Rail Equipment and Systems, 23 March 2005, IEEE, www.ieee.org
IEEE Std. 829-1998	IEEE Standard for Software Test Documentation, 16 September 1998, IEEE, www.ieee.org

2.3.1.2 Vapor Stone Rail Systems Documents

201111185	System Functional Description, RTD Denver EMU, Rev. 4
201111208	Software Quality Assurance Plan, RTD Denver EMU, Rev.1

201111254	Software Verification and Validation Plan, RTD Denver EMU, Rev. 0
201111270	Software Requirement Specification, Generic Door Control Module Application, Rev. 1
201111576	Software Applications Configurations Specifications, Rev. 1
CMM-01	Software Engineering Process (SEP), Rev.3
CMMI-02	Software Engineering Policies, Rev.1
S17-10	Software Integration Test Procedure Template, Rev.2

2.3.1.3 Rotem Documents

REDE112784	PTS Entrance Door for Denver Eagle P3 Project, Rev. 3, 2011-01-13
REDE113762	System Overview for Side Door System Denver EMU, Rev. 0, 201111-04-18

2.3.2 Definitions

Event: An event is an occurrence that is monitored by the system during normal use.
Exception: An exception is a condition outside of specified parameters.
Failure: A failure is an improper condition of items, which requires maintenance or replacement to restore affected items to their normal operation conditions.
Lon Works: Registered trademark of Echelon corp., which presents Echelon distributed network. See http://www.echelon.com

2.3.3 Acronyms

CL	Car Line
CNS	Car Network Simulator
CPLD	Complex Programmable Logic Device
CPU	Central Process Unit
CRC	Cyclic Redundancy Check
CS	Crew Switch
DCB	Door Control Box
DCP	Door Control Panel
DCU	Door Control Unit
EDF	Event Description File
EHS	Emergency Handle Switch
JTAG	Joint Test Action Group
I/O	Input / Output
ID	Identification
IDENT	Identification
IDSW	Identification Switch
IEEE	Institute of Electrical and Electronics Engineers, Inc.
IRAM	Internal Random Access Memory
ITPS	Integration Test Procedure Specification
LED	Light-Emitting Diode
NGA	Neuron Gateway Application
NVRAM	Non-Volatile Random Access Memory
OC	Over Current
OPDF	Operating Parameters Description File
PC	Personal Computer
PSS	Panel Sensing Switch
PTU	Portable Test Application
PWM	Pulse Width Modulation
PWR	Power
RST	Reset
RTD	Regional Transportation District
RTDM	Real-Time Data Monitoring
RTN	Return
S/R	Set / Reset
SACS	Software Applications Configuration Specification
SCI	Software Configuration Item
SDD	Software Design Description
SFD	System Functional Description
SITP	Software Integration Test Plan
SLR	Signal Light Relay
SPI	Serial Peripheral Interface
SPR	Software Problem Report
SRAM	Static Random Access Memory
SRS	System Requirements Specifications
SRTM	Software Requirements Traceability Matrix
SVVP	Software Validation and Verification Plan
TBD	To Be Determined
TL	Train Line
XRAM	On-Chip Extension Random Access Memory
VSRS	Vapor Stone Rail System

3 TEST ITEMS

All the items that make up the DCU Application will be tested during the integration test, the instrumented test, or validated by code walkthrough. The software versions to be tested will be placed under configuration management.

The following documents of the DCU Application will provide the basis for defining the correct operation:

- System Functional Description (SFD)
- Software Requirements Specifications (SRS) +(SACS)
- Software Design Description (SDD)
- Software Validation and Verification Plan (SVVP)
- And this document: Software Integration Test Plan (SITP)

Revision, date and part number of these documents are listed in Section 2.3.1.

Table 3-1 lists corresponding test procedures for the DCU Application:

Table 3-1: Corresponding Test Procedures

Item	Application Executable P/N	Test Procedure Type	Test Procedure Chapter	Test Report
DCU Application	2011L0051	Integration	17.4	Appendix A:
		Instrumented	17.5	Appendix A:

4 FFEATURES TO BE TESTED

An overview of the features to be tested is shown in the project's SRTM. These are listed in this document split into two parts.
Those tests that are to be tested as a part of the full Door system are described in Section 17.4 (Integration Test Procedure).
The features of DCU Application and the Boot Loader Application that cannot be performed without damaging the DCU or using a special setup are described in Section 17.5 (Instrumented Test Procedure).

5 FEATURES NOT TESTED

All the features will be tested except for:

- Event Detection ("NGA CRC Failure" events are impossible to generate on demand, they are simply simulated on the DCU Application side during the instrumented test procedure);
- Logout / Reboot Request (Fatal application error and checksum error are impossible to generate on demand);

Source Code Walkthrough will be performed to validate those functionalities.

6 APPROACH

The test personnel will use the System Functional Description (SFD) and the SRS to prepare all test cases and procedures specifications. This approach will verify the accuracy and comprehensiveness of the information in the documentation in those areas covered by the tests.

The test personnel will run the test procedures and complete the test reports. The test data are to be under configuration management.

The following software programs will be used in the test procedure and if they don't exist, they will need to be developed prior to executing the test procedure:

- PTU Simulator
- Car Network Simulator Application
- PTU Application
- Neuron Gateway Application
- DCU Boot Loader

The PTU Simulator runs on a PC and communicates using the serial port. This application allows receiving commands and data. It also allows sending valid and invalid commands to the DCU Application. This will help testing error handling.

The following features of the DCU Application available for the PTU Application will be tested with the PTU Simulator:

- Valid Login/Logout
- Invalid Login/Physical disconnection
- Command error DCU Application to PTU
- Communication Test
- Static numerical requirements for the serial port
- Virtual Test Switch

The following features of the DCU Application will be tested with the PTU Application:

- Request the code status of the DCU Application
- Request the operating status of the DCU Application
- Start, stop and display Real-Time Monitoring
- Log file download and delete
- Snapshot file download and delete
- Start Self-Test and display results
- Send and receive DCU configuration
- Display the Real-Time clock of the DCU
- Display Door identification
- Download Static RAM Content
- Upload a new version of the DCU Application and Boot Loader
- Request Version of the DCU
- Download Door Movement Information
- System Counters Download and Reset
- Static numerical requirements for the CPU usage

The following features of the DCU Application will be tested by performing a visual inspection of the movement of the Door:

- Door Opening
- Door Closing
- Door behavior with obstruction detection
- Door seek-home operation (with and without obstruction)
- Emergency handles
- Cut-out switch
- Motor temperature sensor
- LED functionality
- Dynamic numerical requirements for departure time

The following features of the DCU Application will be tested with the Neuron Gateway Application and Car Network Simulator:

- Door location configuration
- DCU Time set and retrieve Date and Time
- Start, stop and display Real-Time Monitoring
- Network Integration Requirements

The following features of the DCU Application will be tested using text information generated by the DCU Application compiler:

- Static numerical requirements for the memory usage

The Software Requirements Traceability Matrix will be used to trace all requirements from the SRS to the related test cases. (See Section 17, Software Integration Test Procedure)

7 ITEM PASS/FAIL CRITERIA

The DCU Application software is considered as acceptable if:

- It meets all the criteria defined in the Software Requirements Specifications (SRS) – DCU Application or upon approval of non-conformities detected during the test by the customer.
- After undergoing all the tests, the DCU Application software remains functional without corruption of settings.

8 SUSPENSION CRITERIA AND RESUMPTION REQUIREMENTS

8.1 Suspension Criteria

The following criteria will cause a suspension of all the testing activities:

- Inability to meet a criteria in the SRS;

8.2 Resumption Requirements

When a new version of the DCU Application software is transmitted to the test personnel after a suspension of testing has occurred, the test will be started over again.

9 TEST DELIVERABLES

The following document will be generated by the test personnel and will be placed under the configuration management after test completion.

- This document, DCU Application Software Integration Test Plan
- DCU Application Software Integration Test Report
    - Including “Appendix A: Test Report Sheet” of this document.
- DCU Application Software Validation and Verification Report

10 TESTING TASKS

The testing tasks are defined in the Door System SVVP.

11 ENVIRONMENTAL NEEDS

11.1 Hardware

The testing will be done on a complete DCU board with the Portable Test Unit computer. The materials and equipment used for this Software Integration Test Plan are those called in the specific test procedures and test cases.

11.2 Software

Software loaded in the DCU and in the Portable Test Unit (PTU) will be the latest version. All the other software applications used are those called in the specific test design and the test procedure.

12 RESPONSIBILITIES

The responsibilities are defined in the Door System SVVP.

13 STAFFING AND TRAINING NEEDS

The staffing needs are defined in the Door System SVVP. No training is needed for this project.

14 SCHEDULE

The schedule is defined in the Door System SVVP.

15 RISKS AND CONTINGENCIES

The risks and contingencies are defined in the SQAP.

16 APPROVALS

Approvals names and signatures are included in the cover page of this document.

17 SOFTWARE TEST PROCEDURE

17.1 Purpose

The purpose of Section 17.4 is to to validate the DCU Application that is part of the Door Control Unit (DCU).

The purpose of Section 17.5 is to validate the DCU Application and the DCU Boot Loader that cannot be performed without damaging the DCU or using a special setup.

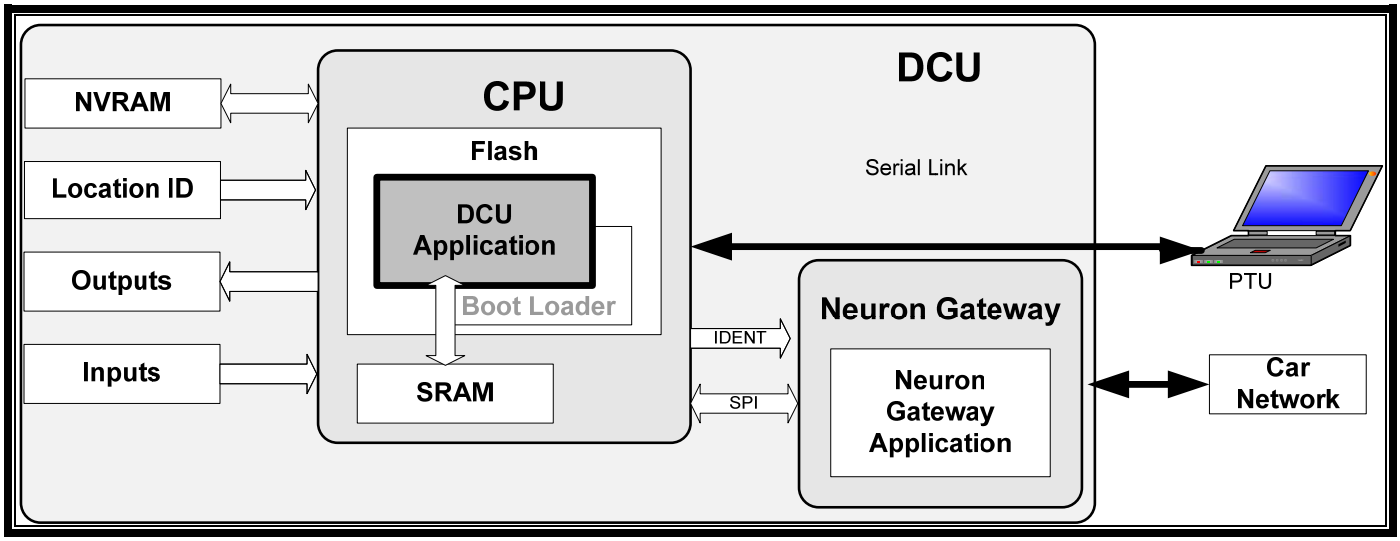

Figure 17-1: External Interface Diagram

17.2 Special Requirements

The following tables list the special equipment, the software and the special files used in the testing:

Table 17-1: Software Configuration Items Required

VSRS Part#	SCI Nomenclature
2011L0051	DCU Application Executable
2011L0052	DCU Neuron Gateway Application Executable
2011L0054	PTU Application Executable
2011L0053	DCU Boot Loader Application Executable
2011L0059	DCU CPLD Binary

Table 17-2: Software Tools and Equipment Required

Description	Version	P/N	Rev.
Software			
PTU Simulator	--	--	--
HiTOP Emulation Software ***– For the instrumented test***	--	--	--
IspVM System	--	--	--
Car Network Simulator	--	--	--
Hardware			
Computer, minimum PENTIUM III, with Windows 2000/XP (for CNC Simulator, NGA Test tool MDS LogD v1.1, PTU simulator application and PTU application)	--	--	--
Computer with MS-DOS (for Pulsar)			
Complete Door System: 1 DCU 1 DCB 1 Door Operators (including lock and motor)	--	--	--
Pulse Generator Test Fixture			
HiTex Development Tools DTrace 167/128k ***– For the instrumented test***	--	--	--

Table 17-3: General Tools Required

Description	Version	P/N	Rev.
PTU cable DB9-PTU PLUG	--	--	--
Banana Cables	--	--	--
Variable Power Supply (Voltage source min. 10A, 0-50V)	--	--	--
DCU Test Fixture	--	--	--
DCU Input Breakout Box	--	--	--
Stop Watch	--	--	--
Measuring Tape	--	--	--
Multimeter	--	--	--
Antistatic Mat	--	--	--
Antistatic Wrist	--	--	--
Bootstrap/Reset Box	--	--	--
DCU with Switch Box for Instrumented Test ***– For the instrumented test***	--	--	--

17.3 Procedure Steps

17.3.1 Log

The result of each test case log in Section 17.4 (Test Procedure) should be recorded in the last column,"P/F, Letter "P" or "F" will indicate Pass or Fail to the test.
Record the test results onto a hardcopy (print) of the Appendix A of this document. Sign and date the document. Copies of Appendix A may be done if test is segmented. Create a new revision of Integration Test Report and append this Integration Test Procedure with the results filled in.
Record the test anomalies that need investigation, if any, as System Problem Reports (SPR) in the ClearQuest database.

17.3.2 Test Setup

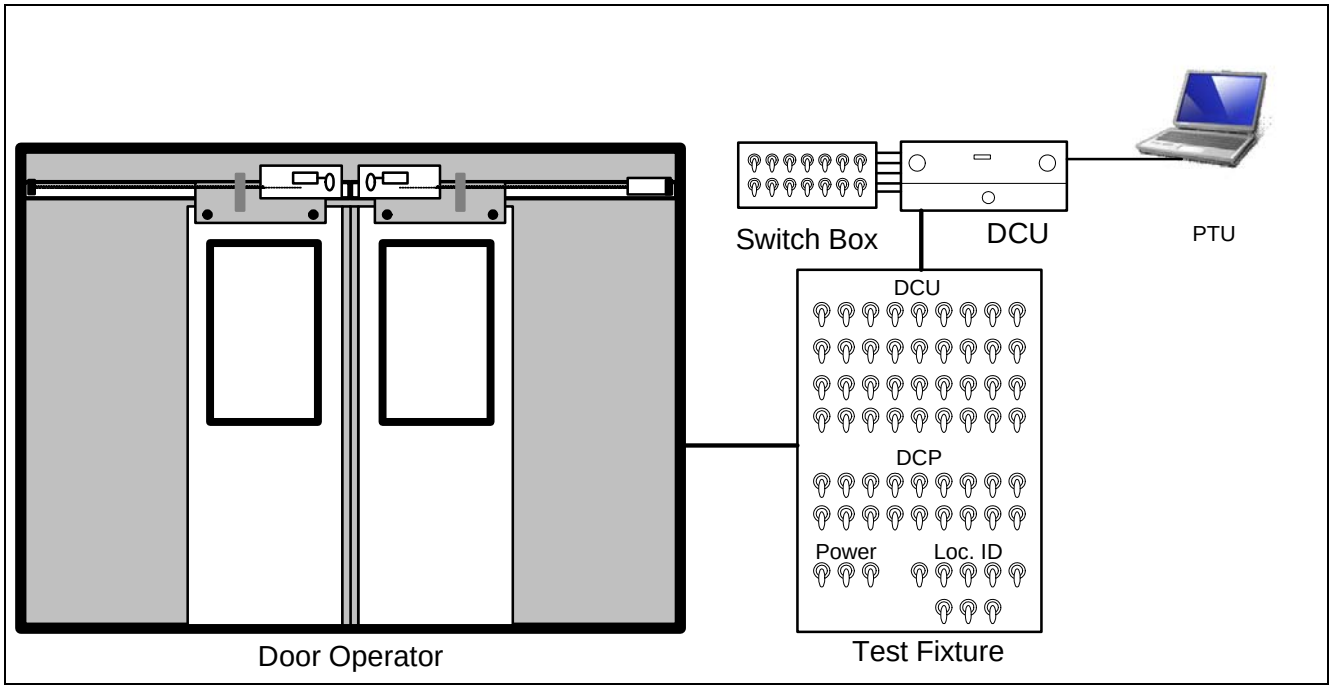

Figure 17-2: Test Setup

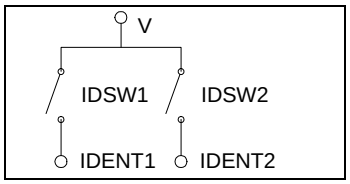

Figure 17-3: Electrical Connections of the Location ID Switches

Table 17-4: Location ID vs. Location ID Switches

Location \ Switch	IDSW1	IDSW2
R 1	0	0
R 2	1	0
L 1	1	1
L 2	0	1

17.3.3 Information about Signals

Signals needed for this test procedure are accessible on the DCU Test Fixture shown in
Figure 17-2 (with switches and banana plugs). When the test case asks the tester to activate a switch, the switch could be on the DCU, on the Door Operator or on the Test Fixture.
When the test case asks to generate a pulse with a specific duration, the tester must use the Pulse Generator with the Pulsar software configured properly. The following figures show the common configuration throughout the test procedure.

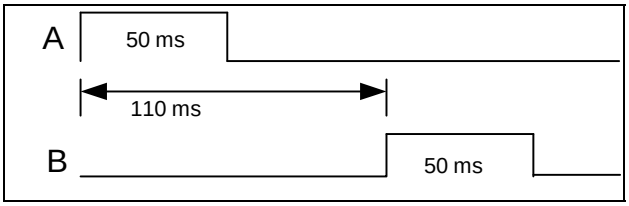

Figure 17-4: Signal Pattern 1

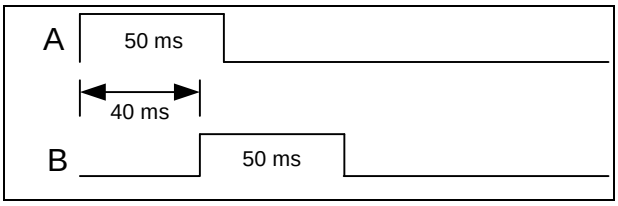

Figure 17-5: Signal Pattern 2

17.3.4 DCU Boot Loader Application Programming

To upload the DCU Boot Loader Application in the DCU, you have to execute the following steps:

1. Turn DCU power OFF.
2. Connect PTU computer and DCU with PTU cable. Make sure that the DCU controller is in the Boot Strap mode using the Bootstrap/Service Pin box that is part of the PTU cable.
3. Start the PTU Application.
4. Select Administration → Other Functions → LoadFlash → Open Flash File to open the DCU Boot Loader Application (H86 format).
5. Select DCU Boot Loader Application and click OK.
6. Select Administration → Other Functions → Load Flash → Send Flash File to start to upload the Boot Loader Application. Follow the uploading prompt of the PTU.
7. After Boot Loader Application process is finished, verify if the “Uploading Successful” information is presented. If the uploading is successful, follow the prompt to remove the PTU cable and reset the DCU. If the uploading is failed, redo steps 6 and 7.
8. The DCU is programmed with the DCU Boot Loader Application.

17.3.5 DCU Application Programming

To upload the DCU Application in the DCU, you have to execute the following steps:

1. Connect the PTU computer and the DCU with the PTU cable.
2. Turn the DCU power ON.
3. Start the PTU Application.
4. Select Administration → Other Functions → SW Download → Open App. File.
5. Select DCU Application and click OK
6. After the file verification, click OK
7. Select Administration → Other Functions → SW Download → Send App. File.
8. Click YES.
9. When the upload is finished successfully, click OK on the prompt window.
10. The DCU is programmed with DCU Application.

17.3.6 Neuron Application Upgrade through the Network

1. Make sure that the PTU cable network connector is connected to the Echelon interface card of the PC.
2. Start the NodeUtil.exe program. The following menu will appear.

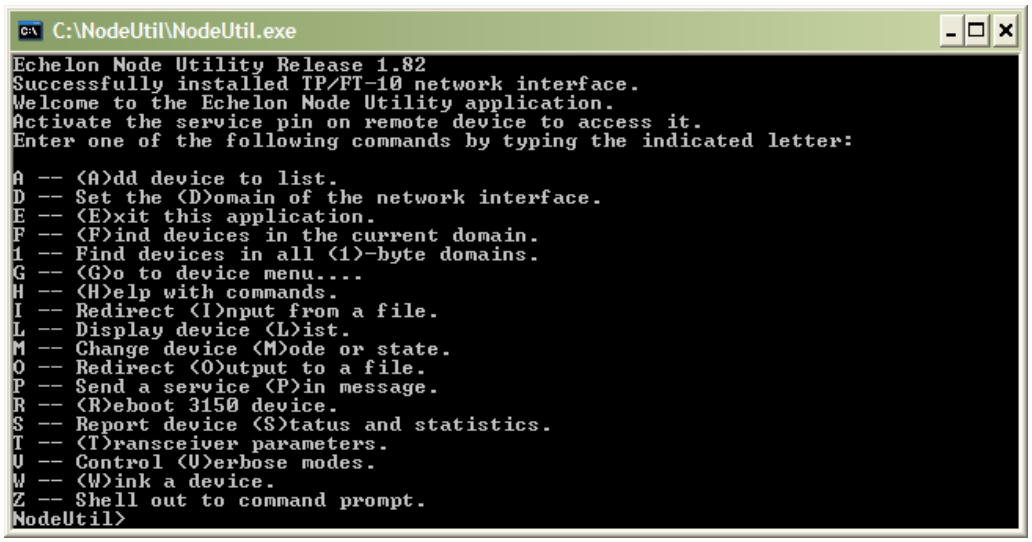

Figure 17-6: NodeUtil Main Menu

3. Send the Service Pin from the DCU using the Bootstrap/Service Pin box that is part of the PTU cable. NodeUtil will receive a message from the DCU.

4. Type “G” to go to the device menu. Type 1 and press [Enter] to choose node 1 (node 1 is the DCU that just sent the service pin). The following menu will appear.

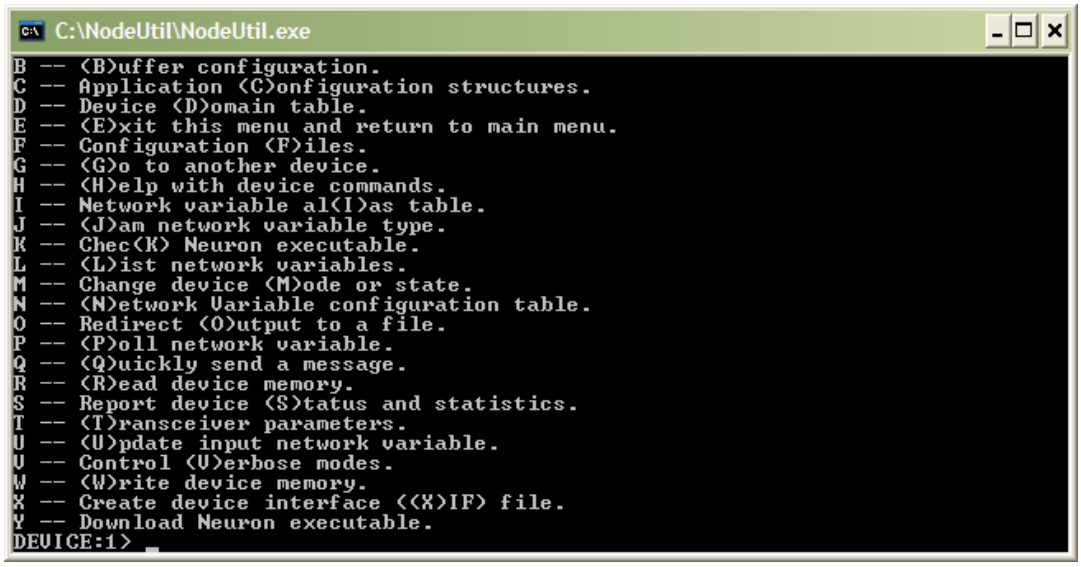

Figure 17-7: NodeUtil Device Menu

5. Type “Y” to Download Neuron Executable. Type the name of the file to program 2011L0052-00Rxx.nxe extracted from the NGA package (2011L0052-00Rxx.zip) and press [Enter]. Note: if the file is not located in the same directory as NodeUtil, type the full path (ex.: c:\folder\sub-folder\2011L0052-00Rxx.nxe).
6. When the file is successfully downloaded, press “Y” to continue.
7. Press “E” twice to exit NodeUtil.

17.3.7 CPLD Erasing and Programming

17.3.7.1 CPLD Erasing

1. Power off the DCU.
2. Connect the CPLD programming cable from the computer to the J8 connector.
3. Power on the DCU.
4. Click on **ispVM System** and choose the **LatticePRO** software.
5. Use the following menu path:
   - File → Open
6. Choose the .wch file you need (2011LXXXX.wch).
7. Select **Part 1** in the *your_file*.wch window.
8. Use the following menu path:
   - Edit → Edit Device
9. The JTAG Part Properties window will appear.
10. In the JTAG Operation field, select option **E = Erase/Initialize Part**.
11. Click on **OK**.
12. Use the following menu path:
    - Project → Process Chain File

13. The LatticePRO Status window will appear.

    Note: If this is the first time the CPLDs is being programmed with this computer, the **Cancel Op** button will change to **Continue** when the chain file is done uploading information. If this is not the first time, two messages will pop up. Click **OK** on all two, then the **Cancel Op** button will change to **Continue** when the chain file is done uploading information.

14. Click on **Continue**.
15. Power off the DCU.
16. Disconnect the cable.
17. Remove the RST jumper.
18. Power on the DCU.

17.3.7.2 CPLD Programming

1. Power off the DCU.
2. Connect the CPLD programming cable from the computer to the J8 connector.
3. Power on the DCU.
4. Click on **ispVM System** and choose the **LatticePRO** software.
5. Use the following menu path:
    - File → Open
6. Choose the .wch file you need (2011LXXXX.wch).
7. Use the following menu path:
    - Project → Process Chain File
8. The LatticePRO Status window will appear.

    Note: If this is the first time the CPLD is being programmed with this computer, the "**Cancel Op"** button will change to "**Continue"** when the chain file is done uploading information. If this is not the first time, three messages will pop up. Click **OK** on all three, then the "**Cancel Op"** button will change to "**Continue"** when the chain file is done uploading information.

9. Click on **Continue**.
10. Power off the DCU.
11. Disconnect the cable.
12. Power on the DCU.

17.3.8 Emulator Setup

Many tests use the emulator to simulate faulty conditions. We use Hitex DProbe-ST10 hardware emulator along with Hitop software to stop the DCU Application while it runs and change the value of specific variables. To start the emulator, execute the following steps.

1. Turn OFF the DCU.
2. Remove the DCU cover.
3. Install the emulator PressOn adaptor, as per its installation manual, on the ST10F276 CPU.
4. Plug the DProbe-ST10 emulator on the PressOn adaptor; stabilize it with the "leg".
5. Turn on the emulator.
6. Turn on the DCU.

17.3.9 Other information about test

In this test procedure, each test section is independent. The initial setup is described in the Section 17.2 (Special Procedural Requirements) and all test cases must be executed sequentially.
All LEDs mentioned in the test procedure are on the DCU.
With PTU Application, connect to the DCU. Open menu item “File” and “Properties”, in “Login option” section, choose DCU Application.
When a test case requires the use of the PTU Application, the tester should make sure that the PTU is connected and logged into the DCU. When the tester closes the PTU Application, he/she does not require saving the project unless it is explicitly mentioned in the procedure.

When a test case requires using an obstacle, the tester must use an object at least 1/4” (6.35 mm) wide by 2” (76.2 mm) tall. The smallest object size placed between Door leaves shall not be smaller than 1/4” (6.35 mm).

Execute all tests with the power supply set to 37.5 Volts DC +/- 0.2V. Verify with the Voltmeter the supply voltage. See Figure 17-2: Test Setup.
When location and side are not specified, the Door ID is R1 (Passenger Car DCU #1).

17.3.10 Start

In the Test Report Sheet [1]write tester name, write time and date the test started. Collect the serial numbers of the used equipment and get the software versions of the tools.

17.3.11 Proceed

Execute instructions from Section 17.4.
It is mandatory that the Test Cases inside a sub-section be done sequentially.

17.3.12 Measure

The way each test case is measured is described in the column “Output Specifications” within sub-sections in Section 17.4. Tools required are listed in the list titled *Special Procedural Requirements* of each sub-section.

17.3.13 Shutdown

It is possible to suspend the test execution. The systems should be powered down while the test is suspended.

17.3.14 Restart

If restarting the test, consider that each sub-section in Section 17.4 has to be done as a whole sequence. Also, be sure that demanded special requirements are fulfilled before restarting.
If the Door Control Unit Application is updated during the execution, the whole test has to be restarted.

17.3.15 Stop

When the test execution is completed, the test system should be restored to its initial state. All the special tools should be disconnected from the system.

17.3.16 Wrap up

Write the “Time & Date Finished” in the Test Report Sheet. Complete software versions used in the test. Add comments, if any. In the *Approvals* part, complete tester, witness(es) name(s), title(s) and the date(s). Update the corresponding Software Test Report file. If required, make copies of the Test Report Sheet.

17.3.17 Contingencies

Each failure, detected during execution of the test, has to be validated with a member of the design team, to determine if this is an error in the procedure or a test failure.
If a test failure is detected, record the anomaly onto a standard Software Problem Report (SPR) form S01-01 using ClearQuest database.

[1] Appendix A:Test Report Sheet

17.4 TC1.1 Integration Test Procedure

The following software configuration items are used in the test procedure: the PTU Application, the PTU Simulator, Car Network Simulator, and Pulsar.

Since the PTU Application and PTU Simulator both use serial communication port (RS-232) and the PTU cable, run only one application at a time to communicate with the DCU. The PTU Simulator should always be in the PTU simulation state by un-checking the DCU checkbox on the tool bar. Make sure the right serial port number is selected before the test.

To use the PTU Application, the PTU Simulator and Pulsar, follow test case instructions.

When requested to download Event Log and verify logged events, only the events added since last download (located at the end of log) should be considered.

17.4.1 TC1.1.1 Self-Test

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.1.1 Continuous Self-Test is started upon startup of the DCU Application	This functionality is tested in the DCU Section 17.5.	N/A	N/A
2	TC1.1.1.2 Self Test execution: SRAM, NVRAM Address, data lines, and pattern test.	This functionality is tested in the DCU Section 17.5.	N/A	N/A
3	TC1.1.1.3 Self Test execution CRC32 DCU Application	This functionality is tested in the DCU Section 17.5.	N/A	N/A
4	TC1.1.1.4 Self Test execution CRC32 DCU Boot Loader Application	This functionality is tested in the DCU Section 17.5.	N/A	N/A
5	TC1.1.1.5 Self-Test execution SPI link.	This functionality is tested in Section 17.4.4 in Table 17-5 Number 6.	N/A	N/A

#	Test Cases	Input Specifications	Output Specifications	P/F
6	TC1.1.1.7 The results of self-tests are stored in DCU SRAM and retrieved by PTU or the Car Network.	1. Using PTU, in Maintenance menu, select Test. All tests will be called automatically. 2. Start Car Network Simulator, and verify Self-Test result in Diagnostic Information bits.	1. DCU returns PTU_SelfTestStatus with all tests results. The tester will see each test result "instantaneously" because no test is executed upon request, only the result of the latest execution is returned. Green checkmarks for success and red 'X' for failure. 2. In Car Network Simulator, Self-Test result is displayed.	
7	TC1.1.1.8 SRAM and NVRAM, failures detected during a self-test shall be logged in the Event Log file.	This test is tested in Section 17.4.4, Table 17-5.	N/A	N/A

17.4.2 TC1.1.2 Initialization

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application and powered ON.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.2.1 DCU Application verifies CRC of following datas and reset if CRC verification fails: Snapshot File, Event Log File, System Counter File	This test is tested in the Test Case #59 of the Table 17-5.	See Output of the Test Case #59 of the Table 17-5.	
2	TC1.1.2.2 DCU Application initializes following output signals to inactive: CPU.PWM.MOTOR, CPU.DIR.OPEN.MOTOR, CPU.DIR.CLOSE.MOTOR, CPU.CUR_LIMIT.NOR.ACT, PAL.ISO, PAS.ISO.	This functionality is tested in the DCU Section 17.5.	N/A	N/A

17.4.3 TC1.1.3 Input Reading

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.3.1 Inputs Filtering, each sense point is filtered to detect transition by comparing previous and current state.	1.Make sure that the location of the DCU is R1. 2.Make sure that Doors are closed and locked. Use PTU to delete the DCU event log: In PTU, click on Maintenance menu and choose “Log Erasure”, click on “Yes”. 3.Open Doors using Door Open CS. (After opening Doors, put back switches to Rig or normal position) 4.Using Pulsar application and pulse generator test fixture, generate 35 ms signals for Door Close CS on the test fixture. Repeat at least 5 times with at least 1 second between pulses. 5.Generate 150 ms signals for Door Close CS on the test fixture.	1.The PTU Application displays R1 on the PTU Control Menu. 2.The Fault LED stays OFF and only event #54, Event Log Erased by PTU is logged. 3.Doors open. 4.Doors do not close. 5.Doors close. .	
2	TC1.1.3.2 Each sense point is polled every 50 ms, converted to its related DCU input signal that is based on the DCU location, and stored in the Temporary Data.	This functionality is tested in the DCU Section 17.5.	N/A	N/A
3	TC1.1.3.3 The transition status is stored in Temporary Data.	This functionality is tested in the DCU Section 17.5.	N/A	N/A
4	TC1.1.3.4 The DCU Application stores all output signals every 50ms in the Temporary Data.	This functionality is tested in the DCU Section 17.5	N/A	N/A

17.4.4 TC1.1.4 Event Detection

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Use PTU Application.*
- *Set location to R1.*
- *Use CNS as defined.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.4.1 The DCU Application monitors the input signals and the system components in order to detect an event. When an event is detected or cleared, the DCU Application updates the specific Event Status accordingly.	1. Configure and start the CNS valid date/time. 2. Execute all test cases included in Table 17-5.	1. The CNS sends the Present Time packet cyclically and the DCU Application should update the time accordingly. Use the PTU to verify that the DCU date/time is updated. 2. See output of Table 17-5.	
2	TC1.1.4.2 The DCU Application stores Event State and Event Log File (either set or reset) in the Non-Volatile memory.	1. Execute Test Case #9 of the Table 17-5. 2. Using PTU, click on Maintenance → Other Functions → Log Download. 3. Reboot the DCU. 4. Using PTU, click on Maintenance → Other Functions → Log Download.	1. See output of Table 17-5. 2. In PTU, the Event Log contains the Event #9, Door Cut-Out Mechanical Lock in Not Normal. 3. DCU is rebooted. 4. In PTU, the Event Log still contains the Event #9, Door Cut-Out Mechanical Lock in Not Normal.	

#	Test Cases	Input Specifications	Output Specifications	P/F
3	TC1.1.4.3 The DCU Application increments event occurrence count in the System Counters File.	1. Using PTU, click on Maintenance → Other Functions → System Counters Download. Write down the number of the following event occurrence: Events #7, 55, and 80. 2. Execute Test Cases #7, #55, and #80 of Table 17-5. 3. Using PTU, click on Maintenance → Other Functions → System Counters Download. Write down the number of the following event occurrence: Event #7, 55, and 80.	1. Occurrence of Events #7:_____ #55:_____ #80:_____ 2. See output of Table 17-5. 3. Occurrences of Events are incremented. #7:_____ #55:_____ #80:_____	
4	TC1.1.4.4 See SACS Table 3-6 “Event List” for specific timing requirement to each event.	Tested in the test cases of the Table 17-5.	See output of Table 17-5.	
5	TC1.1.4.5 The DCU Application puts the operating mode to MODE.DEGRADED.OPEN when any one of “set” Events has attribute of “Disable Open”.	Tested in the test cases of the Table 17-5.	See output of Table 17-5.	
6	TC1.1.4.6 The DCU Application puts the operating mode to MODE.DEGRADED.CLOSE when any one of “set” Events has attribute of “Disable Close”.	Tested in the test cases of the Table 17-5.	See output of Table 17-5.	

#	Test Cases	Input Specifications	Output Specifications	P/F
7	TC1.1.4.7 The DCU Application puts the operating mode to MODE.DEGRADED.SLOW when any one of “set” Events has attribute of “Reduced Speed”.	Tested in the test cases of the Table 17-5.	See output of Table 17-5.	

Table 17-5: TC1.1.5 Events List

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
1	TC1.1.5.1 Watchdog Reset Detected	The DCU logs this event when the processor has been reset by the watchdog timer circuit.	This functionality is tested in the Section 17.5.	N/A	N/A
2	TC1.1.5.2 DCU App. CRC32 Discrepancy	The DCU sets this event when the CRC32 of the application does not match with CRC32 value stored in the Application table. The DCU resets this event when the CRC32 of the application matches with CRC32 value stored in the Application table.	This functionality is tested in the Section 17.5.	N/A	N/A
3	TC1.1.5.3 Boot Loader CRC32 Discrepancy	The DCU sets this event when the CRC32 of the Boot Loader application does not match with CRC32 value stored in the Boot Loader table. The DCU resets this event when the CRC32 of the Boot Loader application matches with CRC32 value stored in the Boot Loader table.	This functionality is tested in the Section 17.5.	N/A	N/A
4	TC1.1.5.4 SRAM Failure	The DCU sets this event when at least one SRAM cell or SRAM addressing does not work properly. The DCU resets this event when the SRAM is fully functional.	This functionality is tested in the Section 17.5.	N/A	N/A
5	TC1.1.5.5 NVRAM Failure	The DCU sets this event when at least one NVRAM cell or NVRAM addressing does not work properly. The DCU resets this event when the NVRAM is fully functional.	This functionality is tested in the Section 17.5.	N/A	N/A

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
6	TC1.1.5.6 SPI Error	The DCU sets this event when communication errors repeated on the Neuron SPI link. The DCU resets this event when the communication is working without error.	1. Use the PTU to delete the DMCU event log: Connect to the controller. Click on Monitoring menu and choose Log Erasure, click on "Yes". Verify that the Door Warning is OFF. 2. Reset the Neuron processor by installing a jumper between Test Connector on the board. Wait ten seconds. 3. Remove the jumper. Using PTU, click on Maintenance → Other Functions → Log Download.	1. Warning LED is OFF. 2. Warning LED goes ON. 3. Warning LED goes OFF. The Event Log contains event #6, SPI Error Detected in set and reset.	
7	TC1.1.5.7 Motor Power not Available	The DCU sets this event when the Motor Power input voltage is below functional range (<22 V). The DCU resets this event when the Motor Power input voltage is within functional range (22-44 V).	1. On the test fixture, turn OFF the Motor Power switch. 2. Put the Motor Power switch to Rig. 3. Using PTU, click on Maintenance → Other Functions → Log Download.	1. Warning LED goes ON. 2. Warning LED goes OFF. 3. In PTU, Event #7, Motor Power not available.	
8	TC1.1.5.8 Motor Power Supply Below 30 V	The DCU sets this event when the Motor Power input voltage is between 22 and 30 Volts. The DCU resets this event when the Motor Power returns over 30 Volts.	1. Lower the voltage on the main power supply down to 28 volts. 2. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 3. Toggle ON/OFF Door Close TL Switch. 4. Using PTU, click on Maintenance → Other Functions → Graph View → Refresh. 5. Put the voltage back to 37.5 volts 6. Using PTU, click on Maintenance → Other Functions → Log Download.	1. The Warning LED flashes once. After 3 seconds from the time power supply generates less than 30V. 2. Doors open slowly due to "Reduced Speed" attribute. 3. Doors close slowly due to "Reduced Speed" attribute. 4. The opening time is 3s. The closing time is 3s and 3s of delay. 5. The Warning LED is off. 6. The Event Log contains event #8, Motor Power Supply Below 30 V.	
9	TC1.1.5.9 Door Cut-Out Mechanical Lock in Not Normal	The DCU sets this event when the Door Cut-Out Mechanical Lock is not in normal. The DCU resets this event when the Door Cut-Out Mechanical Lock signal is in normal.	1. Turn Door Cut-out Mechanical Lock Switch to CUT OFF on the rig. 2. Turn Door Cut-out Mechanical Lock Switch in Cutout position. 3. Using Car Network Simulator, check the diagnostic data section.	1. The Warning LED stays OFF. 2. The Warning LED goes ON. 3. In Car Network Simulator, event 9 is set.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
			4. Turn Door Cut-out Mechanical Lock Switch in Normal position. 5. Using Car Network Simulator, check the diagnostic data section. 6. Using PTU, click on Maintenance → Other Functions → Log Download. 7. On the test fixture, turn the DOOR CUT-OUT MECHANICAL LOOK switch to ON. 8. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches. 9. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches to open Doors. 10. Pull the red Emergency Handle near the Door Operator and push Door panels to fully open position. 11. Toggle ON/OFF Door Close TL Switch to close Doors. 12. Using PTU, click on Maintenance → Other Functions → Graph View → Refresh. 13. On the test fixture, turn the DOOR CUT-OUT MECHANICAL LOOK switch to Rig.	4. The Warning LED goes OFF. 5. In Car Network Simulator, event 9 is reset. 6. In PTU, the Event Log contains the Event #9, Door Panel Latch / Cutout Switch, and the Event #10, Motor Power with Door Panel Latch/Cutout Switch in Cutout, in set and reset states. 7. The Warning LED goes ON. 8. In the PTU, Zero Speed CL (+) and (-) are on. 9. Doors do not open due to “Disable Open” attribute. 10. Doors open manually. 11. Doors close slowly due to “Reduced Speed” attribute. 12. The closing time is 3s and 3 second of delay. 13. The Warning LED goes OFF.	
10	TC1.1.5.10 --	Unused	N/A	N/A	N/A

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
11	TC1.1.5.11 Emergency Handle Activation	The DCU sets this event when the Emergency Handle signal is activated. The DCU resets this event when the Emergency Handle signal is deactivated.	1. Pull the red Emergency Handle. 2. Using Car Network Simulator, check the diagnostic data section. 3. Toggle ON/OFF Door Close TL switch to close doors. 4. Using PTU, click on Maintenance → Other Functions → Log Download. 5. Using Car Network Simulator, check the diagnostic data section.	1. Doors open manually. 2. In Car Network Simulator, event 11 is set. 3. Doors close. 4. In PTU, the Event Log contains event #11, Emergency Handle State Activation Detected in set and reset states. 5. In Car Network Simulator, event 11 is reset.	
12	TC1.1.5.12 --	Unused.	N/A	N/A	N/A
13	TC1.1.5.13 Door Open TLs Mismatch	The DCU sets this event when there is a mismatch between the Door Open (+) TL and the Door Open (-) TL signals. The DCU resets this event when both Door Open (+) and (-) TLs signals are active. Note: The DCU does not set this event if the Door Cut-Out Mechanical Lock is not in Normal.	1. Put Door Open (+) TL switch to OFF. Activate Door Open (-) TL switch on the test fixture. 2. Using Car Network Simulator, check the diagnostic data section. 3. Deactivate Door Open (-) TL switch on the test fixture. Activate and deactivate simultaneously Door Open (+) TL and Door Open (-) TL on the test fixture. 4. Using Car Network Simulator, check the diagnostic data section. 5. Press Close pushbutton on the DCP. Using PTU, click on Maintenance → Other Functions → Log Download.	1. The Door doesn't move. The Warning LED goes ON after 500 ms. 2. In Car Network Simulator, event 13 is set. 3. The Door opens. The Warning LED goes OFF. 4. n Car Network Simulator, event 13 is reset. 5. The Door closes. The Event Log contains the event #13, Door Open TLs Mismatch, in set and reset states.	
14	TC1.1.5.14 Door Unlocking Timeout	The DCU sets this event when the Lock Sensing Switches indicate locked for more than 1.4s after an unlock command. The DCU resets this event when the next time the Lock Sensing Switches indicate unlocked within 1.4s from the unlock command.	1. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches. 2. Press red mushroom button to cut-out power in Door operator. Use the locking plier to lock the solenoid lock in its locked position. Then pull out red mushroom button to bring back the power. 3. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches and wait for 3 seconds. 4. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. Make sure there is no Door command.	1. In the PTU Data Monitoring, Zero Speed CL (+) and (-) are on. 2. The solenoid lock is locked. 3. The motor power is activated. Warning LED goes ON. Doors do not open. 4. Doors stay closed.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
			Take out the locking plier carefully. 5. Open and close Doors toggling ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 6. Using PTU, click on Maintenance → Other Functions → Log Download.	5. Doors open and close. Warning LED goes OFF. 6. In PTU, the Event Log contains event #14, Door Unlocking Timeout, in set and reset states.	
15	TC1.1.5.15 Door Opening Time Longer than Expected	The DCU sets this event when the Door opening exceeds 1s more than the configured opening time. The DCU resets this event when the Door is fully open.	1. In the PTU, configure the opening time to 3 seconds. Then place a wooden obstacle (around 12 to 24 inches) between steel rig frame and the right Door panel's top rear end so that Doors would be blocked while opening. 2. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. When Doors are blocked for about 1 second, take out the wooden obstacle to let it open fully. 3. Using Car Network Simulator, check the diagnostic data section. 4. Toggle ON/OFF Door Close TL Switches. 5. In PTU, click on Maintenance → Other Functions → Graph View → Refresh Data. 6. Open and close Doors. 7. Using PTU, click on Maintenance → Other Functions → Log Download. 8. Using Car Network Simulator, check the diagnostic data section.	1. Wooden obstacle installed. 2. Doors blocked while opening. When an obstacle is removed, Doors open with a delay. The Warning LED goes ON. 3. In Car Network Simulator, the event 15 is set. 4. Doors close slowly due to "Reduce Speed" attribute. 5. The closing time is 3s and 3 second of delay. 6. Doors open and close. The Warning LED goes OFF. 7. In PTU, the Event Log contains event #15, Door Opening Time Longer than Expected, in set and reset states. 8. In Car Network Simulator, the event 15 is reset.	
16	TC1.1.5.16 Door Opening Timeout	The DCU sets this event when the Door opening exceeds 10s more than the configured opening time. The DCU resets this event when the Door is fully open.	1. In the PTU, configure the opening time to 3 seconds. Then place a wooden obstacle (around 12 to 24 inches) between steel rig frame and right Door panel's top rear end so that Doors would be blocked while opening. 2. Toggle ON/OFF Door Open (+) TL and Door Open (-)	1. Wooden obstacle is installed. 2. Doors open little bit and is blocked. After 10	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
			TL switch. Let the Doors blocked for about 11 seconds. 3. Toggle ON/OFF Door Close TL Switches. Take out the wooden obstacle. 4. Open and close Doors. 5. Using PTU, click on Maintenance → Other Functions → Log Download.	seconds. 3. Doors close. 4. Doors open and close. 5. In PTU, the Event Log contains Event #16, “Door Opening Timeout”, in set and reset states.	
17	TC1.1.5.17 Loss of “Closed&Locked” Status w/o Door Command	The DCU sets this event when Doors are unlocked or open without an opening command or an emergency activation. The DCU resets this event when Doors are closed and locked.	**Event with Lock Switch Before Opening** 1. Make sure Doors are closed and locked. On the test fixture, switch ON all SLS switches (Right and Left) at the same time. 2. Using Car Network Simulator, check the diagnostic data section. 3. Turn both SLS switches to Rig. **Event with PSS Before Opening** 4. On the test fixture, turn ON all PSS switches (Right and Left) at the same time. 5. Turn both PSS switches to Rig. 6. Using PTU, click on Maintenance → Other Functions → Log Download. 7. Using Car Network Simulator, check the diagnostic data section.	1. The Warning LED goes ON. 2. In Car Network Simulator, event 17 is set. 3. The Warning LED goes OFF. 4. The Warning LED goes ON. 5. The Warning LED goes OFF. 6. In PTU, the Event Log contains event #17, Loss of “Closed&Locked” Status w/o Cmd, in set and reset states. 7. In Car Network Simulator, event 17 is reset.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
18	TC1.1.5.18 Door Closing Time Longer than Expected	The DCU sets this event when the Door closing exceeds 1s more than the configured closing time. The DCU resets this event when Doors are closed and locked.	1. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches 2. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL switch. 3. On the test fixture, turn ON PSS Right, PSS Left, SLS Right, and SLS Left at the same time. Toggle ON/OFF Door Close TL switch to close Doors. 4. Using Car Network Simulator, check the diagnostic data section. 5. On the test fixture, turn PSS Right, PSS Left, SLS Right, and SLS Left switches to Rig at the same time. Open and close Doors. 6. Using PTU, click on Maintenance → Other Functions → Log Download. 7. Using Car Network Simulator, check the diagnostic data section.	1. In the PTU, Zero Speed CL (+) and (-) are on. 2. Doors open. 3. Doors close. The Warning LED goes ON. 4. In Car Network Simulator, event 18 is set. 5. Doors open and close. The Warning LED goes OFF. 6. In PTU, the Event Log contains event #18, Door Closing Time Longer than Expected, in set and reset states. 7. In Car Network Simulator, event 18 is reset.	
19	TC1.1.5.19 Door Locking Time Longer than Expected	The DCU sets this event when it takes more than 1s from the configured closing time to lock Doors. The DCU resets this event when Doors are closed and locked.	1. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches. 2. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL switches. 3. On the test fixture, turn ON SLS Right and Left switches. Close doors using Door Close TL switch. 4. Using Car Network Simulator, check the diagnostic data section. 5. Turn SLS Right and Left switches to Rig. Open and close Doors using TL switches. 6. Using PTU, click on Maintenance → Other Functions → Log Download. 7. Using Car Network Simulator, check the diagnostic data section.	1. In PTU Data Monitoring, Zero Speed CL (+) and (-) are on. 2. Doors open. 3. Doors close. Approximately 3.6 seconds later, the Warning LED goes ON. 4. In Car Network Simulator, event 19 is set. 5. Doors open and close. The Warning LED goes OFF. 6. In PTU, the Event Log contains event #19, Door Locking Time Longer than Expected, in set and reset. 7. In Car Network Simulator, event 19 is reset.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
20	TC1.1.5.20 Loss of Zero Speed with Door Open	The DCU sets this event when any Zero Speed signal is lost with open or unlocked Doors. The DCU resets this event when both Zero Speed signals are present or Doors are closed and locked.	1. Turn ON Zero Speed CL (+) and (-) on the test fixture **Loss of Zero Speed CL (+) and (-)** 2. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL switches. 3. Turn to OFF Zero Speed (+) and (-) CL on the test fixture. 4. Using Car Network Simulator, check the diagnostic data section while doors are closing. 5. Turn ON Zero Speed (+) CL and (-) CL. **Loss of Zero Speed CL PWR** 6. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL switches. 7. On the test fixture, turn OFF Zero Speed (+) CL. 8. On the test fixture, turn ON Zero Speed (+) CL. **Loss of Zero Speed CL RTN** 9. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL switches. 10. Turn OFF Zero Speed (-) CL. 11. Turn ON Zero Speed (-) CL. 12. Using PTU, click on Maintenance → Other Functions → Log Download. 13. Using Car Network Simulator, check the diagnostic data section.	1. In the PTU Data Monitoring, Zero Speed CL (+) and (-) are on 2. Doors open. 3. The PTU Data Monitoring displays “20” set as Doors are closing. And the PTU Data Monitoring displays “20” reset as Doors are closed and locked. 4. In Car Network Simulator, event 20 is set while doors were closing. 5. The Warning LED is off. 6. Doors open. 7. The PTU Data Monitoring displays “20” set as Doors are closing. And the PTU Data Monitoring displays “20” reset as Doors are closed and locked. 8. The Warning LED is OFF. 9. Doors open. 10. The PTU Data Monitoring displays “20” and “97” set as Doors are closing. And the PTU Data Monitoring displays “20” reset as Doors are closed and locked. 11. The Warning LED is OFF. 12. In PTU, the Event Log contains event #20, Loss of Zero Speed with Door Open, in set and reset states three times. 13. In Car Network Simulator, event 20 is reset.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
21	TC1.1.5.21 PSS Mismatch	The DCU sets this event when there is a mismatch between left and right Panel Sensing Switch signals. The DCU resets this event when both Panel Sensing Switch signals are consistent.	1. Make sure the Door is closed and locked. Turn on the PSS righ switch on the test fixture. 2. Using Car Network Simulator, check the diagnostic data section. 3. On the test fixture, turn off the PSS left switch. 4. Using Car Network Simulator, check the diagnostic data section. 5. Using the PTU, click Log and Refresh.	1. The Warning LED turns on. 2. In Car Network Simulator, event 21 is set. 3. The Warning LED turns off. 4. In Car Network Simulator, event 21 is reset. 5. The log contains only Event #21, PSS Mismatch, in set and reset.	
22	TC1.1.5.22 Door Closing Obstruction	The DCU logs this event when the DCU is unable to close Doors after exceeding the number of configured obstruction retry.	1. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 2. In PTU, select “DCU Configure” from the Administration → Other Function menu. In “Obstruction Recycle” box, select 1 retry; then click “Apply”, and then “Cancel”. 3. Block the right side Door panel 5” away from the fully open position. Toggle ON/OFF Close TL Switches. Let the Doors hit the obstruction. 4. Using Car Network Simulator, check the diagnostic data section. 5. Remove the obstacle, and toggle ON/OFF Door Close TL switch to close the doors. 6. Using Car Network Simulator, check the diagnostic data section. 7. Using the PTU, click Log and Refresh.	1. Doors open. 2. The Door panels are configured to retry to close once upon detection of obstruction. 3. The Door panels move in close direction and hit the obstacle. The Door panel re-opens by 3 inches and retries to close only once. If the obstacle is detected, the door panels fully open. 4. In Car Network Simulator, event 22 is set. 5. Doors close. 6. In Car Network Simulator, event 22 is reset. 7. The log contains Event #22, Door Closing Obstruction, in set and reset.	
23	TC1.1.5.23 Motor Open Supply Error	The DCU sets this event when the motor open supply signal is active without valid logic for more than 2s. The DCU resets this event when the motor open supply signal is inactive.	This functionality is tested in the Section 17.5.	N/A	N/A

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
24	TC1.1.5.24 Solenoid Lock Supply Error	The DCU sets this event when the solenoid lock supply signal is active without a valid logic for more than 0.2s.The DCU resets this event when the solenoid lock supply signal is inactive.	This functionality is tested in the Section 17.5.	N/A	N/A
25	TC1.1.5.25 --	Unused.	N/A	N/A	N/A
26	TC1.1.5.26 --	Unused.	N/A	N/A	N/A
27	TC1.1.5.27 Motor Encoder Phase Lost Error	The DCU sets this event when there are more than 20 transitions on one encoder phase and no transition on the other phase within 5ms. The DCU resets this event whenever there are transitions on both phases.	1. Set Encoder Phase A switch to OFF position. Toggle ON/OFF Door Open (+) and TL Door Open (-) TL switch. 2. Set Encoder Phase A switch to Rig position. Open and close Doors using TL command. 3. Set Encoder Phase /A switch to OFF position. Toggle ON/OFF Door Open (+) and TL Door Open (-) TL switch. 4. Set Encoder Phase /A switch to Rig position. Open and close Doors using TL command. 5. Set Encoder Phase B switch to OFF position. Toggle ON/OFF Door Open (+) and TL Door Open (-) TL switch. 6. Put Encoder Phase B switch in Rig position. Open and close Doors toggling ON/OFF Door Open (+) and TL Door Open (-) TL switch twice. 7. Set Encoder Phase /B switch to OFF position. Toggle ON/OFF Door Open (+) and TL Door Open (-) TL switch. 8. Put Encoder Phase /B switch in Rig position. Open and close Doors using TL command. 9. Using PTU, click on Maintenance → Other Functions → Log Download.	1. The Warning LED goes ON. 2. The Warning LED goes OFF. Doors open slowly. 3. The Warning LED goes ON. Doors do not open. 4. The Warning LED goes OFF. Doors open slowly. 5. The Warning LED goes ON. Doors do not open. 6. The Warning LED goes OFF. Doors open slowly. 7. The Warning LED goes ON. Doors do not open. 8. Doors open slowly. The Warning LED goes OFF. 9. In PTU, the Event Log contains event #27, Motor Encoder Phase Lost Error Detected, in set and reset states four times. Note: Encoder errors may generate false obstructions.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
28	TC1.1.5.28 Motor Encoder Cable/Circuit Error	The DCU sets this event when encoder phase A, phase B, and Index inputs are all on. The DCU resets this event when any one of the encoder signal is off.	1. Turn “Motor Encoder” to OFF on the test fixture. 2. Toggle ON/OFF Door Open (+) and TL Door Open (-) TL switch to open Doors. 3. Turn SW9 back to Rig. 4. Using PTU, click on Maintenance → Other Functions → Log Download.	1. The Fault LED goes ON. 2. Doors do not open due to “Disable Open” attribute. 3. The Fault LED goes OFF. 4. In PTU, the Event Log contains the event #28, Motor Encoder Cable/Circuit Error.	
29	TC1.1.5.29 Motor Control Module Communication Error	The DCU sets this event when the motion control module is unable to handle requests from the Door operator module. The DCU resets this event when the next time a request is accepted.	This functionality is tested in the Section 17.5.	N/A	N/A
30	TC1.1.5.30 Motor Drive Error or Wire	The DCU sets this event when the motor wire is disconnected or the motor does not respond for 0.2s.The DCU resets this event when the motor responds to the DCU command within 0.2s.	1. Turn ON Motor Current switch on the test fixture. Toggle ON/OFF Door Open (+) and TL Door Open (-) TL switch. 2. Using Car Network Simulator, check the diagnostic data section. 3. Turn Motor Current switch to Rig on the test fixture. Push, pull, and push DCS on the DCB. 4. Using PTU, click on Maintenance → Other Functions → Log Download. 5. Using Car Network Simulator, check the diagnostic data section.	1. The Door panel does not open. The Warning LED goes ON. 2. In Car Network Simulator, event 30 is set. 3. The Warning LED goes OFF 4. In PTU, the Event Log contains event #30, Motor Drive Error or Wire Disconnected, in set and reset states. 5. In Car Network Simulator, event 30 is reset.	
31	TC1.1.5.31 Motor Drive Circuit (H-Bridge) Error	The DCU sets this event when the motor current flows with a single active transistor on the motor drive during the test. The DCU resets this event when each transistor passes the test.	This functionality is tested in the Section 17.5.	N/A	N/A
32	TC1.1.5.32 Motor Current Reading Error	The DCU sets this event when the motor current reading is out of the expected range. The DCU resets this event when motor current reading is within range.	This functionality is tested in the Section 17.5.	N/A	N/A

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
33	TC1.1.5.33 Motor Voltage Reading Error	The DCU sets this event when the motor voltage reading is out of expected range. The DCU resets this event when motor voltage reading is within range.	This functionality is tested in the Section 17.5.	N/A	N/A
34	TC1.1.5.34 Normal Current Limit Error	The DCU sets this event when the current limiting circuit does not cap the motor current at the normal current range. The DCU resets this event when the current limiting circuit caps the motor current at the next Door opening.	This functionality is tested in the Section 17.5.	N/A	N/A
35	TC1.1.5.35 High Current Limit Error	The DCU sets this event when the current limiting circuit does not cap the motor current at the high current range. The DCU resets this event when the current limiting circuit caps the motor current at the next Door opening.	This functionality is tested in the Section 17.5.	N/A	N/A
36	TC1.1.5.36 Current Level Selector Error	The DCU sets this event when the DCU is unable to activate the normal current limit. The DCU resets this event when the current limits test is executed successfully.	This functionality is tested in the Section 17.5.	N/A	N/A
37	TC1.1.5.37 Motor OC Shutdown Circuit Failure	The DCU sets this event when the motor over-current shutdown circuit activation or deactivation time is out of tolerances when tested. The DCU resets this event when the current limits test is executed successfully.	1. Turn OFF DCU. On the DCU electronic board, connect pin 1 and 6 of test connector 3 (TC3) with electrical wire or jumper. Turn ON DCU. 2. On the test fixture. Toggle ON/OFF Door Open (+) and TL Door Open (-) TL switch. 3. Remove the jumper. Then reboot DCU. 4. Using PTU, click on Maintenance → Other Functions → Log Download.	1. When DCU turns on, the Motor Over Current Circuit test is performed. The Fault LED goes ON. 2. Doors do not open due to “Disable Open” attribute. 3. After DCU reboots. 4. In PTU, Event #37, Motor OC Shutdown Circuit Failure, is set and reset once. The Fault LED goes OFF.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
38	TC1.1.5.38 Motor OC Shutdown Activated	The DCU sets this event when motor over-current shutdown limit is reached, meaning that excessive current is used for 2s. The DCU resets this event when the motor over-current shutdown circuit is inactive.	This functionality is tested in the Section 17.5.	N/A	N/A
39	TC1.1.5.39 5V Logic Supply Out of Range	The DCU sets this event when the 5V Logic Supply is out of range. The DCU resets this event when the 5V Logic Supply is within range.	This functionality is tested in the Section 17.5.	N/A	N/A
40	TC1.1.5.40 5V Encoder Supply Out of Range	The DCU sets this event when the 5V Encoder Supply is out of range. The DCU resets this event when the 5V Encoder Supply is within range.	This functionality is tested in the Section 17.5.	N/A	N/A
41	TC1.1.5.41 15V Motor Supply Out of Range	The DCU sets this event when the 15V Motor Supply is out of range. The DCU resets this event when the 15V Motor Supply is within range.	This functionality is tested in the Section 17.5.	N/A	N/A
42	TC1.1.5.42 V-Gate Open Supply Out of Range	The DCU sets this event when the V-Gate Open Supply is out of range. The DCU resets this event when the V-Gate Open Supply is within range.	This functionality is tested in the Section 17.5.	N/A	N/A
43	TC1.1.5.43 V-Gate Close Supply Out of Range	The DCU sets this event when the V-Gate Close Supply is out of range. The DCU resets this event when the V-Gate Close Supply is within range.	This functionality is tested in the Section 17.5.	N/A	N/A
44	TC1.1.5.44 Door Closing Visual Alarm OC	The DCU sets this event when the feedback signal is inactive with a Door Closing Visual Light turned on. The DCU resets this event when the feedback signal is active with a light turned on.	1. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL switches. 2. On the test fixture, connect Door Closing Visual Alarm banana plug to Ground using the banana plug wire. Toggle ON/OFF Door Close TL switch. 3. Remove the banana plug wire. Open and close doors using Trainline command switches. 4. Using PTU, click on Maintenance → Other Functions	1. Doors open. 2. The Warning LED goes ON and doors close. 3. Doors open and close. The Warning LED goes OFF. 4. In PTU, the Event Log contains event #44,	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
			→ Log Download.	Door Closing Visual Alarm OC Detected, in set and reset states.	
45	TC1.1.5.45 Door Closing Visual Alarm On w/o Command	The DCU sets this event when the feedback signal is active without a Door Closing Visual Light turned on. The DCU resets this event when the feedback signal is inactive with a light turned off.	This functionality is tested in the Section 17.5.	N/A	N/A
46	TC1.1.5.46 Solenoid Low Side OC	The DCU sets this event when the feedback signal is inactive with a Solenoid Low Side turned on. The DCU resets this event when the feedback signal is active with a Solenoid Low Side turned on.	This functionality is tested in the Section 17.5.	N/A	N/A
47	TC1.1.5.47 Solenoid Low Side On w/o Command	The DCU sets this event when the feedback signal is active without a Solenoid Low Side turned on. The DCU resets this event when the feedback signal is inactive with a Solenoid Low Side turned off.	This functionality is tested in the Section 17.5.	N/A	N/A
48	TC1.1.5.48 Spare Output OC	The DCU sets this event when the feedback signal is inactive with a Spare Output turned on. The DCU resets this event when the feedback signal is active with a Spare Output turned on.	This functionality is tested in the Section 17.5.	N/A	N/A
49	TC1.1.5.49 Spare Output On w/o Command	The DCU sets this event when the feedback signal is active without a Spare Output turned on. The DCU resets this event when the feedback signal is inactive with a Spare Output turned off.	This functionality is tested in the Section 17.5.	N/A	N/A
50	TC1.1.5.50 DCU Temperature Reading Error	The DCU sets this event when the DCU Board Temperature reading is out of expected range. The DCU resets this event when the reading is within range.	This functionality is tested in the Section 17.5.	N/A	N/A

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
51	TC1.1.5.51 Door Open Power TL Stuck	The DCU sets this event when Door Open (+) TL signal is active for more than 20s. The DCU resets this event when the Door Open (+) TL signal becomes inactive.	1. On the test fixture, turn on the Door Open (+) TL switch for more than 20s. 2. Turn off the Door Open (+) TL switch. 3. Using PTU, in Maintenance → Other Functions menu, select “Log Download”.	1. The Warning LED turns on. 2. The Warning LED turns off. 3. In PTU, the Event Log contains event #51, Door Open Power TL Stuck, in set and reset states	
52	TC1.1.5.52 Door Open Return TL Stuck	The DCU sets this event when Door Open (-) TL signal is active for more than 20s. The DCU resets this event when the Door Open (-) TL signal becomes inactive.	1. On the test fixture, turn on the Door Open (-) TL switch for more than 20s. 2. Turn off the Door Open (-) TL switch. 3. Using PTU, in Maintenance → Other Functions menu, select “Log Download”.	1. The Warning LED turns on. 2. The Warning LED turns off. 3. In PTU, the Event Log contains event #52, Door Open Return TL Stuck, in set and reset states	
53	TC1.1.5.53 Door Open Crew Key Stuck	Monitored by L1 and R1 DCU. The DCU sets this event when the Door Open Crew Switch or the Crew Switch Enable signal is activated for more than 20 seconds. The DCU resets this event when the stuck signal is deactivated.	1. Set the location ID to R1, Reboot the DCU. Verify the location ID in the PTU Control Menu. 2. On the test fixture, turn ON CS OPEN PWR and RTN switch for 19 seconds. Then turn OFF both switches. 3. On the test fixture turn CS CLOSE switch ON and OFF to close Doors. 4. On the test fixture, turn ON CS OPEN PWR and RTN switches. And wait for 21 seconds. 5. Using Car Network Simulator, check the diagnostic data section. 6. Toggle ON/OFF Door Close (+) and TL Door Close (-) TL switch to close Doors. 7. Toggle ON/OFF Door Open (+) and TL Door Open (-) TL switch to open Doors. 8. Then turn OFF both CS OPEN PWR and RTN switches. 9. Turn on and off CS OPEN PWR and RTN switches. And turn CS CLOSE ON and OFF to close Doors.	1. The location ID is R1. 2. No fault is generated. Doors Open. 3. Doors close. 4. The Warning LED is ON. 5. In Car Network Simulator, event 53 is set in the diagnostic data. 6. Doors close. 7. Doors do not open due to “Disable Open” attribute. 8. When Door Open CS are off. 9. Doors open and close.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
			10. Using PTU, in Maintenance → Other Functions menu, select "Log Download". 11. Using Car Network Simulator, check the diagnostic data section.	10. In PTU, the Event Log contains event #53, Door Open Crew Switch Open Stuck, in set and reset states. 11. In Car Network Simulator, event 53 is reset in the diagnostic data.	
54	TC1.1.5.54 Event Log Erased by PTU	The DCU logs this event when it receives a request to erase the event log from the PTU.	1. Using PTU, in Maintenance menu, select "Log Erasure". Click "OK". 2. Using PTU, in Maintenance → Other Functions menu, select "Log Download".	1. The Warning LED flashes once. 2. The Warning LED is OFF. The Event Log contains a new pulse of event #54, Event Log Erased by PTU and another of event #55, Event Log Retrieved by PTU.	
55	TC1.1.5.55 Event Log Retrieved by PTU	The DCU logs this event when it receives a request to get the event log from the PTU.	1. Using PTU, in Maintenance → Other Functions menu, select "Log Download".	1. The Warning LED flashes once. In the PTU, the Event Log contains a new pulse of event #55, Event Log Retrieved by PTU.	
56	TC1.1.5.56 --	Unused.	N/A	N/A	N/A
57	TC1.1.5.57 No Time for More Than 10 seconds	The DCU sets this event when it has not received the present time after 10 seconds. The DCU resets this event when it receives the present time.	1. Using the Car Network Simulator, stop the time transmission. 2. Using PTU, in Maintenance → Other Functions menu, select "Log Download".	1. The Warning LED turns on. 2. In the PTU, the Event Log contains event #57, No Time for More than 10s, in set and reset states.	
58	TC1.1.5.58 System Boot-Up	The DCU logs this event after a DCU boot-up.	1. Use PTU to delete the DCU event log. Click on Maintenance menu and choose "Log Erasure", click on "Yes". Verify that the DCU Fault is OFF. 2. Reset the DCU (deactivate and activate DCU power switch). 3. Download Event Log using the PTU and selecting Log Download from the Maintenance → Other Functions menu. Click on Log Download.	1. Fault LED is OFF. 2. After the Door Normal LED starts flashing (meaning DCU Application is running), the Warning LED flashes once. 3. In PTU, the event #58, System Boot-up, has been pulsed 1 time.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
59	TC1.1.5.59 Non-Volatile Data Corrupted	The DCU logs this event when a structure in non-volatile memory has been reset to its default value due to CRC16 mismatch at boot-up.	1. Turn off DCU and close the PTU Application. Turn ON the Bootstrap switch on the PTU DB-9 cable to set the controller in bootstrap mode. Start CTA-5000 DCU CPU Test application; open the CTA-5000.tdb file and select 3 NVRAM pattern tests. Turn on DCU. From the Run menu, select “Run Selected Test”. Wait for the result of the test. Exit the CPU Test Application. Turn off DCU. Remove the PTU DB-9 cable. 2. Turn on DCU. Start the PTU application. Using PTU, click on Maintenance → Other Functions → Log Download.	1. NVRAM is corrupted. 2. In PTU, the Event Log contains more than one “pulse” of the event #59, Non-Volatile Data Corrupted.	
60	TC1.1.5.60 --	Unused.	N/A	N/A	N/A
61	TC1.1.5.61 --	Unused.	N/A	N/A	N/A
62	TC1.1.5.62 Domain Table Failure Detected by Neuron	The DCU sets this event when the Neuron Firmware detects a failure. The DCU resets this event when the Neuron Firmware error is cleared.	This functionality is tested in the Section 17.5.	N/A	N/A
63	TC1.1.5.63 Neuron Application CRC32 Failure	The DCU sets this event when the CRC32 of the Neuron Application does not match with CRC32 value stored in the Application table. The DCU resets this event when the CRC32 of the application matches with CRC32 value stored in the Application table.	This functionality is tested in the Section 17.5.	N/A	N/A
64	TC1.1.5.64 --	Unused.	N/A	N/A	N/A
65	TC1.1.5.65 Door Close TL Stuck	The DCU sets this event when the Door Close TL is active for more than 20s. The DCU resets this event when the Door Close TL becomes inactive.	1. On the test fixture, turn on the Door Close TL switch for 19s. Then turn off the switch. 2. Turn on the Door Close TL switch and wait for 21s. 3. Using Car Network Simulator, check the diagnostic data section. 4. Turn off the switch. 5. Using Car Network Simulator, check the diagnostic data section. 6. Using PTU, in Maintenance → Other Functions menu, select “Log Download”.	1. The Warning LED stays OFF. No fault is generated. 2. The Warning LED turns on. 3. In Car Network Simulator, event 30 is set. 4. The Warning LED turns off. 5. In Car Network Simulator, event 30 is set. 6. The Event Log contains event #65, Door Close TL Stuck, in set and reset states.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
66	TC1.1.5.66 Door Close Crew Switch Stuck	Monitored by L1 and R1 DCU. The DCU sets this event when the Door Close Crew Switch signal is active for more than 20s. The DCU resets this event when the Door Close Crew Switch signal becomes inactive.	1. Set the location ID to R1. Reboot the DCU. Verify the location ID in the PTU Control Menu. 2. On the test fixture, turn ON Door Close CS for 19 seconds. Then turn it OFF. 3. Turn ON Door Close CS. And wait for 21 seconds. 4. Using Car Network Simulator, check the diagnostic data section. 5. Then turn OFF Door Close CS. 6. Using PTU, in Maintenance → Other Functions menu, select “Log Download”. 7. Using Car Network Simulator, check the diagnostic data section.	1. The location ID is Door1. 2. No fault is generated. 3. The Warning LED is ON. 4. In Car Network Simulator, Event #66 is set in the diagnostic data. 5. Warning LED is OFF. 6. In PTU, the Event Log contains event #66, Door Close Crew Switch Close Stuck, in set and reset states. 7. In Car Network Simulator, Event #66 is reset in the diagnostic data.	
67	TC1.1.5.67 Crew Switch Signal at Invalid Location	The DCU sets this event when any one of crew switch inputs is active at a non crew switch location. The DCU resets this event when none of the crew switch inputs are active.	1. Set the location ID to R2. Reboot the DCU. Verify the location ID in the PTU Control Menu. 2. On the test fixture, turn Door Open CS PWR and RTN switches ON. 3. Toggle ON/OFF Door Open (+) and TL Door Open (-) TL switch to open Doors. 4. Turn Door Open PWR and RTN switches OFF. 5. On the test fixture, turn Door Close CS ON. 6. Turn Door Close CS OFF. 7. Repeat the previous test (Steps #2 ~ #6) at location L2. Verify the location ID in the PTU Control Menu. 8. Change the location of the DCU to R1, and try to open and close Doors using CS (either using the key in the CS or in the test fixture). 9. Change the location of the DCU to L1, and try to open and close Doors using CS (either using the key	1. The location ID is R2. 2. The Warning LED goes ON. Doors do not open. 3. Doors do not open due to “Disable Open” attribute. 4. The Warning LED goes OFF. 5. The Warning LED goes ON. 6. The Warning LED goes OFF. 7. Same results as location ID R2 above. 8. Doors open, and then close, the Warning LED stays OFF. 9. Doors open, and then close, the Warning LED stays OFF.	

#	Event Name	Event Description	Test Description		
			Input Specification	**Output Specification**	**P/F**
			in the CS or in the test fixture). 10. Using PTU, click on Maintenance → Other Functions → Log Download.	10. In PTU, the Event Log contains the event #67, Crew Switch Signal at Invalid Location, in set and reset states.	
68	TC1.1.5.68 LSS Mismatch with Door Open	The DCU sets this event when there is mismatch between left and right Lock Sensing Switch signals with unlocked Door. The DCU resets this event when both Lock Sensing Switches signals are consistent with unlocked Door.	*Note: There are two Lock Switches. Proceed below test with #1 switch first, then repeat all with #2 switch, and with both switches at the last.* 1. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches. Make sure Doors are closed. **LS Error Before Opening** 2. On the test fixture, turn ON SLS switch. 3. Using Car Network Simulator, check the diagnostic data section. 4. Using PTU, observe the Motor Current in the Data Monitoring for the test case #7 of the Section 17.4.6. 5. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL switches to attempt to open Doors. 6. Turn SLS switch to Rig. **LS Error After Opening** 7. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL switches to open Doors. 8. Turn ON SLS switch. 9. Toggle ON/OFF Door Open (+) and TL Door Open (-) TL switch. 10. Using PTU, observe the Motor Current in the Data Monitoring for the test case #7 of the Section 17.4.6. 11. Toggle ON/OFF Door Open (+) and TL Door Open (-) TL switch to attempt to open Doors. 12. Turn SLS switch to Rig. 13. Using PTU, click on Maintenance → Other Functions	*Note: In order to pass the test, all tests must pass with every combination of LS.* 1. Doors stay closed and locked. 2. The Warning LED turns on. 3. In Car Network Simulator, Event #68 is set. 4. The Motor Current is applied in the negative value. 5. Doors stay closed due to “Disable Open” attribute. 6. The Warning LED goes OFF. 7. Doors open. 8. Doors stay open. 9. Doors close fully. The Warning LED goes ON. 10. The Motor Current is applied in the negative value. 11. Doors stay closed. 12. The Warning LED goes OFF. 13. In PTU, the Event Log contains event #68	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
			→ Log Download. 14. Using Car Network Simulator, check the diagnostic data section.	LSS Mismatch with Door Open, in set and reset states. 14. In Car Network Simulator, Event #68 is reset.	
69	TC1.1.5.69 LSS Mismatch with Door Closed	The DCU sets this event when there is mismatch between left and right Lock Sensing Switch signals with locked Door. The DCU resets this event when both Lock Sensing Switches signals are consistent with locked Door.	1. Make sure doors are closed and locked. 2. On the test fixture, turn on the Lock Sensing Switch #1. 3. Using Car Network Simulator, check the diagnostic data section. 4. Turn off the Lock Sensing Switch #1. 5. Repeat the test step 1 to 4 with Lock Sensing Switch #2. 6. Using the PTU, click Log and Refresh. 7. Using Car Network Simulator, check the diagnostic data section.	1. Doors are closed. 2. The Warning LED turns on. 3. In Car Network Simulator, event 69 is set in the diagnostic data. 4. The Warning LED turns off. 5. Same results as before. 6. Verify the event log #68, Lock Activated with Door Open, is set and reset once. 7. In Car Network Simulator, event 69 is reset in the diagnostic data.	
70	TC1.1.5.70 Crew Switch Open Signals Mismatch	Monitored by L1 and R1 DCU. The DCU sets this event when the Crew Switch Door Open signal is not consistent with Crew Switch Enable for 8s.The DCU resets this event when Crew Switch Door Open and Crew Switch Enable are consistent.	1. Set the location ID to R1. Reboot the DCU. Verify the location ID in the PTU Control Menu. 2. On the test fixture, for only 7 seconds, turn ON Door Open CS switch. 3. On the test fixture, for only 7 seconds, turn ON Door Enable CS switch. 4. Turn ON Door Open CS switch. And wait for 9s. 5. Turn RIG Door Open CS switch. 6. Turn ON Door Enable CS switch. And wait for 9s. 7. Turn RIG Door Open CS switch. 8. Toggle on and off Door Open TL and Door Unlock TL to clear the event 23 “Motor Open Supply Error”.	1. The location ID is R1. 2. The Warning LED stays OFF. (The fault LED might be on due to event #23) 3. The Warning LED stays OFF. 4. The Warning LED blinks 5. None. 6. The Warning LED blinks. 7. The fault LED might be on due to event #23 8. None.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
			9. Repeat the test (step #2 to #9) for locations L1. Verify the location ID in the PTU Control Menu. 10. Using PTU, click on Maintenance → Other Functions → Log Download.	9. Same output as at location R1. 10. In PTU, the Event Log contains several occurrences of the event #70, Crew Switch Open Signals Mismatch, in set and reset states.	
71	TC1.1.5.71 Door Summary Relay High Error	The DCU sets this event when the DSR High is energized with open and unlocked Door for 3s. The DCU resets this event when signals are consistent. Note: When DCML is not in normal, the DCU won't set this event.	1. Open the door. 2. Activate DSR + monitoring. 3. Using Car Network Simulator, check the diagnostic data section. 4. Deactivate DSR+ monitoring 5. Using Car Network Simulator, check the diagnostic data section. 6. Close the door. Using PTU, click on Maintenance → Other Functions → Log Download.	1. Doors are open. 2. Event is logged 3. In Car Network Simulator, event 71 is set in the diagnostic data. 4. Event is cleared 5. In Car Network Simulator, event 71 is reset in the diagnostic data. 6. In PTU, the Event Log contains event #71, Door Summary Relay High Error, in set and reset states.	
72	TC1.1.5.72 Door Summary Relay Low Error	The DCU sets this event when the DSR Low is energized with open and unlocked Door for 3 seconds. The DCU resets this event when the signals are consistent. Note: When DCML is not in normal, the DCU won't set this event.	1. Open the door. 2. Activate DSR – monitoring 3. Using Car Network Simulator, check the diagnostic data section. 4. Deactivate DSR – monitoring 5. Using Car Network Simulator, check the diagnostic data section. 6. Close the door. Using PTU, click on Maintenance → Other Functions → Log Download.	1. Doors are open. 2. Event is logged 3. In Car Network Simulator, event 72 is set in the diagnostic data. 4. Event is cleared 5. In Car Network Simulator, event 72 is reset in the diagnostic data. 6. In PTU, the Event Log contains event #72, Door Summary Relay Low Error, in set and reset states.	
73	TC1.1.5.73 Zero Speed Carlines Mismatch	The DCU sets this event when there is mismatch between (+) and (-) Zero Speed CL signals for 2s. The DCU resets this event when both Zero Speed signals are consistent.	1. Turn off Zero Speed (+) CL and Zero Speed (-) CL switches. 2. On the test fixture, turn ON Zero Speed (+) CL switch. Wait 3s.	1. In the PTU Data Monitoring, Zero Speed (+) CL and (-) CL are both off. 2. The Warning LED goes ON.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
			3. Using Car Network Simulator, check the diagnostic data section. 4. Turn off Zero Speed (+) CL switch. 5. On the test fixture, turn ON Zero Speed (-) CL switch. Wait 3s. 6. Turn off Zero Speed (-) CL switch. 7. Using PTU, click on Maintenance → Other Functions → Log Download. 8. Using Car Network Simulator, check the diagnostic data section.	3. In Car Network Simulator, event 73 is set. 4. The Warning LED goes OFF. 5. The Warning LED goes ON. 6. The Warning LED goes OFF. 7. In PTU, the Event Log contains event #73, Zero Speed Car Lines Mismatch, in set and reset states. 8. In Car Network Simulator, event 73 is reset.	
74	TC1.1.5.74 Crew Switch w/o Zero Speed	Monitored only by L1 and R1 DCU. The DCU sets this event when any one of Crew Switch signals is active without the presence of Zero Speed signals for 1s. The DCU resets this event when the Zero Speed signals are active or when the Crew Switch signals are inactive.	1. Put simultaneously Zero Speed (+) CL and Zero Speed (-) CL switches to OFF position. Activate the Crew Switch in Open position. 2. Using Car Network Simulator, check the diagnostic data section. 3. Release the Crew Switch. 4. Using Car Network Simulator, check the diagnostic data section. 5. Put simultaneously simultaneously Zero Speed (+) CL and Zero Speed (-) CL switches to on position. Using PTU, click on Maintenance → Other Functions → Log Download.	1. The Warning LED goes ON. 2. In Car Network Simulator, event 74 is set in the diagnostic data. 3. The Warning LED goes OFF. 4. In Car Network Simulator, event 74 is reset in the diagnostic data. 5. The Warning LED goes OFF. The Event Log contains event #74, Crew Switch w/o No-Motion Detected, in set and reset states.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
75	TC1.1.5.75 Door Open PWR TL w/o Zero Speed	The DCU sets this event when the Door Open (+) TL is activated without the presence of Zero Speed signals. The DCU resets this event when the Zero Speed signals are active or when the Door Open (+) TL is inactive.	1. Set the location ID to R1. Reboot the DCU. Verify the location ID in the PTU Control Menu. 2. Turn off Zero Speed (+) CL and Zero Speed (-) CL switches. And turn ON Door Open (+) TL switch. 3. Using Car Network Simulator, check the diagnostic data section. 4. Turn off Door Open TL switch, and toggle On/OFF both Door Open (+) and Door Open (-) TL switches. 5. Using PTU, click on Maintenance → Other Functions → Log Download. 6. Using Car Network Simulator, check the diagnostic data section.	1. The location ID is R1. 2. The Warning LED goes ON. 3. In Car Network Simulator, event 75 is set. 4. The Warning LED goes OFF. 5. In PTU, the Event Log contains event #75, Door Open PWR TL w/o Zero Speed, in set and reset states. 6. In Car Network Simulator, event 75 is reset.	
76	TC1.1.5.76 Door Open RTN TL w/o Zero Speed	The DCU sets this event when the Door Open (-) TL is activated without the presence of Zero Speed signals. The DCU resets this event when the Zero Speed signals are active or when the Door Open (-) TL is inactive.	1. Turn off Zero Speed (+) CL and Zero Speed (-) CL switches. And turn ON Door Open (-) TL switch. 2. Using Car Network Simulator, check the diagnostic data section. 3. Turn off Door Open (-) TL switch, and toggle On/Off both Door Open (+) and Door Open (-) TL switches. 4. Using PTU, click on Maintenance → Other Functions → Log Download. 5. Using Car Network Simulator, check the diagnostic data section.	1. The Warning LED goes ON. 2. In Car Network Simulator, event 76 is set. 3. The Warning LED goes OFF. 4. In PTU, the Event Log contains event #76, Door Open RTN TL w/o Zero Speed, in set and reset states. 5. In Car Network Simulator, event 76 is reset.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
77	TC1.1.5.77 Passenger Door Enable TL Mismatch	The DCU sets this event when there is a mismatch between the (+) and (-) Passenger Door Enable TL signals. The DCU resets this event when both Passenger Door Enable TL signals are same.	1. Activate Passenger Door Enable (+) TL switch for 3 seconds 2. Using Car Network Simulator, check the diagnostic data section. 3. Deactivate Passenger Door Enable (+) TL switch. 4. Activate Passenger Door Enable (-) TL switch. 5. Deactivate Door Enable (-) TL switch. 6. Using Car Network Simulator, check the diagnostic data section. 7. Using PTU, click on Maintenance → Other Functions → Log Download.	1. The Warning LED is ON. 2. In Car Network Simulator, event 77 is set in the diagnostic data. 3. The Warning LED is OFF 4. The Warning LED is ON. 5. The Warning LED is OFF 6. In Car Network Simulator, event 77 is reset in the diagnostic data. 7. In PTU, the Event Log contains two Event #77, Passenger Door Enable TL Mismatch, in set and reset states.	
78	TC1.1.5.78 Passenger Door Enable TL Stuck	The DCU sets this event when the both Passenger Door Enable TL signals are stuck high for more than 20s.The DCU resets this event when both Passenger Door Enable TL signals are inactive.	1. Activate Passenger Door Enable TL (+) for 19 seconds, then turn it OFF. 2. Activate Passenger Door Enable TL (+) for 21s. 3. Using Car Network Simulator, check the diagnostic data section. 4. Turn OFF Passenger Door Enable TL. 5. Using Car Network Simulator, check the diagnostic data section. 6. \Using PTU, click on Maintenance → Other Functions → Log Download.	1. The Warning LED is OFF. 2. The Warning LED is ON. 3. In Car Network Simulator, event 78 is set in the diagnostic data. 4. The Warning LED is OFF. 5. In Car Network Simulator, event 78 is reset in the diagnostic data. 6. In PTU, the Event Log contains Event #78, Passenger Door Enable TL Stuck, in set in reset states.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
79	TC1.1.5.79 Passenger Door Enable TL w/o Zero Speed	The DCU sets this event when either the Passenger Door Enable (+) TL or Enable (-) TL is activated without the presence of Zero Speed signals. The DCU resets this event when both Passenger Door Enable TL signals are inactive or both Zero Speed signals are active.	1. Deactivate Zero Speed (+) CL and Zero Speed (-) CL. 2. Activate Passenger Door Enable (+) TL for 3 seconds. 3. Using Car Network Simulator, check the diagnostic data section. 4. Deativate Passenger Door Enable (+) TL. 5. Activate Passenger Door Enable (-) TL for 3 seconds. 6. Deactivate Passenger Door En (–) TL. 7. Using Car Network Simulator, check the diagnostic data section. 8. Reactivate Zero Speed (+) CL and Zero Speed (-) CL. Using PTU, click on Maintenance → Other Functions → Log Download.	1. None. 2. The Warning LED is ON. 3. In Car Network Simulator, event 79 is set in the diagnostic data. 4. The Warning LED is OFF 5. The Warning LED is ON. 6. The Warning LED is OFF. 7. In Car Network Simulator, event 79 is reset in the diagnostic data. 8. In PTU, the Event Log contains Event #79, Passenger Door Enable TL Stuck, in set in reset states.	
80	TC1.1.5.80 Passenger Door Open Pushbutton Stuck	The DCU sets this event when the Passenger Door Open Pushbutton is stuck high for more than 20s.The DCU resets this event when the Passenger Door Open Pushbutton is inactive.	1. Activate Passenger Door Open Pushbutton switch for 19 seconds then turn it OFF. 2. Activate Passenger Door Open Pushbutton switch for 21 seconds. 3. Using Car Network Simulator, check the diagnostic data section. 4. Turn Passenger Door Open Pushbutton switch OFF. 5. Using Car Network Simulator, check the diagnostic data section. 6. Using PTU, click on Maintenance → Other Functions → Log Download.	1. The Warning LED is OFF 2. The Warning LED is ON. 3. In Car Network Simulator, event 80 is set in the diagnostic data. 4. The Warning LED turns OFF 5. In Car Network Simulator, event 80 is reset in the diagnostic data. 6. In PTU, the Event Log contains Event #80, Passenger Door Enable TL Stuck, in set in reset states.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
81	TC1.1.5.81 Passenger Door Enable Light OC	The DCU sets this event when the feedback signal is inactive with a Passenger Door Enable Light turned on. The DCU resets this event when the feedback signal is active with a light turned on.	1. On the test fixture, turn on Zero Speed (+) and Zero Speed (-) CL switches. 2. On the test fixture, connect Passenger Door Enable Light banana plug to Ground using the banana plug wire. Turn on Passenger Door Enable (+) TL and Passenger Door Enable (-) TL switches. 3. Turn off Passenger Door Enable (+) TL and Passenger Door Enable (-) TL. 4. Using PTU, click on Maintenance → Other Functions → Log Download.	1. None. 2. The Warning LED is ON. 3. The Warning LED is OFF. 4. In PTU, the Event Log contains Event #81, Passenger Door Enable Light OC, in set and reset states.	
82	TC1.1.5.82 Passenger Door Enable Light On w/o Command	The DCU sets this event when the feedback signal is active without a Passenger Door Enable Light turned on. The DCU resets this event when the feedback signal is inactive with a light turned off.	This functionality is tested in the Section 17.5.	N/A	N/A
83	TC1.1.5.83 Door Open TL w/o Mechanical Lock Normal	The DCU sets this event when the Door Open TL signal is activated without the Mechanical Lock in normal. The DCU resets this event when the Door Open TL is inactive or when the Mechanical Lock is in normal.	1. On the test fixture, turn on Zero Speed (+) CL and Zero Speed (-) CL switches. Turn on Door Cut-out Mechanical Lock. 2. Turn on the Door Open (+) TL and (-) TL switches. 3. Using Car Network Simulator, check the diagnostic data section. 4. Turn off Door Open (+) TL and (-) TL switches. 5. Using Car Network Simulator, check the diagnostic data section. 6. Using PTU, click on Maintenance → Other Functions → Log Download.	1. DCU logs event 9 and Warning LED turns on. 2. The Warning LED stays on. 3. In Car Network Simulator, event 83 is set in the diagnostic data. 4. The Warning LED stays on due to event 9. 5. In Car Network Simulator, event 83 is reset in the diagnostic data. 6. The Event Log contains event #83, Door Open TL w/o Mechanical Lock Normal, in set and reset states.	
84	TC1.1.5.84 --	Unused.	N/A	N/A	N/A

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
85	TC1.1.5.85 Invalid CPLD Version	The DCU sets this event when the CPLD returns the invalid version or lower than the minimum expected version. The DCU resets this event when the CPLD returns a version equals or higher than the minimum expected version.	1. Reprogram the CPLD using 2007L0123 Version 0 with the Lattice software as seen Section 17.3.7.2 for details. 2. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 3. Reprogram the CPLD with the latest released revision using the Lattice software as seen Section 17.3.7.2 for details. 4. Using PTU, click on Maintenance → Other Functions → Log Download.	1. The Fault LED is ON. 2. Doors do not open due to “Disable Open” attribute. 3. The Fault LED is OFF. 4. In PTU, the Event Log contains event #85, Invalid CPLD Version Detected, in set and reset states.	
86	TC1.1.5.86 CPLD Test Version	The DCU sets this event when the CPLD returns an unofficial version. The DCU resets this event when the CPLD returns an official release version.	1. Reprogram the CPLD using Section 17.3.7.2 with the Lattice software as seen Section 17.3.7.2 for details. 2. Reprogram the CPLD using Section 17.3.7.2 with the Lattice software. 3. Using PTU, click on Maintenance → Other Functions → Log Download.	1. The Warning LED is ON. 2. The Warning LED is OFF. 3. In PTU, the Event Log contains event #86, CPLD Test Version Detected, in set and reset states.	
87	TC1.1.5.87 Analog Reference Error	The DCU sets this event when the analog reference value is more than 5% off. The DCU resets this event when the analog reference vale is less than 3% off.	This functionality is tested in the Section 17.5.	N/A	N/A
88	TC1.1.5.88 Door Closing Audible Alarm OC	The DCU sets this event when the feedback signal is inactive with a Door Closing Audible Alarm turned on. The DCU resets this event when the feedback signal is active with an alarm turned on.	1. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL switches. 2. On the test fixture, connect Door Closing Audible Alarm banana plug to Ground using the banana plug wire. Toggle ON/OFF Door Close TL switch. 3. Remove the banana plug wire. Open and close doors using Trainline command switches. 4. Using PTU, click on Maintenance → Other Functions → Log Download.	1. Doors open. 2. The Warning LED goes ON and doors close. 3. Doors open and close. The Warning LED goes OFF. 4. In PTU, the Event Log contains Event #88, Door Closing Audible Alarm OC, in set and reset states.	

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
89	TC1.1.5.89 Door Closing Audible On w/o Command	The DCU sets this event when the feedback signal is active without a Door Closing Audible Alarm turned on. The DCU resets this event when the feedback signal is inactive with an alarm turned off.	This functionality is tested in the Section 17.5.	N/A	N/A
90	TC1.1.5.90 No Open Interlock Feedback with Output On	The DCU sets this event when the feedback signal is inactive with an Open Interlock turned on. The DCU resets this event when the feedback signal is active with an Open Interlock turned on.	This functionality is tested in the Section 17.5.	N/A	N/A
91	TC1.1.5.91 Open Interlock Feedback On w/o Output On	The DCU sets this event when the feedback signal is active without an Open Interlock turned on. The DCU resets this event when the feedback signal is inactive with an Open Interlock turned off.	This functionality is tested in the Section 17.5.	N/A	N/A
92	TC1.1.5.92 Solenoid High Side OC	The DCU sets this event when the feedback signal is inactive with a Solenoid High Side turned on. The DCU resets this event when the feedback signal is active with a Solenoid High Side turned on.	This functionality is tested in the Section 17.5.	N/A	N/A
93	TC1.1.5.93 Solenoid High Side ON w/o Command	The DCU sets this event when the feedback signal is active without a Solenoid High Side turned on. The DCU resets this event when the feedback signal is inactive with a Solenoid High Side turned off.	This functionality is tested in the Section 17.5.	N/A	N/A

#	Event Name	Event Description	Test Description		
			Input Specification	Output Specification	P/F
94	TC1.1.5.94 Motor Rotation with Door Locked	The DCU sets this event when the motor is able to turn with Door locked and closed. The DCU resets this event when the motor does not turn with locked and closed Door.	1. Shut down the DCU (with mushroom button on rig). On the fixture, put to OFF (center) position the following switches: • SLS1 (Lock SW #1) • SLS2 (Lock SW #2) • PSS #1 • PSS #2 2. Turn back ON the DCU, and hold the door's bottom with an obstacle (your feet) during 4 seconds. 3. On the fixture, activate the Door Close CS switch. 4. On the fixture, activate the Door Open CS switch. 5. On the test fixture, put back to down position SLS and PSS switches. 6. Press OPEN button on the DCP. 7. Press CLOSE button on the DCP. 8. Using PTU, click on Maintenance → Other Functions → Log Download.	1. None. 2. The door pushes lightly during around 4 seconds. 3. The Warning LED is ON. 4. The door does NOT move. 5. None. 6. The door opens. 7. The Warning LED is OFF. 8. In PTU, the Event Log contains Event #94, Motor Rotation with Door Locked, in set and reset states.	

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Note: Go back to the beginning of this section to complete test cases 1 and 2 after execution of Table 17-5.

17.4.5 TC1.1.6 Door Open Sequence

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Use the PTU Application with the administrator access*
- *Make sure that the location of the DCU under test is R1.*
- *Make sure the Doors are closed and locked.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.6.2 Valid Door Open Trainlines	1. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches. 2. Using test fixture, generate signal like signal pattern 1. (Figure 17-4) Door Open (+) TL is A and Door Open (+) TL is B. 3. Using test fixture, generate signal like signal pattern 2. Door Open (+) TL is A and Door Open (-) TL is B. 4. Pull and Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 5. Using test fixture, generate signal like signal pattern 1. Door Open (-) TL is A and Door Open (+) TL is B. 6. Using test fixture, generate signal like signal pattern 2. Door Open (-) TL is A and Door Open (+) TL is B. 7. Pull and Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches.	1. In PTU Data Monitoring, Zero Speed CL (+) and (-) are on. 2. Doors stay closed. 3. Doors open. 4. Doors close. 5. Doors stay closed. 6. Doors open. 7. Doors close.	
2	TC1.1.6.3 Doors open according to configured time limit. Opening time is adjustable from 2.0 to 3.0 seconds in 0.5 seconds increments; 2.0 is default	1. In PTU, select “DCU Configure” from the Administration → Other Function menu. In “Opening Time” combo box, select 2.0 seconds then in “Delay Before Opening” box, select 0 then click “Apply”, and then “Cancel Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. When Doors are open, select Graph Download from the Maintenance → Other Functions menu in PTU. 2. Toggle ON/OFF Door Close TL switch.	1. Doors open. Verify that the opening travel time is between 1.8 and 2.2 seconds. Opening time (2.0): _____ 2. Doors close.	

#	Test Cases	Input Specifications	Output Specifications	P/F
		3. In PTU, select “DCU Configure” from the Administration → Other Function menu. In “Opening Time” box, select 2.5 seconds; click “Apply”, and then “Cancel”. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. When Doors are open, click Refresh button on the PTU Graph window. 4. Toggle ON/OFF Door Close TL switch. 5. In PTU, select “DCU Configure” from the Administration → Other Function menu. In “Opening Time” combo, select 3.0 seconds; click “Apply”, and then “Cancel”. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. When Doors are open, click Refresh button on the PTU Graph window.	3. Doors open. Verify that the opening travel time is between 2.3 and 2.7 seconds and the delay before Door movement is between 0.0 and 0.50 second. Opening time (2.5): _____ 4. Doors close. 5. Doors open. Verify that the opening travel time is between 2.8 and 3.2 seconds. Opening time (3.0): _____	
3	TC1.1.6.4 Valid Door Open Condition.	1. Turn OFF the Zero Speed (+) and (-) CL switches on the test fixture to disable the condition to open. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 2. Turn ON the Zero Speed (+) and (-) CL switches on the test fixture to enable the condition to open. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 3. Toggle ON/OFF Door Close TL switch. 4. Activate the Door Open CS by using appropriate crew key. 5. Activate the Door Close CS by using appropriate crew key.	1. Doors stay closed and locked. 2. Doors open. 3. Doors close and lock. 4. Doors open. 5. Doors close and lock.	

#	Test Cases	Input Specifications	Output Specifications	P/F
4	TC1.1.6.5 Valid Door Open Crew Switch Signal.	1. Make sure Door Open and Door Enable CS switches are on RIG position. On the test fixture create a CS mismatch by turning ON the Door Open CS first. Then after at least 9s, turn ON Door Enable CS. 2. Using PTU, click on Maintenance → Other Functions → Log Download. 3. Repeat step 1 with 7 sec apart.	1. Doors open. 2. In the PTU, the Event Log contains one Set and Reset of Event #70, Crew Switch Open Signals Mismatch. 3. Doors open.	
5	TC1.1.6.6 Opening Sequence With Passenger Alerting Light and Passenger Audible Signal	Tested in Test Case #8 of Section 17.4.6.	See output of Test Case #8 of Section 17.4.6.	

17.4.6 TC1.1.7 Door Close Sequence

Special Procedural Requirements:

- *Continue the test from the Section 17.4.5 of page 56.*
- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Use the PTU Application with the administrator access*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.7.1 Doors close according to configured time limit. Closing time is adjustable from 2.0 to 3.0 seconds in 0.5 seconds increments; 2.5 is default. Delay before closing is adjustable from 0 to 5 seconds in 1 second increments; 3.0 is default.	1. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches. 2. In PTU, select “DCU Configure” from the Administration → Other Function menu. In “Closing Time” combo box, select 2.0 seconds; in “Delay Before Closing” combo box, select 0; then click “Apply”, and then “Cancel”. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. When the doors are FULLY opened, toggle ON/OFF Door Close TL switch. When Doors are closed, click Refresh button on the PTU Graph window. 3. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 4. In PTU, select “DCU Configure” from the Administration → Other Function menu. In “Closing Time” combo box, select 2.5 seconds; in “Delay Before Closing” combo box, select 1.0 then click “Apply”, and then “Cancel”. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. When the doors are FULLY opened, Toggle ON/OFF Door Close TL switch. When Doors are closed, click Refresh button on the PTU Graph window. 5. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 6. In PTU, select “DCU Configure” from the Administration → Other Function menu. In “Closing Time” combo box, select 3.0 seconds; in “Delay Before Closing” combo box, select 2.0 then click “Apply”, and then “Cancel”. Toggle ON/OFF Door Open (+) TL and Door Open (-)	1. In PTU, Zero Speed (+) CL and Zero Speed (-) CL are on. 2. Doors close. The closing light flashes. Verify that the closing travel time is between 1.8 and 2.2 seconds and the delay before Door movement is between 0 and 0.25 second. Closing time (2.0): _____ Delay before closing (0.0): _____ 3. Doors open. 4. Doors close. The closing light flashes. Verify that the closing travel time is between 2.3 and 2.7 seconds and the delay before Door movement is between 0.7 and 1.2 second. Closing time (2.5): _____ Delay before closing (1.0): _____ 5. Doors open. 6. Doors close. The closing light flashes. Verify that the closing travel time is between 2.8 and 3.2 seconds and the delay before Door movement is between 1.8 and 2.2 second. Closing time (3.00): _____	

#	Test Cases	Input Specifications	Output Specifications	P/F
		TL Switches. When the doors are FULLY opened, Toggle ON/OFF Door Close TL switch. When Doors are closed, click Refresh button on the PTU Graph window. 7. In PTU, select “DCU Configure” from the Administration → Other Function menu. In “Closing Time” combo box, select 2.5 seconds; click “Apply”, and then “Cancel”. Redo the previous test for all other values of “Delay Before Closing”. Verify that the result is within 0.2 second of configured value. 8. In PTU, select “DCU Configure” from the Administration → Other Function menu. Click on “Cancel” to configure as default parameter values.	Delay before closing (2.0): _____ 7. Delay before closing time: 3: _____ 4: _____ 5: _____ 8. The DCU is now configured with default parameter values.	
2	TC1.1.7.2 If Doors have never been locked since last power-up, position encoder count is considered unknown. TC1.1.7.3 When the Door Position is unknown from the power-up, and Input.DoorOpenTL.ISO and Input.DoorUnlockTL.ISO are inactive, the DCU Application closes Doors slowly, in seek-home mode TC1.1.7.4 When above condition occurs, the DCU Application flashes Output.PassAlertLight.ISO and Output.PassAudibleSignal.ISO.	1. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 2. Cycle DCU power. 3. Toggle ON/OFF Door Close switch.	1. Doors open. 2. The Door stays open. 3. Doors close slowly (in seek-home mode) with flashing Door Closing Visual Alarm and Door Closing Audible Warning.	

#	Test Cases	Input Specifications	Output Specifications	P/F
3	TC1.1.7.5 Valid Door Close Condition	**Door Close from Loss of Both Zero Speed (+) and (-) CL.** 1. On the test fixture, turn ON both Zero Speed (+) and (-) CL switches. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 2. Turn OFF both Zero Speed (+) and (-) CL switches. 3. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. **Door Close from activation of the Door CloseTrainline** 4. On the test fixture, turn ON both Zero Speed (+) and (-) CL switches. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 5. Toggle ON/OFF the Door Close (+) TL switch on the test fixture. 6. Toggle ON/OFF the Door Close (+) TL switch. **Door Close from Loss of Zero Speed (+)** 7. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 8. On the test fixture, turn OFF Zero Speed (+) CL switch. 9. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 10. Turn ON the Zero Speed (+) CL **Door Close from Loss of Zero Speed (-)** 11. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 12. Turn OFF Zero Speed (-) CL switch. 13. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. Turn ON both Zero Speed (+) and (-) switches.	1. Doors open fully. 2. Doors close and lock. 3. Doors stay closed. 4. Doors open fully. 5. Doors close and lock. 6. Doors stay closed. 7. Doors open. 8. Warning LED goes ON. Doors close and lock. 9. Doors stay closed. 10. The doors stay closed. 11. Doors open. 12. Warning LED goes ON. Doors close and lock. 13. Doors stay closed.	

#	Test Cases	Input Specifications	Output Specifications	P/F
4	TC1.1.7.6 Door Close Crew Switch	**Normal Door Close CS Command** 1. On the test fixture, turn both Door Open (+) and (-) TL switches ON and OFF. 2. On the test fixture, turn Door Close CS ON and OFF.	1. Doors open. 2. Doors close without Door Closing Visual Alarm and Door Closing Audible Alarm.	
5	TC1.1.7.8 Closing Sequence With Door Closing Visual Alarm and Door Closing Audible Alarm.	**DCVA and DCAA with Close TL** 1. Make sure the Zero Speed (+) and (-) CL switches are on 2. In PTU, configure the "DCU Configuration" to have to default parameters. 3. Toggle On/Off Open (+) and (-) TL switches to open doors. 4. Toggle On/Off Close TL switch to close doors. **DCVA and DCAA with CS Close** 5. Toggle On/Off CS Open and CS Enable switches to open doors. 6. Toggle On/Off CS Close switch to close doors.	1. In PTU Data monitoring, Zero Speed (+) and (-) CL are on. 2. DCU is configured with default configuration parameter values. 3. Doors open. 4. DCVA and DCAA starts to flash and sound for 3 seconds, then the doors close. 5. Doors open. 6. Doors closes immediately without the sound and alarm from the DCVA and DCAA.	
6	TC1.1.7.9 The DCU Application drives the motor in close direction until the DO.LS-L.ISO, DO.LS-R.ISO, DO.PSS-L.ISO, and DO.PSS-R.ISO are deactivated	See Test Case 68 and 69 of the Table 17-5.	See output of the Test Case 68 and 69 of the Table 17-5**.**	

17.4.7 TC1.1.38 Passenger Door Open

Special Procedural Requirements:
- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Use the PTU Application with the administrator access*

Measurements:
- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.38.1 Passenger Door Open Operation	1. On the test fixture, turn on the Zero Speed (+) CL and Zero Speed (-) CL switches. 2. Turn on the both Passenger Door Enable TL (+) and (-) TL switches. 3. Push Passenger Door Open pushbutton. 4. Wait for 5 minutes. 5. Push Passenger Door Open pushbutton. 6. On the test fixture, toggle On/Off Door Close TL switch. 7. Push Passenger Door Open pushbutton. 8. On the test fixture, turn off the Zero Speed (+) CL and Zero Speed (-) CL switches.	1. Zero Speed signals are on. 2. Passenger Door Enable Light turns on. 3. Doors open. 4. After 5mins, the Door Closing Visual Alarm and Audible Alarm flash for 3s, then doors close. 5. Doors open. 6. The Door Closing Visual Alarm and Audible Alarm flash for 3s, then doors close. 7. Doors open. 8. Doors close.	
2	TC1.1.38.2 Passenger Door Open Duration Configuration	1. On the test fixture, turn on the Zero Speed (+) CL and Zero Speed (-) CL switches. 2. Turn on the both Passenger Door Enable TL (+) and (-) TL switches. 3. In PTU, select “DCU Configure” from the Administration → Other Function menu. In “Door open duration for Passenger Door Enable” box, select 0min, then click “Apply”, and then “Cancel”. 4. Push Passenger Door Open pushbutton.	1. Zero Speed signals are on. 2. Passenger Door Enable Light turns on. 3. DCU is configured with 0mins door open duration. 4. Doors open and close.	

#	Test Cases	Input Specifications	Output Specifications	P/F
		5. In PTU, change the “Door open duration for Passenger Door Enable” parameter value from 1 to 10mins. And record the door open duration time. 6. Using the PTU, configure the “Door open duration for Passenger Door Enable” to default value of 5mins.	5. Door open duration for Passenger Door Enable: 1min: _____ 2mins: _____ 3mins:_____ 4min: _____ 5mins: _____ 6mins:_____ 7min: _____ 8mins: _____ 9mins:_____ 10mins:_____	

17.4.8 TC1.1.8 Obstruction Detection

Special Procedural Requirements:

- *Continue the test from Section 17.4.5 of page 56.*
- *Use a calibrated flat bar, 1/4-inch (6.35 mm) wide and 3 inches (76.2 mm) high.*
- *Use a calibrated round bar, 3/8-inch (9.53 mm) in diameter.*
- *Use the PTU Application with the administrator*
- *Obstacle shall always be perpendicular to the Doors.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.8.1 DCU Application detects a flat bar, 1/4-inch (6.35 mm) wide and 3 inches (76.2 mm) high, held rigidly between and perpendicular to the Door panel, as a hand might be held to stop the Doors.	1. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches 2. Hold the flat bar rigidly and perpendicular between Door panels about half height. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. Let the Door panel hit the obstruction and re-open, then remove the bar. 3. Repeat the previous test (Step #1 ~ #2) at the top of Door seal. 4. Repeat the obstruction test (Step #1 ~ #2) at the bottom of Door seal.	1. Doors open. 2. The Door panel moves in close direction and hits the bar. It re-opens by 3 inches and retries to close. The obstacle is removed and the obstructed Door panel closes and locks. 3. Same as previously. 4. Same as previously.	
2	TC1.1.8.2 DCU Application detects an object, 3/8-inch (9.53 mm) in diameter, held rigidly between and perpendicular to the Door panels at all locations along the length of the Door nosing seal.	1.Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 2.Hold the cylindrical bar rigidly and perpendicular between Door panels about half height. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. Let the Door panel hit the obstruction and re-open then remove the bar. 3.Repeat the previous sequence at the top of Door seal. 4.Repeat the obstruction sequence at the bottom of Door seal.	1.Doors open. 2.The Door panel moves in close direction and hits the bar. It re-opens by 3 inches and retries to close. The obstacle is removed and the obstructed Door panel closes and locks. 3.Same as previously. 4.Same as previously.	

#	Test Cases	Input Specifications	Output Specifications	P/F
3	TC1.1.8.3 When DCU Application detects an obstruction it controls the motor to open by 3 inches and retry to close. The reopening distance is adjustable from 0.0 to 25.0 inches in 1 inch increment; 3.0 inch is default.	1. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 2. In PTU, select "DCU Configure" from the Administration → Other Function menu. In "Obstruction Reopen Distance" combo box, select 0 inch; then click "Apply", and then "Cancel". 3. Put a rigid obstacle on the Door bottom track. Install the measuring tape to easily measure re-opening distance. Toggle ON/OFF Door Close TL switch. Let the doors hit the obstruction. Measure the re-opening distance then remove the obstacle. (for the 0 inch re-open distance test, Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches before removing the obstacle.). When the doors are Closed & Locked, Toggle ON/OFF Door Open (+) and (-) TL switches. 4. Repeat the test sequence for Obstruction Reopen Distance parameter of 3, 5, 10, 15, 20, and 25 inches. 5. In PTU, select "DCU Configure" from the Administration → Other Function menu. In "Obstruction Reopen Distance" combo box, select 3 inches; then click "Apply", and then "Cancel".	1. Doors open. 2. The Obstruction Reopen Distance is set to 0 inch. 3~4. The Door panels move in close direction and hit the obstacle. Verify the tolerance of Door panel re-open distance is within 0.5 inch. The obstacle is removed and the obstructed Door panel closes and locks. 0": _____ 15": _____ 3": _____ 20": _____ 5": _____ 25": _____ 10": _____ 5. The DCU is now configured with default Obstruction Reopen Distance of 3 inches.	
4	TC1.1.8.4 When the maximum number of retries is exceeded, the DCU Application reopens the Door fully. The number of retries is adjustable from 0 to 5; 3 is the default value.	1. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 2. In PTU, select "DCU Configure" from the Administration → Other Function menu. In "Obstruction Recycle" combo box, select 1 retry; then click "Apply", and then "Cancel". 3. Insert an obstruction at the right side door panel 5" away from the fully open position. Toggle ON/OFF Door Close TL switch. Let the Doors hit the obstruction.	1. Doors open. 2. The Door panels are configured to retry to close once upon detection of obstruction. 3. The Door panels move in close direction and hit the obstacle. The door panels re-open by 3 inches and retry to close only once. If the obstacle is detected, the door panels fully open.	

#	Test Cases	Input Specifications	Output Specifications	P/F
		4. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches to open the Door then remove the obstacle. 5. Toggle ON/OFF Door Close TL switch to close doors. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 6. Insert the 3/8" diameter round bar in the middle of the Door panel to block Doors from locking. Toggle ON/OFF Door Close TL switch. 7. Repeat the test steps 2, 3, 4, and 5 by changing the "Obstruction Recycle" to 0, 2, 3, 4 and 5 retries in PTU.	4. Doors open. 5. Doors close and open. 6. The door panels move in close direction and hit the obstacle. The door panels re-open by 3 inches and retries to close only once. If the obstacle is detected, the foor panels fully open. 7. Doors reopen after 0 retries (0 retries): _____ Doors reopen after 2 retries (2 retries): _____ Doors reopen after 3 retries (3 retries): _____ Doors reopen after 4 retries (4 retries): _____ Doors reopen after 5 retries (5 retries): _____	
5	TC1.1.8.5 The DCU Application has threshold obstruction time adjustable from 0.0 to 3.0 seconds by 0.1 second increment (default value: 0.0s). TC1.1.8.6 The DCU Application stops Doors against obstacle for the time defined in the threshold obstruction time parameter when it detects obstruction .	Tested in the test case 1 of the Section 17.4.28.	See output of the Test Case 1 of the Section 17.4.28.	

#	Test Cases	Input Specifications	Output Specifications	P/F
6	TC1.1.8.7 Changing position of obstruction .	1. In PTU, select "DCU Configure" from the Administration → Other Function menu, and select 5 obstruction retries. Click "Apply". 2. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches. 3. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 4. Put obstacle in the middle of the Doorway. Toggle ON/OFF Door Close TL. 5. Once Doors reopen, move obstacle object few inches toward opening direction. 6. Within few obstruction recycle, try to move obstacle toward fully opening position. 7. Remove the obstruction from the doorway.	1. DCU is configured with 5 obstruction recycles. 2. In PTU Data Monitoring, Zero Speed CL (+) and (-) are turned on. 3. Doors open. 4. Doors close and hit obstacle. Doors reopen by 3". 5. Doors reattempt to close but hit obstacle and reopen 3" from newly obstruction detected position. 6. As obstruction moves closer to fully opening position, Doors' reopening distance is limited to fully opening position. Then Doors reattempt to close. 7. Doors close and lock.	
6	TC1.1.8.8 Door Closing Visual Alarm and Door Closing Audible Alarm follow normal closing pattern of light and sound during Obstruction Recycle	1. In PTU, select "DCU Configure" from the Administration → Other Function menu, and click "Cancel" to have default configuration value. 2. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches. 3. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches to open Doors. 4. Put obstruction in the doorway. Then close Doors by toggling ON/OFF Door Close TL switch. 5. Remove the obstruction from the Doorway.	1. DCU is configured with default parameter value. 2. In PTU Data Monitoring, Zero Speed CL (+) and (-) are turned on. 3. Doors open. 4. Door Closing Visual Alarm and Door Closing Audible Alarm flash and doors close. As Doors recycle due to obstruction, Door Closing Audible Alarm and Door Closing Visual Alarm flash. 5. Doors close and lock. Door Closing Visual Alarm and Door Closing Audible Alarm turn off.	

#	Test Cases	Input Specifications	Output Specifications	P/F
7	TC1.1.8.9 Obstruction recycling after start of the DCU.	1. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches. 2. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches to open Doors. 3. Place obstruction in the doorway. Turn off DCU power. 4. Turn on the DCU power. Toggle ON/OFF Door Close TL. 5. Push obstruction toward fully opening position. 6. Remove the obstruction from the Doorway.	1. In PTU Data Monitoring, Zero Speed CL (+) and (-) are turned on. 2. Doors open. 3. DCU power is turned off. 4. DCU turns on. Doors close slowly. As Doors hit obstruction, they recycle without reopening. 5. Doors are pushed to opening direction by obstruction. Then Doors reattempt to close. 6. Doors close and lock.	
9	TC1.1.8.11 Obstruction Recycle after emergency reset command from Crew Switch.	1. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches. 2. Make sure DCU is configured to 5 retries. 3. Pull and let go the red emergency handle. 4. Push Door panels manually toward opening direction. 5. Place an obstruction in the middle of the Doorway. Toggle ON/OFF Door Close CS on the test fixture. 6. Remove the obstruction from the Doorway, and toggle ON/OFF Door Close CS on the test fixture.	1. In PTU Data Monitoring, Zero Speed (+) CL and (-) CL are turned on. 2. In PTU, DCU is configured to 5 retries. 3. Emergency door open is activated. 4. Doors are pushed open. 5. Doors close and hit obstruction. Doors recycle for 5 times. Door Closing Visual Alarm and Door Closing Audible Alarm stay off. 6. Doors close and lock.	

17.4.9 TC1.1.9 Emergency State

Special Procedural Requirements:

- *Continue the test from the Section 17.4.5 of page 56.*
- *DCU's must be programmed with the DCU Application under test and powered ON.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.9.1 Emergency State can be initiated at any time even without Zero Speed Car Line signals.	1. If Doors are open, close Doors. 2. Turn off Zero Speed (+) CL and Zero Speed (-) CL switches. Then pull down the Emergency Handle. Using PTU, click on Maintenance → Other Functions → Log Download. 3. Toggle ON/OFF Door Close TL. 4. Pull the Emergency Handle again. 5. Toggle ON/OFF Door Close CS.	1. Doors close. 2. After releasing the Emergency Handle, it goes back to its original position. The Door panels open 1.5 inch manually and stay in manual mode. The Event Log contains event #11, Emergency Handle State Activation, in set state. 3. Doors close. 4. Doors open slightly. 5. Doors close.	
2	TC1.1.9.2 Emergency State is reset to normal operation when Doors are in Closed&Locked condition.	1. Pull the Emergency Handle. 2. Toggle ON/OFF Door Close TL switch. Using PTU, click on Maintenance → Other Functions → Log Download. 3. Reactivate the Emergency State by pulling down the Emergency Handle. 4. Activate the Door Close CS. Using PTU, click on Maintenance → Other Functions → Log Download. 5. Reactivate the Emergency State by pulling down the Emergency Handle 6. Push Door panels in closing direction until closed and locked. Using PTU, click on Maintenance → Other Functions → Log Download.	1. Doors open slightly. 2. Doors close. The Event Log contains event #11 in set and reset states. 3. The Door panels open 1.5 inch manually and stay in manual mode. 4. Doors close. The Event Log contains event #11 in set and reset states. 5. Doors open 1.5 inch manually and stay in manual mode. 6. Doors close. The Event Log contains event #11 in set and reset states.	

#	Test Cases	Input Specifications	Output Specifications	P/F
3	TC1.1.9.3 Emergency Handle Pulled while Doors Opening from Door Open (+) TL and Door Open (-) TL	1. Turn on Zero Speed (+) CL and Zero Speed (-) CL switches. 2. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches to open Doors. 3. While Doors are opening, pull the Emergency Handle. 4. Close Doors by toggling ON/OFF Door Open (+) TL and Door Open (-) TL Switches.	1. In PTU Data Monitoring, Zero Speed CL (+) and (-) are turned on. 2. Doors open. 3. Doors stop. 4. Doors close.	
4	TC1.1.9.4 Emergency Handle Pulled while Opening or Closing from Crew Switch	1. Toggle ON/OFF Door Open CS switch on the test fixture. 2. While Doors are opening, pull the Emergency Handle. 3. Toggle ON/OFF Door Close CS ON on the test fixture. 4. While Doors are closing, pull the Emergency Handle. 5. Turn Door Close CS ON and OFF on the test fixture.	1. Doors open. 2. Doors stop to open and stay still. 3. Doors close. 4. Doors stop to close and stay still. 5. Doors resume closing.	

17.4.10 TC1.1.10 Network Interface

17.4.10.1 TC1.1.10.1 Time

Special Procedural Requirements:

#	Test Cases	Input Specifications	Output Specifications	P/F

17.4.10.2 TC1.1.10.2 SW version

Special Procedural Requirements:

Measurements:

#	Test Cases	Input Specifications	Output Specifications	P/F

17.4.10.3 TC1.1.10.3 DCU Status

Special Procedural Requirements:

Measurements:

#	Test Cases	Input Specifications	Output Specifications	P/F

17.4.10.4 TC1.1.10.4 SW Upgrade

Special Procedural Requirements:

Measurements:

#	Test Cases	Input Specifications	Output Specifications	P/F

17.4.11 TC1.1.11 Data Storage

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Use the PTU Application with the administrator access.*
- Close PTU application when using PTU Simulator.

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.11.1 Upon detection of an event, the DCU Application stores the specific event information along with a timestamp in the Event Log file, which is stored in NVRAM.	Tested in Section 17.4.4.	See Section 17.4.4 output.	
2	TC1.1.11.2 A maximum of 10 entries per fault number is stored for a single day.	1. Using the PTU, send the date “2007/08/05 13:00:00”: 2. Chose one Set/Reset type and one Pulse type events from Table 17-5 (refer to Event Description File in PTU Tools menu for event type). Generate more than 10 occurrences of the events (repeat the test steps 11 times, skip Event Log download). 3. Using PTU, click on Maintenance → Other Functions → Log Download.	1. The date and time are set to 2007/08/05 13:00: xx. 2. See chosen events output. 3. The events are logged exactly 10 times for the current date (2007/08/05).	
3	TC1.1.11.3 The Event Log file is able to store 1500 entries .	1. Change Location ID to R1.. 2. Reboot the DCU. Connect the pulse generator to test fixture as following with following signal timings: Signal 1: Door Open (+) TL A=0 B=2000 Signal 2: Zero Speed (+) CL C=0 D=2000 Signal 3:Open CS E=0 F=2000 Signal 4: Door Close TL G=1000 H=8500 3. Using the PTU, change the date to 2004/01/01 00:00:00. Make sure Doors are closed and locked. Connect the PTU cable and establish a session with the DCU. 4. On the pulse generator computer, press “F6” to generate continuous signals for 90s. And press “F6” one more	1. Location ID is set to R1. 2. The pulse generator is connected to the test fixture. 3. PTU is connected to DCU. 4. The DCU has a lot of events.	

#	Test Cases	Input Specifications	Output Specifications	P/F
		time to stop. 5. Download Event Log using the PTU and selecting Log Download from the Maintenance → Other Functions menu. Identify the last index number of the Log file. 6. Change to the next date on the DCU using the PTU. Generate new faults by pressing “F6” on the pulse generator. After 90s, press “F6” one more time to stop. Download Event Log using the PTU and selecting Log Download from the Maintenance → Other Functions menu. Verify in the Log file the index of the last fault. 7. Calculate the number of time you will have to change the date and time to be at one step of getting 1500 faults. 8. Change the DCU date to the next day. Use the PTU to set the DCU time. Press “F6” on the pulse generator. 9. Repeat previous step until the number of fault is close to 1500. (E times) Press “F6” to stop the pulse generator. 10. Download Event Log using the PTU and selecting Log Download from the Maintenance → Other Functions menu. 11. Disconnect the pulse generator from the test fixture. 12. Change the DCU date to the next day. Use the PTU to set the DCU time. Generate faults up to 1499 by activating or resetting Emergency Handle only. Event #55, Event Log Retrieved by PTU will be the 1500th event. Change date if more than 10 faults have to be generated. 13. Download Event Log using the PTU and selecting Log Download from the Maintenance → Other Functions menu.	5. Number of records in the Log file. Last index number : _____ (A) Faults remaining. (1500 – A) : _____ (B) 6. Number of new faults generated. Last index number : _____ (C) Faults generated. (C – A): _____ (D) 7. Number of days (B-D) / (Dx10)): _____ (E) 8. Faults are generated. 9. Faults are generated. 10. Verify if the expected number of faults has been recorded. 11. Pulse generator disconnected. 12. Get the last index and calculate the missing faults to get 1499 faults. 13. Verify if the number of faults expected has been recorded. Verify that you have index 1 to 1500. Write down the first and second events (index	

#	Test Cases	Input Specifications	Output Specifications	P/F
			#1 and #2 of the Log file: #1(a) : _____ #2(a) : _____	
		14. Add one new fault by downloading the Log File with the PTU. Selecting Log Download from the Maintenance → Other Functions menu.	14. Fault is generated. You still have index 1 to 1500 but the oldest record is lost and new one is added. Write down the first and second fault (index #1 and #2) of the Log file: #1(b) : _____ #2(b) : _____	
		15. Verify that previous index #2(a) is now index #1(b) 16. Change Location ID to R1.	15. Previous index #2(a) is now index #1(b)	
4	TC1.1.11.4 In case of an overflow of the Event Log file, the new data entries replace the entries that are stored at the beginning of the Event Log file.	Tested in previous test case.		

#	Test Cases	Input Specifications	Output Specifications	P/F
5	TC1.1.11.5 When a fault with the snapshot attribute occurs, the set of snapshot samples located in the temporary data, the next 20 snapshot samples and the information identifying the event are stored as a Snapshot File in NVRAM.	1. Using PTU, in Maintenance → Other Functions menu, select “Snapshot Erasure”. Select all snapshots and click Delete button. Click “Yes”. In Maintenance menu, select “Log Erasure”. Click “Yes”, “Ok”, and “Ok”. Reboot DCU. 2. On the test fixture, deactivate Door Unlock TL and momentarily activate Door Open TL for more than 3 seconds. 3. Using PTU, in Maintenance → Other Functions menu, select “Log Download”. 4. Using PTU, in Maintenance → Other Functions menu, select “Snapshot Download”. Click on the snapshot. 5. On the test fixture, put Door Open TL switch to normal position by deactivating it.	1. All Snapshots Files are deleted. The Event Log File is deleted. 2. The Warning LED goes ON. 3. There are two events other than the “Log Erased” and “Log Retrieved” events. The first is event #58, System Boot-Up. The second is event #13, Door Open / Unlock TL Mismatch in set state. 4. There is only one snapshot file, associated with event #13 Door Open / Unlock TL Mismatch.	
6	TC1.1.11.6 Door Movement Information is stored in two different files: one for opening and one for closing sequence.	1. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. When Doors are open, Toggle ON/OFF Door Open (+) and TL Door Open (-) TL switch. Using PTU, in Maintenance → Other Functions menu, choose “Graph Download”.	1. Doors open and close. The Door Movement Information files are downloaded. A total of two files are downloaded: latest opening and closing movements.	
7	TC1.1.11.7 Door Movement Information includes Door Speed, Door Position, PWM (motor command), Motor Current, Position Error, and Position Error Slope.	1. Verify the graphical display of a Door Movement Information file.	1. The top graph contains five curves: Theoretical Speed, Speed, Position, PWM, and Current. The bottom graph contains four curves, Error, Error Slope, Error Max, and Error Min, Max Positive, Max Negative.	

17.4.12 TC1.1.12 PTU Login Request

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.12.1 Upon reception of the PTU_LoginReq with correct argument from the PTU, the DCU Application starts accepting other PTU commands and sends the PTU_Login message to the PTU.	1. Start PTU Simulator (after closing PTU Application). Uncheck “DCU” checkbox. Select PTU_DCUCodeStatusReq with argument 01. Send the command. 2. Select PTU_LoginReq command; remove one byte from the argument and send the command. 3. Select PTU_DCUCodeStatusReq with argument 01. Send the command. 4. Select PTU_LoginReq command and send the command. 5. Select PTU_DCUCodeStatusReq with argument 01. Send the command.	1. The DCU reject the request and replies PTU_Logout command with status PTU_NOTLOGGED: 07 xx xx 00 **49 00** xx xx. 2. The DCU accepts the login request and returns PTU_Login message with error status PTU_LOG_ERROR: 34 xx xx 00 **47 01** 52 31 36 30 2C 20 44 6F 6F 72 20 4D 6F 6E 69 74 6F 72 69 6E 67 20 61 6E 64 20 43 6F 6E 74 72 6F 6C 20 55 6E 69 74 2C 20 56 61 70 6F 72 xx xx. 3. The DCU reject the request and replies PTU_Logout command with error status PTU_NOTLOGGED: 07 xx xx 00 **49 00** xx xx. 4. The DCU accepts the login request and returns PTU_Login message with status PTU_LOG_DCUAPPLICATION: 34 xx xx 00 **47 03** 52 31 36 30 2C 20 44 6F 6F 72 20 4D 6F 6E 69 74 6F 72 69 6E 67 20 61 6E 64 20 43 6F 6E 74 72 6F 6C 20 55 6E 69 74 2C 20 56 61 70 6F 72 xx xx. 5. The DCU accepts the request and returns PTU_DCUCodeStatus message: 0D xx xx 00 **4B 01** xx xx xx xx xx xx xx xx.	

17.4.13 TC1.1.13 PTU Logout / Reboot Request

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- Close PTU application when using PTU Simulator.

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.13.1 Upon reception of the PTU_LogoutReq command, the DCU Application sends PTU_Logout and waits for the PTU to login prior to accept any other PTU command	1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. Send PTU_LogoutReq with parameter “00”. 3. Send PTU_LogoutReq with parameter “00”.	1. The DCU replies PTU_Login message. 2. The DCU replies PTU_Logout command with parameter PTU_LOGOUT_NORMAL: 07 xx xx 00 **49 03** xx xx. 3. The DCU reject the request and replies PTU_Logout command with status PTU_NOTLOGGED: 07 xx xx 00 **49 00** xx xx.	
2	TC1.1.13.2 Upon reception of the PTU_LogoutReq command with reboot argument, the DCU Application discontinues all operations and reboots.	1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. Send PTU_LogoutReq with parameter “01”.	1. The DCU replies PTU_Login message with status PTU_LOG_DCUAPPLICATION: 34 xx xx 00 **47 03** 52 31 36 30 2C 20 44 6F 6F 72 20 4D 6F 6E 69 74 6F 72 69 6E 67 20 61 6E 64 20 43 6F 6E 74 72 6F 6C 20 55 6E 69 74 2C 20 56 61 70 6F 72 xx xx. 2. The DCU replies PTU_Logout command with parameter PTU_LOG_REBOOT_DCU_APP: 07 xx xx 00 **49 04** xx xx. The DCU reboots (running LED goes off).	

17.4.14 TC1.1.14 PTU Location ID Request

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Car Network Simulator.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.14.1 Upon reception of the PTU_Location_IDReq command from the PTU, the DCU Application sends PTU_Location_ID message to the PTU.	1. On the test fixture, set the location ID to R1 and reboot the DCU. 2. Using PTU Simulator, send PTU_LoginReq to DCU. 3. Select PTU_Location_IDReq and send the command. 4. Change Location ID on the test fixture to R2. Reboot DCU. 5. Send PTU_LoginReq to DCU. 6. Select PTU_Location_IDReq and send the command. 7. Repeat steps 4) to 6) for all other Location IDs. 8. Put back the DCU to location R1. Reboot DCU.	1. DCU is set to location R1. 2. The DCU replies PTU_Login message. 3. The DCU returns PTU_Location_ID with status R1 (31 00). 4. The DCU reboots, and it is set to location R2. 5. The DCU replies PTU_Login message. 6. The DCU returns PTU_Location_ID with status P2 (31 02). 7. See previous steps outputs, but expect different Location ID. 8. DCU is set to location ID R1.	

17.4.15 TC1.1.15 PTU Time Request

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.15.1 Upon reception of the PTU_TimeReq command from the PTU, the DCU Application sends PTU_TimeStatus message with the current Local Time value to the PTU.	1. Using PTU Application, update the date and time to 2000/01/01 00:00:00. 2. Using PTU Simulator, login first and select PTU_TimeReq and send the command.	1. The time set to 2000/01/01 00:00:00. 2. DCU updates time and date. The DCU returns PTU_TimeStatus 0A xx xx 00 41 **38 6D 43 80** xx xx.	

17.4.16 TC1.1.16 PTU SelfTest Request

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.16.1 Upon reception of the PTU_SelfTestReq command from the PTU, the DCU Application sends PTU_SelfTestStatus message containing self-tests result to the PTU.	1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. Select PTU_SelfTestReq and send the command.	1. The DCU is logged-in. 2. The DCU returns PTU_SelfTestStatus: 08 xx xx 00 43 **FF** x**F** xx xx.	

17.4.17 TC1.1.17 PTU Code Status Request

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.17.1 Upon reception of the PTU_DCUCodeStatusReq command from the PTU, the DCU Application sends PTU_DCUCodeStatus message containing code size and CRC32 of DCU Application and DCS Boot Loader Application to the PTU.	1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. Select PTU_DCUCodeStatusReq with parameter 01 and send the command. Validate against DCU Boot Loader size (S) and CRC32 (C) found in the zip package. 3. Select PTU_DCUCodeStatusReq with parameter 02 and send the command. Validate against DCU Application size (S) and CRC32 (C) found in the zip package.	1. The DCU is logged-in. 2. The DCU returns PTU_DCUCodeStatus: 0D xx xx 00 4B 01 SS SS SS SS CC CC CC CC xx xx. SS SS SS SS = _____ CC CC CC CC = _____ 3. The DCU returns PTU_DCUCodeStatus: 0D xx xx 00 4B 02 SS SS SS SS CC CC CC CC xx xx.	

17.4.18 TC1.1.18 PTU Real Time Monitoring

Special Procedural Requirements:
- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:
- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.18.1 Upon request for real-time monitoring from the PTU (PTU_DataReq), the DCU Application transmits monitoring data to the PTU (PTU_Data).	***Preparation*** 1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. Select PTU_DataReq, and send the command. Verify the packet format. 3. Close the PTU Simulator and start the PTU Application. Open the Browser window from the Maintenance→Other Functions menu. ***Motor Current Test*** 4. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches and look at the Motor Current. ***Door Panel Opening Position Test*** 5. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches and look at the Door Panel Opening Position. ***DCU Temperature Test*** 6. Look at the DCU Temperature value. ***Lock Switch #1, Lock Switch #2, Panel Sensing Switch Left, and Panel Sensing Switch Right Test*** 7. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches to open Doors.	***Preparation*** 1. The DCU is logged-in. 2. The DCU returns PTU_Data: 0E xx xx 00 0E xx xx xx xx xx xx xx xx xx xx xx xx xx xx. 3. The Browser window is displayed in the PTU Application. ***Motor Current Test Result*** 4. The Motor Current increases and decreases following the Doors movement. ***Door Panel Opening Position Test Result*** 5. The Door Panel Opening Position value (in %) varies following the actual Door position. ***DCU Temperature Test Result*** 6. The DCU Temperature value is about room temperature (around 70 - 90 °F). ***Lock Switch #1, Lock Switch #2, Panel Sensing Switch Left, and Panel Sensing Switch Right Test Result*** 7. The following values are ON: Lock Switch #1, Lock Switch #2 Panel Sensing Switch Left Panel Sensing Switch Right	

#	Test Cases	Input Specifications	Output Specifications	P/F
		8. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches to close Doors.	8. The following values are OFF: Lock Switch #1, Lock Switch #2 Panel Sensing Switch Left Panel Sensing Switch Right	
		Zero Speed (+) CL Test 9. Using the test fixture, generate 35 ms signals for Low Speed Interlock + CL. Repeat at least 5 times with at least 1 second between pulses.	***Zero Speed (+) CL Test Result*** 9. The Zero Speed (+) CL value does not change.	
		10. Generate 150 ms signals for Low Speed Interlock + CL.	10. The Low Zero Speed (+) CL value follows.	
		Zero Speed (-) CL Test 11. On the test fixture, turn ON the Zero Speed CL PWR.	***Zero Speed (-) CL Test Result*** 11. The Zero Speed (-) CL value is on.	
		12. On the test fixture, turn OFF the Zero Speed CL PWR.	12. The Zero Speed (-) CL value is off.	
		Door Open (+) TL Test 13. Using the test fixture, generate 35 ms signals for Door Unlock TL. Repeat at least 5 times with at least 1 second between pulses.	***Door Open (+) TL Test Result*** 13. The Door Open (+) TL value does not change.	
		14. Generate 150 ms signals for Door Unlock TL.	14. The Door Open (+) TL value follows.	
		Door Open (-) TL Test 15. Using the test fixture, generate 35 ms signals for The Door Open (-) TL. Repeat at least 5 times with at least 1 second between pulses.	***Door Open (-) TL Test Result*** 15. The Door Open (-) TL value does not change.	
		16. Generate 150 ms signals for The Door Open (-) TL.	16. The Door Open (-) TL value follows.	
		Passenger Door Enable (+) TL Test 17. Using the test fixture, generate 35 ms signals for Passenger Door Enable (+) TL. Repeat at least 5 times with at least 1 second between pulses.	***Passenger Door Enable (+) TL Test Result*** 17. Passenger Door Enable (+) TL value does not change.	
		18. Generate 150 ms signals for Passenger Door Enable (+) TL.	18. The Passenger Door Enable (+) TL value follows.	

#	Test Cases	Input Specifications	Output Specifications	P/F
		Passenger Door Enable (-) TL Test 19. Using the test fixture, generate 35 ms signals for Passenger Door Enable (-) TL. Repeat at least 5 times with at least 1 second between pulses. 20. Generate 150 ms signals for Passenger Door Enable (+) TL. ***Door Close TL Switch Test*** 21. Using the test fixture, generate 35 ms signals for Door Close TL Switch. Repeat at least 5 times with at least 1 second between pulses. 22. Generate 150 ms signals for Door Close TL Switch. ***Door Cut-Out Mechanical Lock Test*** 23. On the test fixture, turn ON “Door Cut-Out Mechanical Lock” switch. 24. On the test fixture, turn RIG “Door Cut-Out Mechanical Lock” switch. ***Emergency Handle Switch Test*** 25. On the test fixture, turn ON Emergency Handle switch (EHS). 26. On the test fixture, turn RIG Emergency Handle switch (EHS). ***Door Summary Relay (+)Test*** 27. Using the test fixture, generate 35 ms signals for Door Summary Relay (+). Repeat at least 5 times with at least 1 second between pulses. 28. Generate 150 ms signals for Door Summary Relay (+). ***Door Summary Relay (-) Test*** 29. Using the test fixture, generate 35 ms signals for Door Summary Relay (-). Repeat at least 5 times with at least 1 second between pulses.	***Passenger Door Enable (-) TL Test Result*** 19. The Passenger Door Enable (-) TL value is does not chane. 20. The Door Passenger Door Enable (-) TL value follows. ***Door Close TL Switch Test Result*** 21. The Door Close TL Switch value does not change. 22. The Door Close TL Switch value follows. ***Door Cut-Out Mechanical Lock Test Result*** 23. The Door Cut-Out Mechanical Lock value is on. 24. The Door Cut-Out Mechanical Lock value is off. ***Emergency Handle Test Result*** 25. The Emergency State value is on. 26. The Emergency State value is off. ***Door Summary Relay (+) Test Result*** 27. The Door Summary Relay (+) value does not change. 28. The Door Summary Relay (+) value follows. ***Door Summary Relay (-) Test*** 29. The Door Summary Relay (-) value does not change.	

#	Test Cases	Input Specifications	Output Specifications	P/F
		30. Generate 150 ms signals Door Summary Relay (-). ***Encoder Phase A Test*** 31. Activate Emergency Handle switch on the Door operator. Then move Door panels manually in opening and closing direction to observe the Encoder Phase A value. ***Encoder Phase B Test*** 32. Move Door panels manually in opening and closing direction to observe the Encoder Phase B value. ***Encoder Index Test*** 33. Move Door panels manually in opening and closing direction to observe the Encoder Index value. 34. Activate Manual Reset Switch on the Door operator to reset the emergency state. ***DCU Temperature Sensor Test*** 35. The DCU Temperature Sensor Health is tested in the DCU Instrumented Test Procedure Chapter. ***Feedback Door Open Supply Test*** 36. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. When Doors are open, push Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. ***Over Current High Limit and Over Current Normal Limit Input Test*** 37. Delete the log to force a current limits test. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. (This step may require at least 5 attempts to see OV High Current Limit signal on). ***Logic Supply Test*** 38. The 5V Logic Supply is tested in the DCU Instrumented Test Procedure Chapter.	30. The Door Summary Relay (-) value follows. ***Encoder Phase A Test Result*** 31. The Encoder Phase A value follows. ***Encoder Phase B Test Result*** 32. The Encoder Phase B value follows. ***Encoder Index Test Result*** 33. The Encoder Index value follows. 34. Doors close and emergency state is reset. ***DCU Temperature Sensor Test Result*** 35. Tested in the DCU Instrumented Test Procedure Chapter. ***Feedback Door Open Supply Test Result*** 36. The Feedback Door Open Supply is on while Doors are open, and 0 when Doors are closed & locked. ***Over Current High Limit and Over Current Normal Limit Input Test Result*** 37. The Over Current High Limit input and Over Current Normal Limit input values are on while the test is performed. ***Logic Supply Test Result*** 38. The Voltage Monitoring 5V Logic Supply value stays on.	

#	Test Cases	Input Specifications	Output Specifications	P/F
		Motor Power Supply Test 39. Toggle the Motor Power switch on the test fixture.	***Motor Power Supply Test Result*** 39. The Voltage Monitoring 5V Encoder Supply, Voltage Monitoring 15V Motor Supply, Voltage Monitoring V-Gate Open Supply, and Voltage Monitoring V-Gate Close Supply values follow Motor Power.	
		Motor Control Activation Test 40. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches.	***Motor Control Activation Test Result*** 40. While the motor is running, the Motor Control Feedback value is 1.	
		Spare Output Test 41. The Spare Output is tested in DCU Instrumented Test Procedure Chapter.	***Spare Output Test Result*** 41. The Spare Output Feedback value stays off.	
		Running LED Test 42. Look at the Running LED on the DCU.	***Running LED Test Result*** 42. The Running LED value follows the DCU LED.	
		Warning LED Test 43. Generate an exception by deactivating the Low Speed Interlock CL + switch on the test fixture. Clear it by reactivating the Low Speed Interlock CL + switch.	***Warning LED Test Result*** 43. The Warning LED value follows the DCU LED and is ON while the exception is active.	
		Fault LED Test 44. Generate a failure by deactivating the Encoder Power switch on the test fixture. Clear it by reactivating the Encoder Power switch.	***Fault LED Test Result*** 44. The Fault DCU LED value follows the DCU LED and is ON while the failure is active.	
		Door Closing Visual Alarm and Door Closing Audible Alarm Test 45. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches.	***Door Closing Visual Alarm and Door Closing Audible Alarm Test Result*** 45. Doors open.	
		46. Toggle ON/OFF Door Close TL switch.	46. While Doors are closing, Door Closing Visual Signal, Door Closing Visual Alarm Feedback, Door Closing Audible Alarm Signal, and Door Closing Audible Alarm Feedback values toggle.	
		Motor Power Voltage Test 47. Adjust the variable power supply output voltage.	***Motor Power Voltage Test Result*** 47. The Motor Power Voltage follows the power	

#	Test Cases	Input Specifications	Output Specifications	P/F
			supply value (±2 Volts).	
		CPU Open Interlock Power Out 48. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches to open Doors. 49. Toggle ON/OFF Door Close TL Switche to close Doors.	***CPU Open Interlock Power Out Result*** 48. While Doors are opening, CPU Open Interlock Power Out is on. 49. Doors close.	

17.4.19 TC1.1.19 PTU Event Synchronization Request

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.19.1 Upon reception of the PTU_EventSyncRq request from the PTU, the DCU Application sends PTU_Event with the number of an active event (or no data if no event is active) over Serial Link.	1. Make sure the Fault and Warning LEDs are OFF. Using PTU Simulator, send PTU_LoginReq to DCU. 2. Select PTU_EventSyncRq with argument 01 and send the command. 3. Activate only Door Open (+) TL switch on test fixture. 4. Select PTU_EventSyncRq with argument 01 and send the command. 5. Select PTU_EventSyncRq with argument 00 and send the command **twice**. 6. Deactivate Door Open (+) TL switch.	1. The DCU is logged-in. 2. The DCU returns PTU_Event with empty data: 06 xx xx 00 08 xx xx. 3. Warning LED goes ON. 4. The DCU returns PTU_Event with data: 12 xx xx 00 08 00 **0D** xx xx xx xx xx xx xx xx xx xx xx xx. 5. The DCU returns PTU_Event with empty data: 06 xx xx 00 08 xx xx. 6. The Warning LED goes OFF.	

17.4.20 TC1.1.20 PTU Erase Event Log File

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.20.1 Upon reception of the PTU_FileDeleteReq request with Event Log File index argument from the PTU, the DCU Application erases all entries in Event Log file and sends PTU_FileDeleteStatus over Serial Link.	1. Activate Emergency State by pulling down the Emergency Handle. Then close doors by toggling ON/OFF the Door Close TL switch. 2. Using the PTU, click on Monitoring→Other Functions→Log Download. 3. Close the PTU. Using the PTU Simulator, send PTU_LoginReq to DCU. 4. Select PTU_FileDeleteReq, write 03 as the parameter, and send the command. 5. Close PTU Simulator, start PTU. Connect to DCU and select Log Download from the Maintenance → Other Functions menu.	1. An event is logged. 2. The log contains many events (the exact content is not relevant for this test). 3. The DCU is logged-in. 4. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 03 xx xx. 5. There are only event #54, Event Log Erased by PTU, and event #55, Event Log Retrieved by PTU, in the Log file.	

17.4.21 TC1.1.21 PTU Erase Snapshot File

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.21.1 Upon PTU_FileDeleteReq request with a Snapshot File index argument from the PTU, the DCU Application erases the selected Snapshot File and sends PTU_FileDeleteStatus over Serial Link.	1. Generate 10 snapshots producing event #74 (See Table 17-5). 2. On the test fixture, turn ON the “DOOR CUT-OUT MECHANICAL LOOK” Left switch. Then turn RIG “DOOR CUT-OUT MECHANICAL LOOK” Left switch. 3. Generate 10 snapshots producing event #13 (See Table 17-5). 4. Using the PTU, click on Monitoring→Other Functions→Log Download. 5. Close the PTU. Using the PTU Simulator, send PTU_LoginReq to DCU. 6. Select PTU_FileDeleteReq, write 06 as the parameter, and send the command. 7. Select PTU_FileDeleteReq, write 07 as the parameter, and send the command. 8. Select PTU_FileDeleteReq, write 08 as the parameter, and send the command. 9. Select PTU_FileDeleteReq, write 09 as the parameter, and send the command. 10. Select PTU_FileDeleteReq, write 0A as the parameter, and send the command. 11. Select PTU_FileDeleteReq, write 0B as the parameter, and send the command.	1. 10 Snapshots are created in DCU. 2. First 10 snapshots are copied over to the last 10 places. 3. 10 more snapshots are created in DCU. 4. The log contains many snapshots (the exact content is not relevant for this test). 5. The DCU is logged-in. 6. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 06 xx xx. 7. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 07 xx xx. 8. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 08 xx xx. 9. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 09 xx xx. 10. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 0A xx xx. 11. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 0B xx xx.	

#	Test Cases	Input Specifications	Output Specifications	P/F
		12. Select PTU_FileDeleteReq, write 0C as the parameter, and send the command. 13. Select PTU_FileDeleteReq, write 0D as the parameter, and send the command. 14. Select PTU_FileDeleteReq, write 0E as the parameter, and send the command. 15. Select PTU_FileDeleteReq, write 0F as the parameter, and send the command. 16. Select PTU_FileDeleteReq, write 10 as the parameter, and send the command. 17. Select PTU_FileDeleteReq, write 11 as the parameter, and send the command. 18. Select PTU_FileDeleteReq, write 12 as the parameter, and send the command. 19. Select PTU_FileDeleteReq, write 13 as the parameter, and send the command. 20. Select PTU_FileDeleteReq, write 14 as the parameter, and send the command. 21. Select PTU_FileDeleteReq, write 15 as the parameter, and send the command. 22. Select PTU_FileDeleteReq, write 16 as the parameter, and send the command. 23. Select PTU_FileDeleteReq, write 17 as the parameter, and send the command. 24. Select PTU_FileDeleteReq, write 18 as the parameter, and send the command.	12. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 0C xx xx. 13. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 0D xx xx. 14. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 0E xx xx. 15. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 0F xx xx. 16. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 10 xx xx. 17. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 11 xx xx. 18. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 12 xx xx. 19. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 13 xx xx. 20. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 14 xx xx. 21. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 15 xx xx. 22. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 16 xx xx. 23. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 17 xx xx. 24. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 18 xx xx.	

#	Test Cases	Input Specifications	Output Specifications	P/F
		25. Select PTU_FileDeleteReq, write 19 as the parameter, and send the command. 26. Close PTU Simulator, start PTU. Connect to DCU and select Snapshot Download from the Maintenance → Other Functions menu.	25. The DCU returns PTU_FileDeleteStatus: 07 xx xx 00 4E 19 xx xx. 26. All Snapshot Files are empty. No Snapshot is downloaded.	

17.4.22 TC1.1.22 PTU Transfer Event Log File

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.22.1 Upon reception of PTU_FileReq with Event Log File index argument from the PTU, the DCU Application sends the PTU_FileStat to the PTU and prepares file transfer of the Event Log File.	1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. In PTU Simulator, select PTU_FileReq, write 00 00 03 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will open the file for sending. 3. Select the PTU_NextPackReq and send the command three times. 4. Select PTU_FileReq, write 02 00 03 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will close the file.	1. The DCU is logged-in. 2. The DCU returns PTU_FileStat with status FS_XFER_UNDERWAY: 22 xx xx 00 **04 04** 00 00 1C 00 03 xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx. 3. Because the Event Log file contents only few items, on the first packet the DCU returns PTU_LastPack: xx xx xx 00 **22** […] xx xx. 4. The DCU returns PTU_FileStat with status FS_XFER_OK: 22 xx xx 00 **04 00** 00 00 1C 00 03 xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx.	

17.4.23 TC1.1.23 PTU Transfer Snapshot File

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.23.1 Upon reception of PTU_FileReq with a PTU Snapshot File index argument from the PTU, the DCU Application sends the PTU_FileStat to the PTU and prepares file transfer of the PTU Snapshot File.	1. Using PTU, in Maintenance → Other Functions menu, select Snapshot Erasure, [Select All], [Delete], and [Cancel]. Close PTU. 2. Generate a Door Open TL w/o Door Unlock TL error (Event #13) by turning on the Door Open TL switch for only one second on the test fixture. 3. Using PTU Simulator, send PTU_LoginReq to DCU. 4. In PTU Simulator, select PTU_FileReq, write 00 00 06 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will open the file for sending. 5. Select the PTU_NextPackReq and send the command. 6. If the previous answer from the DCU was PTU_NextPack (fifth byte is 21), repeat the previous step. Otherwise, select PTU_FileReq, write 02 00 06 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will close the file.	1. All snapshots are deleted. 2. The Warning LED will be ON during the mismatch. 3. The DCU is logged-in. 4. The DCU returns PTU_FileStat with status FS_XFER_UNDERWAY: 22 xx xx 00 **04 04** 00 00 1C 00 06 xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx. 5. The DCU returns PTU_NextPack: C6 xx xx 00 **21** […] xx xx or PTU_LastPack xx xx xx 00 **22** […] xx xx. 6. The DCU returns PTU_FileStat with status FS_XFER_OK: 22 xx xx 00 **04 00** 00 00 1C 00 06 xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx.	

17.4.24 TC1.1.24 PTU Transfer Event Description File and Operating Parameters Description File

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.24.1 Upon reception of PTU_FileReq with an Event Description File index argument from the PTU, the DCU Application sends the PTU_FileStat to the PTU and prepares the file transfer of the Event Description File.	1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. In PTU Simulator, select PTU_FileReq, write 00 00 02 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will open the file for sending. 3. Select the PTU_NextPackReq and send the command. 4. If the previous answer from the DCU was PTU_NextPack (fifth byte is 21), repeat the previous step. Otherwise, select PTU_FileReq, write 02 00 02 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will close the file.	1. The DCU is logged-in. 2. The DCU returns PTU_FileStat with status FS_XFER_UNDERWAY: 22 xx xx 00 **04 04** 00 00 1C 00 02 xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx. 3. The DCU returns PTU_NextPack: C6 xx xx 00 **21** […] xx xx or PTU_LastPack xx xx xx 00 **22** […] xx xx. 4. The DCU returns PTU_FileStat with status FS_XFER_OK: 22 xx xx 00 **04 00** 00 00 1C 00 02 xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx.	
2	TC1.1.24.2 Upon reception of PTU_FileReq with an Operating Parameters Description File index argument from the PTU, the DCU Application sends the PTU_FileStat to the PTU and prepares the file transfer of the Operating Parameters Description File.	1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. In PTU Simulator, select PTU_FileReq, write 00 00 04 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will open the file for sending. 3. Select the PTU_NextPackReq and send the command. 4. If the previous answer from the DCU was PTU_NextPack (fifth byte is 21), repeat the previous step. Otherwise, select PTU_FileReq, write 02 00 04 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will close the file.	1. The DCU is logged-in. 2. The DCU returns PTU_FileStat with status FS_XFER_UNDERWAY: 22 xx xx 00 **04 04** 00 00 1C 00 04 xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx 01 02 xx xx. 3. The DCU returns PTU_NextPack: C6 xx xx 00 **21** […] xx xx or PTU_LastPack xx xx xx 00 **22** […] xx xx. 4. The DCU returns PTU_FileStat with status FS_XFER_OK: 22 xx xx 00 **04 00** 00 00 1C 00 04 xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx 01 02 xx xx.	

17.4.25 TC1.1.25 PTU Transfer Door Movement Information File

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.25.1 Upon reception of PTU_FileReq with a Door Open Movement Information File index argument from the PTU, the DCU Application sends the PTU_FileStat to the PTU and prepares file transfer of the Door Movement Information File.	**Open:** 1. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. When Doors are fully open Toggle ON/OFF Door Close TL switch. 2. Using PTU Simulator, send PTU_LoginReq to DCU. 3. In PTU Simulator, select PTU_FileReq, write 00 00 1A 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will open the file for sending. 4. Select the PTU_NextPackReq and send the command. 5. If the previous answer from the DCU was PTU_NextPack (fifth byte is 21), repeat the previous step. Otherwise, select PTU_FileReq, write 02 00 1A 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will close the file.	1. Doors open, and then close. 2. The DCU is logged-in. 3. The DCU returns PTU_FileStat with status FS_XFER_UNDERWAY: 22 xx xx 00 **04 04** 00 00 1C 00 1A xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx. 4. The DCU returns PTU_NextPack: C6 xx xx 00 **21** […] xx xx or PTU_LastPack xx xx xx 00 **22** […] xx xx. 5. The DCU returns PTU_FileStat with status FS_XFER_OK: 22 xx xx 00 **04 00** 00 00 1C 00 1A xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx.	

#	Test Cases	Input Specifications	Output Specifications	P/F
2	TC1.1.25.2 Upon reception of PTU_FileReq with a Door Close Movement Information File index argument from the PTU, the DCU Application sends the PTU_FileStat to the PTU and prepares file transfer of the Door Movement Information File.	**Close:** 1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. In PTU Simulator, select PTU_FileReq, write 00 00 1B 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will open the file for sending. 3. Select the PTU_NextPackReq and send the command. 4. If the previous answer from the DCU was PTU_NextPack (fifth byte is 21), repeat the previous step. Otherwise, select PTU_FileReq, write 02 00 1B 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will close the file.	1. The DCU is logged-in. 2. The DCU returns PTU_FileStat with status FS_XFER_UNDERWAY: 22 xx xx 00 **04 04** 00 00 1C 00 1B xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx. 3. The DCU returns PTU_NextPack: C6 xx xx 00 **21** […] xx xx or PTU_LastPack xx xx xx 00 **22** […] xx xx. 4. The DCU returns PTU_FileStat with status FS_XFER_OK: 22 xx xx 00 **04 00** 00 00 1C 00 1B xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx.	

17.4.26 TC1.1.26 PTU Transfer Memory Content

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.26.1 Upon reception of PTU_FileReq with a Memory Dump index argument from the PTU, the DCU Application sends the PTU_FileStat to the PTU and prepares file transfer of the memory content.	1. Using the PTU Simulator, send PTU_LoginReq to DCU. **XRAM1 Memory Dump** 2. In PTU Simulator, select PTU_FileReq, write 04 00 01 00 00 E0 00 00 00 07 FF as the parameter, and send the command. This will retreive directory entry. 3. In PTU Simulator, select PTU_FileReq, write 00 00 01 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will open the file for sending. 4. Select the PTU_NextPackReq and send the command.	1. The DCU is logged-in. **XRAM1 Memory Dump Result** 2. The DCU returns PTU_FileStat with status FS_LOOKUP_OK: 22 xx xx 00 **04 01** xx 00 1F 00 01 53 52 41 4D 20 4D 65 6D 6F 72 79 20 44 75 6D 70 00 00 **07 FF** xx xx xx xx. 3. The DCU returns PTU_FileStat with status FS_XFER_UNDERWAY: 22 xx xx 00 **04 04** 00 00 1F 00 01 00 00 00 00 [...] xx xx xx xx. 4. The DCU returns PTU_NextPack: C6 xx xx 00 **21** […] xx xx or PTU_LastPack xx xx xx 00 **22**	

#	Test Cases	Input Specifications	Output Specifications	P/F
			[…] xx xx.	
		5. If the previous answer from the DCU was PTU_NextPack (fifth byte is 21), repeat the previous step. Otherwise, select PTU_FileReq, write 02 00 1B 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will close the file.	5. The DCU returns PTU_FileStat with status FS_XFER_OK: 22 xx xx 00 **04 00** xx 00 1F 00 01 xx [...] xx xx xx xx xx.	
		IRAM Memory Dump 6. In PTU Simulator, select PTU_FileReq, write 04 00 01 00 00 F6 00 00 00 07 FF as the parameter, and send the command. This will retreive directory entry.	**IRAM Memory Dump Result** 6. The DCU returns PTU_FileStat with status FS_LOOKUP_OK: 22 xx xx 00 **04 01** xx 00 1F 00 01 53 52 41 4D 20 4D 65 6D 6F 72 79 20 44 75 6D 70 00 00 **07 FF** xx xx xx xx.	
		7. In PTU Simulator, select PTU_FileReq, write 00 00 01 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will open the file for sending.	7. The DCU returns PTU_FileStat with status FS_XFER_UNDERWAY: 22 xx xx 00 **04 04** xx 00 1F 00 01 00 00 00 00 00 00 [...] xx xx xx xx.	
		8. Select the PTU_NextPackReq and send the command.	8. The DCU returns PTU_NextPack: C6 xx xx 00 **21** […] xx xx or PTU_LastPack xx xx xx 00 **22** […] xx xx.	
		9. If the previous answer from the DCU was PTU_NextPack (fifth byte is 21), repeat the previous step. Otherwise, select PTU_FileReq, write 02 00 1B 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will close the file.	9. The DCU returns PTU_FileStat with status FS_XFER_OK: 22 xx xx 00 **04 00** xx 00 1F 00 01 xx [...] xx xx xx xx.	
		XRAM2 Memory Dump First 256 Bytes 10. In PTU Simulator, select PTU_FileReq, write 04 00 01 00 0F 00 00 00 00 00 FF as the parameter, and send the command. This will retreive directory entry.	**XRAM2 Memory Dump Result** 10. The DCU returns PTU_FileStat with status FS_LOOKUP_OK: 22 xx xx 00 **04 01** xx 00 1F 00 01 53 52 41 4D 20 4D 65 6D 6F 72 79 20 44 75 6D 70 00 00 **00 FF** xx xx xx xx.	
		11. In PTU Simulator, select PTU_FileReq, write 00 00 01 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will open the file for sending.	11. The DCU returns PTU_FileStat with status FS_XFER_UNDERWAY: 22 xx xx 00 **04 04** 00 00 1F 00 01 00 00 00 00 00 [...] xx xx xx xx.	
		12. Select the PTU_NextPackReq and send the command.	12. The DCU returns PTU_NextPack: C6 xx xx 00 **21** […] xx xx or PTU_LastPack xx xx xx 00 **22**	

#	Test Cases	Input Specifications	Output Specifications	P/F
		13. If the previous answer from the DCU was PTU_NextPack (fifth byte is 21), repeat the previous step. Otherwise, select PTU_FileReq, write 02 00 1B 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will close the file. **XRAM2 Memory Dump Last 256 Bytes** 14. In PTU Simulator, select PTU_FileReq, write 04 00 01 00 0F FF 00 00 00 00 FF as the parameter, and send the command. This will retreive directory entry. 15. In PTU Simulator, select PTU_FileReq, write 00 00 01 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will open the file for sending. 16. Select the PTU_NextPackReq and send the command. 17. If the previous answer from the DCU was PTU_NextPack (fifth byte is 21), repeat the previous step. Otherwise, select PTU_FileReq, write 02 00 01 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will close the file. **NVRAM Memory Dump First 256 Bytes** 18. In PTU Simulator, select PTU_FileReq, write 04 00 01 00 12 00 00 00 00 00 FF as the parameter, and send the command. This will retreive directory entry. 19. In PTU Simulator, select PTU_FileReq, write 00 00 01 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will open the file for sending. 20. Select the PTU_NextPackReq and send the command.	[…] xx xx. 13. The DCU returns PTU_FileStat with status FS_XFER_OK: 22 xx xx 00 **04 00** xx 00 1F 00 01 xx [...] xx xx xx. **XRAM2 Memory Dump Result** 14. The DCU returns PTU_FileStat with status FS_LOOKUP_OK: 22 xx xx 00 **04 01** xx 00 1F 00 01 53 52 41 4D 20 4D 65 6D 6F 72 79 20 44 75 6D 70 00 00 **00 FF** xx xx xx xx. 15. The DCU returns PTU_FileStat with status FS_XFER_UNDERWAY: 22 xx xx 00 **04 04** xx 00 1F 00 01 00 00 00 [...] xx xx xx xx. 16. The DCU returns PTU_NextPack: C6 xx xx 00 **21** […] xx xx or PTU_LastPack xx xx xx 00 **22** […] xx xx. 17. The DCU returns PTU_FileStat with status FS_XFER_OK: 22 xx xx 00 **04 00** xx 00 1F 00 01 xx [...] xx xx xx. **NVRAM Memory Dump Result** 18. The DCU returns PTU_FileStat with status FS_LOOKUP_OK: 22 xx xx 00 **04 01** xx 00 1F 00 01 53 52 41 4D 20 4D 65 6D 6F 72 79 20 44 75 6D 70 00 00 **00 FF** xx xx xx xx. 19. The DCU returns PTU_FileStat with status FS_XFER_UNDERWAY: 22 xx xx 00 **04 04** xx 00 1F 00 01 00 00 00 00 00 [...] xx xx xx xx. 20. The DCU returns PTU_NextPack: C6 xx xx 00 **21** […] xx xx or PTU_LastPack xx xx xx 00 **22** […] xx xx.	

#	Test Cases	Input Specifications	Output Specifications	P/F
		21. If the previous answer from the DCU was PTU_NextPack (fifth byte is 21), repeat the previous step. Otherwise, select PTU_FileReq, write 02 00 01 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will close the file.	21. The DCU returns PTU_FileStat with status FS_XFER_OK: 22 xx xx 00 **04 00** xx 00 1F 00 01 xx [...] xx xx xx.	
		NVRAM Memory Dump Last 256 Bytes 22. In PTU Simulator, select PTU_FileReq, write 04 00 01 00 12 7F 00 00 00 00 FF as the parameter, and send the command. This will retreive directory entry.	**NVRAM Memory Dump Result** 22. The DCU returns PTU_FileStat with status FS_LOOKUP_OK: 22 xx xx 00 **04 01** xx 00 1F 00 01 53 52 41 4D 20 4D 65 6D 6F 72 79 20 44 75 6D 70 00 00 **00 FF** xx xx xx xx.	
		23. In PTU Simulator, select PTU_FileReq, write 00 00 01 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will open the file for sending.	23. The DCU returns PTU_FileStat with status FS_XFER_UNDERWAY: 22 xx xx 00 **04 04** xx 00 1F 00 01 [...] xx xx xx.	
		24. Select the PTU_NextPackReq and send the command.	24. The DCU returns PTU_NextPack: C6 xx xx 00 **21** […] xx xx or PTU_LastPack xx xx xx 00 **22** […] xx xx.	
		25. If the previous answer from the DCU was PTU_NextPack (fifth byte is 21), repeat the previous step. Otherwise, select PTU_FileReq, write 02 00 01 00 00 00 00 00 00 00 00 as the parameter, and send the command. This will close the file.	25. The DCU returns PTU_FileStat with status FS_XFER_OK: 22 xx xx 00 **04 00** xx 00 1F 00 01 xx [...] xx xx xx.	
		Memory Dump All Internal Memory 26. Close the PTU Simulator, and start the PTU Application. Click on the menu Maintenance→Other Functions →Memory Dump. Then click “Dump” on the new window.	**Memory Dump All Internal Memory Result** 26. The File Transfer Progress window displays the % of the transfer. Then the Memor Dump window displays all internal memory content.	
		Wrong Address Memory Dump 27. In PTU Simulator, (login first using PTU_LoginReq) select PTU_FileReq, write 04 00 01 00 00 FE FF 00 00 00 FF as the parameter, and send the command. This will retreive directory entry.	**Wrong Address Memory Dump Result** 27. The DCU returns PTU_FileStat with status FS_ACCESS_UNAVAIL: 22 xx xx 00 **04 09** xx 00 1F 00 01 53 52 41 4D 20 4D 65 6D 6F 72 79 20 44 75 6D 70 00 00 **00 00** xx xx xx xx.	

17.4.27 TC1.1.27 PTU Version Request

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.27.1 Upon reception of PTU_VersionReq with a Version argument from the PTU, the DCU Application sends the PTU_Version message containing Event Description File and Operating Description File versions to the PTU.	1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. Select PTU_VersionReq, write 01 00 as the parameter, and send the command.	1. The DCU is logged-in. 2. The DCU returns PTU_Version: 0A xx xx 00 10 EE ee OO oo xx xx. EE = EDF major, ee = EDF minor, OO = OPDF major, oo = OPDF minor. EDF version: ____.____ OPDF version: ____.____	
2	TC1.1.27.2 Upon reception of PTU_VersionReq with a SciVersion argument from the PTU, the DCU Application sends one or all SCI versions to the PTU using the PTU_SciVersion message.	1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. Select PTU_VersionReq, write 02 00 as the parameter, and send the command. 3. Select PTU_VersionReq, write 02 01 as the parameter, and send the command. 4. Select PTU_VersionReq, write 02 02 as the parameter, and send the command. 5. Select PTU_VersionReq, write 02 03 as the parameter, and send the command.	1. The DCU is logged-in. 2. The DCU returns PTU_SciVersion including the three DCU SCIs: 42 xx xx 00 11 **04 01 43 50 4C 44** 00 00 xx xx xx xx xx xx xx 00 **03 03 58 4D 4C 2D 44 43 55** xx xx xx **02 01 44 43 55 20 41 50 50** 00 00 00 00 00 xx xx xx xx xx xx xx xx xx xx **01 01 44 43 53 20 42 4F 4F 54** 00 00 00 00 xx xx xx xx xx xx xx xx xx xx xx. 3. The DCU returns PTU_SciVersion including the one DCU SCI: 42 xx xx 00 **11 01 01 44 43 53 20 42 4F 4F 54** 00 00 00 00 xx xx xx xx xx xx xx xx xx xx xx. 4. The DCU returns PTU_SciVersion including the one DCU SCI: 42 xx xx 00 **11 02 01 44 43 55 20 41 50 50** 00 00 00 00 00 xx xx xx xx xx xx xx xx xx xx xx. 5. The DCU returns PTU_SciVersion including the one DCU SCI: 42 xx xx 00 **11 03 03 58 4D 4C 2D 44 43 55** 00 00 xx xx xx xx xx xx xx xx xx xx	

#	Test Cases	Input Specifications	Output Specifications	P/F
		6. Select PTU_VersionReq, write 02 04 as the parameter, and send the command.	xx xx. 6. The DCU returns PTU_SciVersion including the one DCU SCI: 42 xx xx 00 **11 04 01 43 50 4C 44** 00 00 xx xx xx xx xx xx xx xx xx xx xx xx.	

17.4.28 TC1.1.28 PTU Adjust Door Parameters

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Use the PTU Application with the administrator access*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.28.1 Upon reception of the PTU_DCUConfig request with update parameter from the PTU, the DCU Application validates and updates the Configuration File and sends PTU_DCUConfigStatus message with the current Door parameters to the PTU.	1. Valid *Opening time* is tested in section 17.4.5. 2. Valid *Closing time* and *Delay before closing* values are tested in section 17.4.6. 3. Valid *Obstruction recycle* is tested in Section 17.4.8. 4. Valid *Obstruction reopening distance* is tested in Section 17.4.8. 5. In PTU, select “DCU Configure” from the Administration → Other Function menu. In “Delay before reclosing” box, select 0s, then click “Apply”, and then “Cancel”. 6. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches to open Doors. Put obstruction on the Doorway and toggle ON/OFF Door Close TL Switches. 7. Doors reopen 3” (default *Reopen distance*) and stop to close again. As soon the doors stop before closing again, start the stopwatch. Stop the stopwatch when the doors start to reclose. Remove the obstruction to allow doors to close. Repeat the time measurement by changing the *Delay before reclosing* value to 1, 2, 3, 4, and 5s.	1. See Section 17.4.5 outputs. 2. See Section 17.4.6 outputs. 3. See Section 17.4.8 outputs. 4. See Section 17.4.8 outputs. 5. DCU is configured to 0s Delay before reclosing value. 6. Doors close and hit an obstruction. 7. Doors reopen after hitting an obstacle and wait for the delay (*Delay before reclose*). Then doors reclose. 0: _____ 1: _____ 2: _____ 3: _____	

#	Test Cases	Input Specifications	Output Specifications	P/F
			4: _____ 5: _____	
		8. Valid *Delay before closing* is tested in Section 17.4.6.	8. See Section 17.4.6 outputs.	
		9. Valid *Door open duration for Passenger Door Enable* is tested in Section 17.4.7.	9. See Section 17.4.6 outputs.	
		10. On the PTU, change the DCU Configure values to following value: 1 Opening time: 2.5s 2 Closing time: 3s 3 Obstruction recycle: 5 4 Obstruction reopening distance: 10” 5 Delay before reclosing: 5s 6 Delay before closing: 2s 7 Door open duration for Passenger Door Enable: 10mins Click on Apply. Reboot the DCU. Reconnect with PTU and verify configuration parameters.	10. The DCU Configure values are same after the reboot.	
		11. On the PTU, change the DCU Configure values to default: 1 Opening time: 2s 2 Closing time: 2.5s 3 Obstruction recycle: 3 4 Obstruction reopening distance: 3” 5 Delay before reclosing: 2s 6 Delay before closing: 3s 7 Door open duration for Passenger Door Enable: 5mins Click on Apply.	11. The DCU Configure values are set to default.	
2	TC1.1.28.2 Upon reception of the PTU_DCUConfig request with invalid parameter from the PTU, the DCU Application does not update the Configuration File	1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. Select PTU_DCUConfig, write 02 01 00 0C as the parameter (set the Opening time to 1.2 sec.), and send the command.	1. The DCU is logged-in. 2. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 01 00 14** xx xx.	

#	Test Cases	Input Specifications	Output Specifications	P/F
	and sends PTU_DCUConfigStatus message with the current Door parameters to the PTU.	3. Select PTU_DCUConfig, write 02 01 00 1A as the parameter (set the Opening time to 2.6 sec.), and send the command. 4. Select PTU_DCUConfig, write 02 02 00 13 as the parameter (set the Closing time to 1.9 sec.), and send the command. 5. Select PTU_DCUConfig, write 02 02 00 1F as the parameter (set the Closing time to 3.1 sec.), and send the command. 6. Select PTU_DCUConfig, write 02 03 00 06 as the parameter (set the Obstruction recycle to 6), and send the command. 7. Select PTU_DCUConfig, write 02 03 01 00 as the parameter (set the Obstruction recycle to 256), and send the command. 8. Select PTU_DCUConfig, write 02 04 00 1C as the parameter (set the Obstruction reopen distance to 28”), and send the command. 9. Select PTU_DCUConfig, write 02 04 FF FF as the parameter (set the Obstruction reopen distance to 65535”), and send the command. 10. Select PTU_DCUConfig, write 02 05 00 06 as the parameter (set the Delay before reclosing to 6), and send the command. 11. Select PTU_DCUConfig, write 02 05 FF FF as the parameter (set the Delay before reclosing to 65535), and send the command. 12. Select PTU_DCUConfig, write 02 06 00 06 as the parameter (set the Delay before door closing to 6), and send the command.	3. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 01 00 14** xx xx. 4. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 02 00 19** xx xx. 5. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 02 00 19** xx xx. 6. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 03 00 03** xx xx. 7. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 03 00 03** xx xx. 8. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 04 00 03** xx xx. 9. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 04 00 03** xx xx. 10. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 05 00 02** xx xx. 11. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 05 00 02** xx xx. 12. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 06 00 03** xx xx.	

#	Test Cases	Input Specifications	Output Specifications	P/F
		13. Select PTU_DCUConfig, write 02 06 FF FF as the parameter (set the Delay before door closing to 65535.), and send the command. 14. Select PTU_DCUConfig, write 02 07 00 0B as the parameter (set the Door open duration for Passenger Door Enable to 11mins), and send the command. 15. Select PTU_DCUConfig, write 02 07 00 FF as the parameter (set the Door open duration for Passenger Door Enable to 255mins), and send the command.	13. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 06 00 03** xx xx. 14. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 07 00 05** xx xx. 15. The DCU returns PTU_DCUConfigStatus: 0A xx xx 00 **4D 02 07 00 05** xx xx.	

17.4.29 TC1.1.30 PTU Communication Test

Special Procedural Requirements:
- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:
- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.30.1 Upon reception of PTU_ComTestReq from the PTU, the DCU Application shall send the PTU_ComTestStatus to the PTU.	1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. Select PTU_ComTestReq and send the command.	1. The DCU is logged-in. 2. The DCU returns PTU_ComTestStatus: 06 xx xx 00 50 xx xx.	

17.4.30 TC1.1.31 DCU to PTU Error

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.31.1 DCU sends PTU_DcutoPTAError command to the PTU Application to inform it that even if the packet is valid, the DCU Application does not recognize the command received or the data format.	1. Using PTU Simulator, send PTU_LoginReq to DCU. 2. Select PTU_VersionReq, enter FF 00 as parameter and send the command.	1. The DCU is logged-in. 2. The DCU returns PTU_DcutoPTAError with parameters CMD_NOTSUPPORTED and PTU_VersionReq: 08 xx xx 00 **32 02 0F** xx xx.	

17.4.31 TC1.1.32 PTU System Counters

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- Close PTU application when using PTU Simulator.
- Use Car Network Simulator.

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.32.1 Upon reception of the PTU_SystemCountersReq message from PTU with parameter 0 (retrieve), the DCU Application shall send the PTU_SystemCounters message to the PTU.	1. Using the PTU, select Maintenance→Other Functions→Counters Download. 2. Using the PTU, click on “Reset Doors Cycles”. 3. Using the PTU, select Maintenance→Other Functions→Counters Download. 4. Fully open and close the Door twice using the DCB Door Control Box toggle. In PTU, select Maintenance→Other Functions→Counters Download. 5. Activate the Emergency Handle. Reset it. Using the PTU, select Maintenance→Other Functions→Counters Download.	1. In PTU, the system counters are downloaded and displayed. All counters should not be 0. 2. In DCU, the system counters are reset. 3. In PTU, the system counters are downloaded and displayed. All counter values are 0. 4. In PTU, the Door Cycles Counter has a value of 2. 5. In PTU, the Event #11 counter has a value of 1.	
2	TC1.1.32.2 Upon the reception of the PTU_SystemCountersReq message from PTU with parameter 1, the DCU Application shall reset all counters, update the counter reset date and send the PTU_SystemCounters message to the PTU.	Tested in previous test case.	See Output of previous test.	

17.4.32 TC1.1.33 DCU Status LED Function

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.33.1 The DCU Application flashes the DCU.LED.RUN twice a second when it is operating.	Make sure that the DCU is turned on. Then Observe the Running LED on the DCU box.	The green Running LED flashes twice per second.	
2	TC1.1.33.2 The DCU Application turns on the DCU.LED.WARN when event with exception level attribute is set. The DCU Application turns off the DCU.LED.WARN when there is no event set with exception level attribute.	Tested in the test cases of the Table 17-5.	See output of Table 17-5.	
3	TC1.1.33.3 The DCU Application turns on the DCU.LED.FAULT when event with failure level attribute is set. The DCU Application turns off the DCU.LED.FAULT when there is no event set with failure level attribute.	Tested in the test cases of the Table 17-5.	See output of Table 17-5.	

17.4.33 TC1.1.35 Watchdog Function

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.35.1 The DCU Application reboots using the CPU register WDT.Reset when any one of safety related threads/processes does not respond for 6 seconds.	Tested in test case #1 of Table 17-5.	See Output test case #1 of Table 17-5	

17.4.34 TC1.1.36 Static Numerical Requirements

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Close PTU application when using PTU Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.36.1 The system has an interface with a PTU via the RS232 interface with operating parameters of 115200 baud rate, 8 data bit, no parity, 1 start bit, 1 stop bit.	1. In PTU Simulator status bar, change the speed to 38400. Sent PTU_LoginReq. 2. Change back the speed to 115200. Send PTU_LoginReq.	1. The DCU does not answer. 2. The DCU is logged in (login status is green).	
2	TC1.1.36.2 DCU Application uses less than 90% of the Volatile memory. The DCU Application uses less than 45% of the Non-Volatile memory.	Get DCUAPP.m66 file from the Visual SourceSafe, and open it with notepad to calculate the free memory. Look into section “MEMORY MAP OF MODULE: OUTPUT/DCUAPP.ABS (?C_STARTUP)” in the file to calculate free memory. For each memory component listed below, add used memory length from the DCUAPP.m66 and subtract it from	Amount of IFlash memory used (Byte): A: _____	

#	Test Cases	Input Specifications	Output Specifications	P/F
		memory component size to calculate the free memory. Refer to example Table 17-6 to help calculation. Memory — Location — Size IFlash — 0x000000~0x8FFFF — 0x8FFFF XFlash — 0x090000~0xEFFFF — 0x5FFFF SRAM — 0x0F0000~0x100000 — 0x10000 NVRAM — 0x120000~0x127F00 — 0x07F00 External SRAM — 0x300000~0x400000 — 0x100000 External Flash — 0x400000~0x5FFFFF — 0x200000 Compare calculated values with the requirement.	Amount of XFlash memory used (Byte): B: _____ Amount of SRAM memory used (Byte): C: _____ Amount of External SRAM memory used (Byte): D: _____ Amount of NVRAM memory used (Byte): E: _____ Amount of External Flash memory used (Byte): F: _____ Percentage of volatile memory used: (C + D) / (SRAM size + External SRAM size) = G: _____ Percentage of non volatile memory used: (A + B + E + F) / (IFlash size + XFlash size + NVRAM size + External Flash size) = H: _____ The value at G is less than 90% and the value at H is less than 45%.	

Note: The SRAM content is lost at each power-down and there is no shared memory between DCU Boot Loader and DCU Applications. NVRAM is used by the Boot Loader Application only for the memory check function and this zone of memory is also used by the DCU Application. Then, DCU Boot Loader memory requirements have no impact on DCU Application memory usage.

Table 17-6: Memory Mapping Example

Memory Type	Start	End	Size (Bytes)	Module
SRAM (Internal IRAM & XRAM)	00E002H	00E50DH	1292	SERIAL
	00E50EH	00E510H	3	SPI
	00E511H	00E511H	1	AR166_CONFIG
	00E512H	00E53CH	43	?SELFTEST
	00E53DH	00E53DH	1	?IO
	00E53EH	00E548H	11	?PID
	00E54AH	00E54BH	2	?PTU
	00F600H	00F8B3H	692	PID
	00F8B4H	00F98BH	216	IO
	00F98CH	00F9FBH	112	SPI
	00F9FCH	00F9FFH	4	DOP
	00FA00H	00FBFFH	512	EM STACK
	00FC00H	00FC1FH	32	C_MAINREGISTERS
	00FC20H	00FC35H	22	MONIT
	00FC36H	00FC3BH	6	AR166_CONFIG
	00FC3CH	00FC41H	6	AR_RESOURCE
	00FC42H	00FC43H	2	INT
	00FD00H	00FDB5H	182	IO
	00FDB6H	00FDBBH	6	MONIT
	00FDBCH	00FDE7H	44	DOP
	00FDE8H	00FDEBH	4	PID
	0F0000H	0F01FFH	512	INTERRUPT_DCU
	0F0220H	0F0220H	1	AR166_CONFIG
	0F0226H	0F0ABDH	2200	AR166_CONFIG
	0F0ABEH	0F0D6DH	688	LDTS
	0F0D6EH	0F0E39H	204	MONIT
	0F0E3AH	0F0ED3H	154	DOP
	0F0ED4H	0F0F59H	134	AR_RESOURCE
	0F0F5AH	0F0FCAH	113	PID
	0F0FCBH	0F0FCBH	1	RTC
	0F0FCCH	0F1021H	86	IO
	0F1022H	0F104FH	46	SPI
	0F1050H	0F1073H	36	PTU
	0F1074H	0F1083H	16	SERIAL
	0F1084H	0F1093H	16	CRC
	0F1094H	0F10A3H	16	FLOG
	0F10A4H	0F10B2H	15	SELFTEST

Memory Type	Start	End	Size (Bytes)	Module
	0F10B4H	0F10B9H	6	MAIN
	0F10BAH	0F10BCH	3	WATCHDOG
	0F10BEH	0F10BFH	2	INT
	0F10C0H	0F15BFH	1280	PTE
	0F15C0H	0F19BFH	1024	IO
	0F19C0H	0F1DBFH	1024	MONIT
	0F1DC0H	0F21BFH	1024	DOP
	0F21C0H	0F25BFH	1024	SERIAL
	0F25C0H	0F29BFH	1024	LDTS
	0F29C0H	0F2DBFH	1024	SPI
	0F2DC0H	0F31BFH	1024	PID
	0F31C0H	0F362EH	1135	PTU
	0F3630H	0F39E7H	952	FLOG
	0F39E8H	0F3A67H	128	AR_RESOURCE
	0F3A68H	0F3A9DH	54	IO
	0F3D00H	0F3EFFH	512	C_USERSTACK
	0F3F00H	0F7D8FH	16016	PID
	0FBC10H	0FC112H	1283	SERIAL
	0FC114H	0FC5E3H	1232	FLOG
			37202	**TOTAL**
			32430	**FREE**
NVRAM (External NVRAM & XFLASH	120000H	12748FH	29840	?HD?LDTS_EVENT
	127600H	127889H	650	?FD?FLOG
	127CE0H	127CE9H	10	?FD?DOP
	127D00H	127D46H	71	?FD?WATCHDOG
	127F00H	127F1DH	30	?FD?SELFTEST
	127FEEH	127FF1H	4	?FD?RTC
			30605	**TOTAL**
			329843	**FREE**

17.4.35 TC1.1.37 Dynamic Numerical Requirements

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Start the DCU and wait for the Car Network Simulator.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.1.37.1 Departure test does not take more than 1 minute.	1. Turn DCU power off. Unplug PTU cable. Simultaneously turn DCU power on and start a stopwatch Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches repeatedly. Stop the stopwatch when the Door starts to open.	1. Boot time: _____ < 1 minutes.	
2	TC1.1.37.2 DCU Application uses less than 50% of available processing time	1. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. Toggle ON/OFF Door Close TL Switch. Generate one obstruction, and then let the Door close. Using PTU, click “Memory Dump” button on the toolbar. Dump the memory using parameter “Start Address=F0000” and “Length=FF”. When the memory is dumped, read the value written at address 0xF0220.	1. Write down the CPU usage found at address 0xF0220. CPU usage: _____ <= 0x32 (50%)	
3	TC1.1.37.3 The communication with the car Network shall be operational within 60 seconds after the power on .	1. While the Car Network Simulator is running, reboot the DCU. 2. Wait for 60 seconds 3. Using the PTU, select Maintenance→Other Functions→Ethernet Information.	1. DCU is rebooted. 2. Delay is expired. 3. IP Address displayed in PTU Application is same as the one from the Car Network Simulator traces.	

17.5 TC1.2 Instrumented Test Procedure

The following software tools are used in the test procedure: PTU Application, PTU Simulator, Car Network Simulator.

The PTU and PTU Simulator share one serial communication port (RS-232) and PTU cable. There is only one application, either PTU or PTU Simulator, running and communicating to the DCU at a given time. The PTU Simulator should always be in PTU simulation state by unchecking DCU checkbox on the tool bar. Make sure the right serial port number is selected before to use in the test.

17.5.1 TC1.2.1 DCU Application

17.5.1.1 TC1.2.1.1 Self-test

Special Procedural Requirements:

- *DCU must be programmed with DCU Application under test and powered ON.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.2.1.1.1 Continuous self-tests are started upon startup of DCU Application	Make sure the emulator is correctly plugged in and powered on. Start Hitop Application. Open DCU Application project. Go to selftest.c file and set a breakpoint (click on the left margin) at the line “WATCHDOG_Trig (WDT_TASK_SELF_TEST);" (is line 256 in version 12 of the file). Start the emulator by pressing F5 and wait for the program to stop at breakpoint. Press CTRL+B to remove the breakpoint. Reset the emulator by pressing ALT+R.	The emulator is started and quickly stops at the breakpoint, meaning that the Selftest_TaskMemory task is called at startup of application.	

#	Test Cases	Input Specifications	Output Specifications	P/F
2	TC1.2.1.1.2 Self Test execution CRC DCU Application	*Note: Event Detection of Event #2 in Section 17.5.1.3 test is performed along with this procedure.* 1. In Hitop main window, select the selftest.c file into the Source window. Search into SELFTEST.C the code line “switch (Status.ucFlashAppCrc)”, right after “Status.ucFlashAppCrc = SELFTEST_FlashAppCrc ();” (around line 260 for version 12 of selftest.c module). Put cursor on the line and press CTRL+B to insert breakpoint. 2. Start the emulator by pressing F5. 3. Press CTRL+B to remove the breakpoint. With the mouse, go to Disassembly tab, and press “F10” twice. Then double click on the “Z” to change it to “1” in Register window. 4. Start the emulator by pressing F5, and pay attention to DCU board whether it reboots. 5. With the PTU, from the menu tool bar, select “Maintenance → Other Functions → Log Download”. 6. Reset the emulator by pressing ALT+R.	1. The left column of the line turns red to indicate the breakpoint is inserted. 2. The buttons change to to indicate emulator running. The emulator stops. 3. Pressing “F10” twice, the arrow goes to the line “JMPR CC_Z, ^#270”. 4. After one second, the DCU reboots. The buttons change to to indicate emulator is running. 5. Verify that the Event #2 DCU Application CRC32 Discrepancy, has been set one time.	

#	Test Cases	Input Specifications	Output Specifications	P/F
3	TC1.2.1.1.3 Self Test execution CRC Boot Loader	*Note: Event Detection of Event #3 in Section 17.5.1.3 test is performed along with this procedure.* 1. In Hitop main window, select the selftest.c file into the Source window. Search into SELFTEST.C the code line "switch (Status.ucFlashBootCrc)", right after "Status.ucFlashBootCrc = SELFTEST_FlashBootCrc ();" (around line 290 for version 12 of selftest.c module). Put cursor on the line and press CTRL+B to insert breakpoint. 2. Start the emulator by pressing F5. 3. Press CTRL+B to remove the breakpoint. With the mouse, go to Disassembly tab, and press "F10" twice. Then double click on the "Z" to change it to "1" in Register window. 4. Start the emulator by pressing F5, and pay attention to DCU board whether it reboots. 5. With the PTU, from the menu tool bar, select "Maintenance → Other Functions → Log Download". 6. Reset the emulator by pressing ALT+R.	1. The left column of the line turns red to indicate the breakpoint is inserted. 2. The buttons change to to indicate emulator running. The emulator stops. 3. Pressing "F10" twice, the arrow goes to the line "JMPR CC_Z, ^#301". 4. After one second, the DCU reboots. The buttons change to to indicate emulator is running. 5. Verify that the error #3, Boot Loader Application CRC32 Discrepancy, has been set one time.	
4	TC1.2.1.1.4 Self Test execution: SRAM, NVRAM Address, data lines, and pattern test.	*Note: Event Detection of Event #4 and Event #5 in Section 17.5.1.3 test is performed along with this procedure.* **SRAM** 1. In Hitop main window, select the selftest.c file into the Source window. Search into SELFTEST.C the code line" if (! (uiStatus & BIT_SRAM_ADDR_SELFTEST))", (around line 377 for version 12 of selftest.c module). Put cursor on the line and press CTRL+B to insert breakpoint. 2. Start the emulator by pressing F5.	1. The left column of the line turns red to indicate the breakpoint is inserted. 2. The buttons change to to indicate emulator running. The emulator stops.	

#	Test Cases	Input Specifications	Output Specifications	P/F
		3. Press CTRL+B to remove the breakpoint. Go to the Disassembly window. Press F10 once. 4. With the mouse, click on the value field of R4 in Register window and write 0x0EF7. Press enter to validate the new value. 5. Start the emulator by pressing F5, and observe whether the DCU reboots. 6. With the PTU, from the menu tool bar, select "Maintenance → Other Functions → Log Download". 7. Reset the emulator by pressing ALT+R. **NVRAM** 8. In Hitop main window, select the selftest.c file into the Source window. Search into SELFTEST.C the code line"if (! (uiStatus & BIT_NVRAM_ADDR_SELFTEST))", (around line 392 for version 12 of selftest.c module). Put cursor on the line and press CTRL+B to insert breakpoint. 9. Start the emulator by pressing F5. 10. Press CTRL+B to remove the breakpoint. Select the assembly window and press F10 once. 11. With the mouse, click on the value field of R4 in Register window and write 0x0FDF. Press enter to validate the new value. 12. Start the emulator by pressing F5, and observe whether the DCU reboots.	3. Assembly code is shown and the program goes two assembly lines further. The arrow goes to the line "JB R4.3, ^#388". 4. The value changes to 0EF7 and becomes red. 5. The buttons change to to indicate emulator is running. After one second, the DCU reboots. 6. Verify that the Event #4, SRAM Failure, has been set one time. 7. Emulator Resets. 8. The left column of the line turns red to indicate the breakpoint is inserted. 9. The buttons change to to indicate emulator running. The emulator stops. 10. Assembly code is shown and the program goes two assembly lines further. The arrow goes to the line "JB R4.5, ^#403". 11. The value changes to 0FDF and becomes red. 12. The buttons change to to indicate emulator is running. After one second, the DCU reboots.	

#	Test Cases	Input Specifications	Output Specifications	P/F
		13. With the PTU, from the menu tool bar, select "Maintenance → Other Functions → Log Download". 14. Reset the emulator by pressing ALT+R.	13. Verify that the Event #5, NVRAM Failure, has been set one time. .	

17.5.1.2 TC1.2.1.2 Initialization

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Use PTU Application.*
- *Set location to R1, as seen in* Figure 17-3
- *Reboot the DCU.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F

1	TC1.2.1.2.1 DCU Application initializes following output signals to inactive: CPU.PWM.MOTOR, CPU.DIR.OPEN.MOTOR, CPU.DIR.CLOSE.MOTOR, CPU.CUR_LIMIT.NOR.ACT, PAL.ISO, and PAS.ISO.	1. In Hitop main window, open io.c file into the Source window. Search into the function void IO_Init(void) in the io.c the code lines:
		
		// Reset Output Structure
		memset(&IO_OutCpu, (char)0x00, sizeof(IO_OutCpu));
		(around line 1534 for version 10 of io.c module).
		
		2. In Hitop main window, open io.h file into the Source window. Search into the io.c the code lines:
		
		typedef struct _CPU_SignalsOut
		{
		…
		unsigned char Dir_Open_Mot_Iso_Out;
		unsigned char Dir_Close_Mot_Iso_Out;
		unsigned char PWM_Motor_Out;
		unsigned char NorCurLim_Act_Out;
		unsigned char Door_Closing_Visual_Alarm_Act_Out;
		unsigned char Door_Closing_Audible_Alarm_Act_Out;
		….
		} CPU_SignalsOut;
		(around line 758 for version 3 of io.h module).
			1. The function Io_init() resets output structure.
			
			2. The CPU_SignalsOut structure contains following variables: CPU.PWM.MOTOR, CPU.DIR.OPEN.MOTOR, CPU.DIR.CLOSE.MOTOR, CPU.CUR_LIMIT.NOR.ACT, Output.PassAlertLight.ISO, and Output.PassAudibleSignal.ISO	

17.5.1.3 TC1.2.1.3 Event Detection

Special Procedural Requirements:

- *DCU must be programmed with the DCU Application under test and powered ON.*
- *Use PTU Application.*
- *Set location to R1, as seen in* Figure 17-3*.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.2.1.3.1 Upon detection of an event, the DCU Application stores the specific event information along with a timestamp in the Event Log file.	1. Connect to the Ethernet Simulator and start the Ethernet Simulator application. Adjust the date and time to the current PC value and then use the simulator to send out the present time packet. 2. Execute all test cases included in Table 24-1.	1. The Ethernet Simulator sends the Present Time packet cyclically and the DCU Application should update the time accordingly. 2. See output.	

The description of the “Event Name” is shown in Table 17-5.

Table 17-7: Events list

Nbr	Event Name / Description	Test Description		
		Input Specification	Output Specification	P/F
1	TC1.2.1.3.2 Watchdog Reset Detected	1. In Hitop main window, select the io.c file into the Source window. Search into io.c the code line “return (bWatchDogReset);” (around line 1739 for version 10 of io.c module). Put cursor on the line and press CTRL+B to insert a breakpoint. 2. Select the bWatchDogReset variable and drag it to the watch window at the bottom of the Hitop window. 3. Start the emulator by pressing F5. 4. Press CTRL+B to remove the breakpoint. Modify the value of bWatchDogReset to “01”. Press enter to validate the new value. 5. Start the emulator by pressing F5.	1. The left column of the line turns red to indicate the breakpoint inserted. 2. The variable bWatchDogReset is monitored. 3. The buttons change to to indicate emulator running. The emulator stops. 4. The value changes to 1 (0x01) and becomes red. 5. The buttons change to to indicate emulator running.	

Nbr	Event Name / Description	Test Description		
		Input Specification	Output Specification	P/F
		6. With the PTU, from the menu tool bar, select Maintenance → Other Functions → Log Download. 7. Reset the emulator by pressing ALT+R.	6. Verify that the error #1, Watchdog Reset Detected, has been pulsed one time. 7.	
2	TC1.2.1.3.3 DCU App. CRC32 Discrepancy	Tested in section 17.5.1.1.	N/A	N/A
3	TC1.2.1.3.4 Boot Loader CRC32 Discrepancy	Tested in section 17.5.1.1.	N/A	N/A
4	TC1.2.1.3.5 SRAM Failure	Tested in section 17.5.1.1.	N/A	N/A
5	TC1.2.1.3.6 NVRAM Failure	Tested in section 17.5.1.1.	N/A	N/A
23	TC1.2.1.3.7 Motor Open Supply Error	**Make sure the Door is closed and locked.** 1. On the Switch Box, turn on the SW16. 2. Move back the switch to its normal position. 3. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The DCU Fault LED turns on. 2. The DCU Fault LED turns off. 3. The log contains event #23, Motor Open Supply Error Detected, in set and reset states.	
24	TC1.2.1.3.8 Solenoid Lock Supply Error	**Make sure the Door is closed and locked.** 1. On the Switch Box, turn on the SW23. 2. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 3. Move back the switch to its normal position. And Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 4. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The DCU Fault LED turns on. 2. The Doors do not move. 3. The DCU Fault LED turns off. 4. The log contains event #24, Solenoid Lock Supply Error, in set and reset states.	

Nbr	Event Name / Description	Test Description		
		Input Specification	Output Specification	P/F
29	TC1.2.1.3.9 Motor Control Module Communication Error	1. In Hitop main window, select the dop.c file into the Source window. Search into dop.c the code line “if (ucStt_Error != FALSE)” (around line 1041 for version 25 of dop.c module). Put cursor on the line and press CTRL+B to insert a breakpoint. 2. Select “ucStt_Error” and drag it into the watch window. 3. Start the emulator by pressing F5. 4. Press CTRL+B to remove the breakpoint. With the mouse, click on the “value” field of “ucStt_Error” in Watch window and write 0x01. Press enter to validate the new value. 5. Start the emulator by pressing F5. 6. With the PTU, from the menu tool bar, select Maintenance → Other Functions → Log Download. 7. Reset the emulator by pressing ALT+R.	1. The left column of the line turns red to indicate the breakpoint inserted. 2. The variable “ucSTT_Error” is monitored. 3. The buttons change to to indicate emulator running. The emulator stops. 4. The value changes to 1 (0x01) and becomes red. 5. The buttons change to to indicate emulator running. And the DCU Fault LED turns on. 6. Verify that the error #29, Motor Control Module Communication Error, in set and reset states.	
31	TC1.2.1.3.10 Motor Drive Circuit (H-Bridge) Error	***Caution: Never activate two switches connected to SW1, SW2, SW3, or SW4 simultaneously.*** 1. On the Switch Box, turn on the SW1. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches to open Doors. 2. Turn off the SW1. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches to close Doors, then pull toggle to open, and then push toggle to close. 3. On the Switch Box, turn on the SW4. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches.	1. The Doors do not move. The Door Fault LED goes ON. 2. The Door Fault LED goes OFF. 3. The Doors do not move. The Door Fault LED goes ON.	

Nbr	Event Name / Description	Test Description		
		Input Specification	Output Specification	P/F
		4. Turn off the SW4. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches, then pull toggle to open, and then push toggle to close. 5. On the Switch Box, turn on the SW3. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches to open Doors. 6. Turn off the SW3. Push toggle to close Doors, then pull toggle to open, and then push toggle to close. 7. On the Switch Box, turn on the SW2. **Do not try to open the Door.** 8. Turn off the SW2.Push toggle to close Doors. 9. Connect the PTU; select Maintenance → Other Functions → Log Download.	4. The Door Fault LED goes OFF. 5. The Doors do not move. The Door Fault LED goes ON. 6. The Door Fault LED goes OFF. 7. The Door Fault LED goes ON. 8. The Door Fault LED goes OFF. 9. The log contains three occurrences of event #31, Motor Drive Circuit Error, event #23, Motor Open Supply Error Detected, in set and reset states.	
32	TC1.2.1.3.11 Motor Current Reading Error	1. On the Switch Box, turn on the SW12. 2. Turn off the SW12. 3. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The Door Fault LED goes ON. 2. The Door Fault LED goes OFF. 3. The log contains event #32, Motor Current Reading Error Detected, in set and reset states.	
33	TC1.2.1.3.12 Motor Voltage Reading Error .	1. On the Switch Box, turn on the SW18. 2. Turn off the SW18. 3. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The Door Fault LED goes ON. 2. The Door Fault LED goes OFF. 3. The log contains event #33, Motor Voltage Reading Error Detected, in set and reset states.	

Nbr	Event Name / Description	Test Description		
		Input Specification	Output Specification	P/F
34	TC1.2.1.3.13 Normal Current Limit Error	1. On the Switch Box, turn on the SW17. Reset the DCU. Wait for the green LED to start blinking and send a close command. 2. Turn off the SW17. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches, and then toggle to close Doors. Let the Door lock. 3. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The Door Fault LED goes ON. 2. The Door Fault LED goes OFF,. 3. The log contains event #34, Normal Current Limit Error Detected, in set and reset states.	
35	TC1.2.1.3.14 High Current Limit Error	1. On the Switch Box, turn on the SW7. Reset the DCU. Wait for the green LED to start blinking and send a close command. 2. Turn off the SW7. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches to open Doors, and then toggle ON/OFF Door Close TL to close Doors. Let the Door lock. 3. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The Door Fault LED goes ON. 2. The Door Fault LED goes OFF. 3. The log contains event #35, High Current Limit Error Detected, in set and reset states.	
36	TC1.2.1.3.15 Current Level Selector Error	1. On the Switch Box, turn on the SW21. Reset the DCU. Wait for the green LED to start blinking and Toggle ON/OFF Door Close TL. 2. Turn off the SW21.Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 3. Toggle ON/OFF Door Open (+) TL and Door Open (-) TL Switches. 4. With the PTU, select Maintenance → Other Functions → Log Download.	1. The Door Fault LED goes ON. 2. The Doors open. 3. The Door Fault LED goes OFF. 4. The log contains event #36, Current Level Selector Error, in set and reset states.	

Nbr	Event Name / Description	Test Description		
		Input Specification	Output Specification	P/F
38	TC1.2.1.3.16 Motor OC Shutdown Activated	1. On the Switch Box, turn on the SW19. 2. With the Car Network Simulator, observe the diagnostic data section. 3. Turn off the SW19. 4. With the Car Network Simulator, observe the diagnostic data section. 5. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The Door Warning LED goes ON. 2. In the Car Network Simulator, event 38 is set. 3. The Door Warning LED goes OFF. 4. In the Car Network Simulator, event 38 is reset. 5. The log contains event #38, Motor Over Current Shutdown Activated, in set and reset states.	
39	TC1.2.1.3.17 5V Logic Supply Out of Range	1. On the Switch Box, turn on the SW20. 2. Turn off the SW20. 3. With the PTU, select Maintenance → Other Functions → Log Download.	1. The Door Warning LED goes ON. 2. The Door Warning LED goes OFF. 3. The log contains event #39, 5V Logic Supply Out of Range, in set and reset states.	
40	TC1.2.1.3.18 5V Encoder Supply Out of Range	1. On the Switch Box, turn on the SW11. 2. Turn off the SW11. 3. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The Door Warning LED goes ON. 2. The Door Warning LED goes OFF. 3. The log contains event #40, 5V Encoder Supply Out of Range, in set and reset states.	
41	TC1.2.1.3.19 15V Motor Supply Out of Range	1. On the Switch Box, turn on the SW9. 2. Turn off the SW9. 3. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The Door Warning LED goes ON. 2. The Door Warning LED goes OFF. 3. The log contains event #41, 15V Motor Supply Out of Range, in set and reset states.	
42	TC1.2.1.3.20 V-Gate Open Supply Out of Range	1. On the Switch Box, turn on the SW5. 2. Turn off the SW5. 3. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The Door Warning LED goes ON. 2. The Door Warning LED goes OFF. 3. The log contains event #42, High Side Open Drive Supply Out of Range, in set and reset states.	

Nbr	Event Name / Description	Test Description		
		Input Specification	Output Specification	P/F
43	TC1.2.1.3.21 V-Gate Close Supply Out of Range	1. On the Switch Box, turn on the SW6. 2. Turn off the SW6. 3. Wait until the DCU Fault LED turns ON then put back the switch to its normal position. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The Door Warning LED goes ON. 2. The Door Warning LED goes OFF. 3. The log contains event #43, High Side Close Drive Supply Out of Range, in set and reset states.	
45	TC1.2.1.3.22 Door Closing Visual Alarm On w/o Command	1. On the Switch Box, turn on the SW15. 2. Turn off the SW15. 3. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The Door Warning LED goes ON. 2. The Door Warning LED goes OFF. 3. The log contains event #45, Door Closing Visual AlarmOn without Command, in set and reset states.	
47	TC1.2.1.3.23 Solenoid Low Side On w/o Command	1. On the Switch Box, turn on the SW14. 2. Turn off the SW14. 3. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The Door Warning LED goes ON. 2. The Door Warning LED goes OFF. 3. The log contains event #47, Door Closing Audible Warning On without Command, in set and reset states.	

Nbr	Event Name / Description	Test Description		
		Input Specification	Output Specification	P/F
48	TC1.2.1.3.24 Spare Output OC Over current occurred on the Spare Sourcing Output.	1. In Hitop main window, select the io.c file into the Source window. Search into io.c the code line “if (bSpare)” (around line 3060 for version 10 of io.c module). Put cursor on the line and press CTRL+B to insert a breakpoint. 2. Select “bSpare” and drag it into the watch window. 3. Start the emulator by pressing F5. When it stops, with the mouse, click on the “value” field of “bSpare” in Watch window and write 1. Press enter to validate the new value. ***Repeat the previous step 3 times. The third time, press CTRL+B to remove the breakpoint***. 4. Start the emulator by pressing F5. 5. With the PTU, select Maintenance → Other Functions → Browser to observe the Spare Output and Spare Output Feedback values. 6. With the PTU, from the menu tool bar, select Maintenance → Other Functions → Log Download. 7. Reset the emulator by pressing ALT+R.	1. The left column of the line turns red to indicate the breakpoint inserted. 2. The variable “bSpare” is monitored. 3. The buttons change to to indicate emulator running. The emulator stops. The value changes to 1 and becomes red. 4. The buttons change to to indicate emulator running. 5. The Spare Output and Spare Output Feedback values from the Data Monitoring change from 0 to 6. Verify that the Event #48, Spare Output Over Current, has been set once.	
49	TC1.2.1.3.25 Spare Output On without Command The Physical spare output is activated without command from DCU Application.	1. On the Switch Box, turn on the SW13. 2. Turn off the SW13. 3. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The Door Warning LED goes ON. 2. The Door Warning LED goes OFF. 3. The log contains event #49, Spare Output On without Command, in set and reset states.	

Nbr	Event Name / Description	Test Description		
		Input Specification	Output Specification	P/F
50	TC1.2.1.3.26 DCU Temperature Reading Error Detected The DCU Temperature reading circuit is out of tolerances.	1. On the Switch Box, turn the R1 to counter-clock wise. 2. Turn the R1 to clock wise until room temperature. 3. With the PTU, select Maintenance → Other Functions → Log Download.	1. The Door Fault LED goes ON. 2. The Door Fault LED goes OFF. 3. The log contains event #50, DCU Temperature Reading Error Detected, in set and reset states.	
87	TC1.2.1.3.27 Analog Reference Error The analog reference is more than 5% off.	1. On the Switch Box, turn on the SW8. 2. Turn off the SW8. 3. Connect the PTU; select Maintenance → Other Functions → Log Download.	1. The Door Fault LED goes ON. 2. The Door Fault LED turns off. 3. The log contains event #87, Analog Reference Error Detected, in set and reset states.	

17.5.1.4 TC1.2.1.4 Logout / Reboot Request

Special Procedural Requirements:

- *DCU must be programmed with DCU application under test and powered ON.*

Measurements:

- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.2.1.4.1 Upon reception of the PTU_LogoutReq command with reboot argument or an occurrence of either a fatal application error or checksum error, the DCU Application discontinues all operations and reboots.	“Checksum error” is tested in section 17.5.1.1.		

17.5.1.5 TC1.2.1.8 PTU Real Time Monitoring

Special Procedural Requirements:
- *DCU must be programmed with DCU application under test and powered ON.*

Measurements:
- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.2.1.8.1 Upon request for real-time monitoring from the PTU (PTU_DataReq), the DCU Application transmits monitoring data to the PTU (PTU_Data).	1. Start the PTU Application, and open the Browser window from the Maintenance → Other Functions menu. ***DCU Temperature Sensor Test*** 2. Turn the R1 knob to counter-clockwise until Warning LED is ON. 3. Turn the R1 knob to room temperature (77oF). ***5V Logic Supply Test*** 4. Turn ON SW20. 5. Turn OFF SW20. ***Spare Output Test and Spare Output Feedback Test*** Spare Output Test is tested in Section 17.5.1.3 at Event # 48 in the Table 17-7.	1. The Browser window is displayed in the PTU Application. 2. The DCU Temperature Sensor signal is OFF on the Browser window. 3. The DCU Temperature Sensor signal is ON on the Browser window. 4. The 5V Logic Supply signal is OFF on the Browser window. 5. The 5V Logic Supply signal is ON on the Browser window.	

17.5.1.6 TC1.2.1.10 Input Reading

Special Procedural Requirements:
- *None*

Measurements:
- *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.2.1.10.1 Each sense point is polled every 50 ms, converted to its related DCU input signal that is based on the DCU location, and stored in Temporary Data.	1. In Hitop main window, open io.h file into the Source window. Search into io.h the code lines: // Constant of execution of IO task in ms #define IO_TASK_DELAY 50 (around line 339 for version 4 of io.h module). 2. For this timing to be respected, the operating system ticker period needs to be smaller or equal to the desired period and the desired period needs to be a multiple of the ticker period. In Hitop main window, open DCUAPPAR166_Config.c file into the Source window. Search into AR166_Config.c the code line "#define OS_TICK 5000" (around line 73).	1. The IO task delay is set to 50 milliseconds. 2. The ticker period is less than or equal to 50000 microseconds. 50000 is a multiple of the ticker period.	
2	TC1.2.1.10.2 The transition status is stored in Temporary Data.	1. In Hitop main window, open io.c file into the Source window. Search into io.c the code lines: // Input Structures SignalsIn idata IOInput, idata IOTrans, bdata IOInputLocal, bdata IOTransLocal; (around line 283 for version 10 of io.c module).	1. Both Transition structures (IOTrans and IOTrans Local) are stored in idata or bdata inside processor's on-chip memory.	

17.5.2 TC1.2.2 DCU Boot Loader Application

17.5.2.1 TC1.2.2.1 Fault LED & Self Test

Special Procedural Requirements:

- *DCU must be programmed with DCU Boot Loader Application software only.*

Measurements:

- *Self-explanatory*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	TC1.2.2.1.1 Static RAM Error (Memory cell, address, and data lines error)	1. Make sure the emulator is correctly plugged in and powered on. Start Hitop application. Open the DCU Boot Loader project. Go to ram.c file and set a breakpoint (click on the left margin) at the line "MemBackupZone.AddressLine.ucCommand = SRAM_ADD_CMD;" (was line 338 in version 3 of the file). 2. Start the emulator by pressing F5 and wait for the program to stop at breakpoint. 3. When the emulator stops at breakpoint, using "go to cursor" jump to line "pucMemoryPosition = (unsigned char *)ulMemoryBase;" (was line 367 in version 3 of the file). 4. When emulator stops, press F10 twice until the line "*pucMemoryPosition = ucData; " is highlighted. Then select the variable ucData and drag it to the watch window. Change its value to 3 (it was 1). Press enter to validate the new value. 5. Press F5 to run the emulaltor. And observe the Fault LED on the DCU board. 6. Reset the emulator by pressing ALT+R.	1. The breakpoint is inserted. 2. The buttons change to to indicate emulator running. 3. The emulator runs until line "pucMemoryPosition = (unsigned char *)ulMemoryBase;" and stops. 4. The value changes to 3 and becomes red. 5. The buttons change to to indicate emulator is running. The red Fault LED blinks three times to indicate an SRAM failure.	

#	Test Cases	Input Specifications	Output Specifications	P/F
2	TC1.2.2.1.2 NVRAM Error (Memory cell, address, and data lines error. Data preserved during testing)	1. Open the DCU Boot Loader project. Go to ram.c file and set a breakpoint (click on the left margin) at the line “MemBackupZone.AddressLine.ucCommand = NVRAM_ADD_CMD;” (is line 343 in version 3 of the file). 2. Start the emulator by pressing F5 and wait for the program to stop at breakpoint. 3. When the emulator stops at breakpoint, using “go to cursor” jump to line “pucMemoryPosition = (unsigned char *)ulMemoryBase;” (was line 367 in version 3 o the file). 4. When emulator stops, press F10 twice until the line “*pucMemoryPosition = ucData; “ is highlighted. Then select the variable ucData and drag it to the watch window. Change its value to 3 (it was 1). Press enter to validate the new value. 5. Press F5 to run the emulaltor. And observe the Fault LED on the DCU board. 6. Reset the emulator by pressing ALT+R.	1. The breakpoint is inserted. 2. The buttons change to to indicate emulator running. 3. The emulator runs until line “pucMemoryPosition = (unsigned char *)ulMemoryBase;” and stops. 4. The value changes to 3 and becomes red. 5. The buttons change to to indicate emulator is running. The red Fault LED blinks five times to indicate an NVRAM failure.	

#	Test Cases	Input Specifications	Output Specifications	P/F
3	TC1.2.2.1.3 External SRAM Error (Memory cell, address, and data lines error)	1. Make sure the emulator is correctly plugged in and powered on. Start Hitop application. Open the DCU Boot Loader project. Go to ram.c file and set a breakpoint (click on the left margin) at the line “MemBackupZone. AddressLine .ucCommand = EXT_SRAM_ADD_CMD;” (was line 348 in version 3 of the file). 2. Start the emulator by pressing F5 and wait for the program to stop at breakpoint. 3. When the emulator stops at breakpoint, using “go to cursor” jump to line “pucMemoryPosition = (unsigned char *)ulMemoryBase;” (was line 367 in version 3 o the file). 4. When emulator stops, press F10 twice until the line “*pucMemoryPosition = ucData; “ is highlighted. Then select the variable ucData and drag it to the watch window. Change its value to 3 (it was 1). Press enter to validate the new value. 5. Press F5 to run the emulaltor. And observe the Fault LED on the DCU board. 6. Reset the emulator by pressing ALT+R.	1. The breakpoint is inserted. 2. The buttons change to to indicate emulator running. 3. The emulator runs until line “pucMemoryPosition = (unsigned char *)ulMemoryBase;” and stops. 4. The value changes to 3 and becomes red. 5. The buttons change to to indicate emulator is running. The red Fault LED blinks seven times to indicate an External SRAM failure.	

#	Test Cases	Input Specifications	Output Specifications	P/F
4	TC1.2.2.1.4 IRAM Error (Memory cell, address, and data lines error)	1. Make sure the emulator is correctly plugged in and powered on. Start Hitop application. Open the DCU Boot Loader project. Go to iram.c file and set a breakpoint (click on the left margin) at the line “MOVB RL2, [R3];” (was line 277 in version 1 of the file). 2. Start the emulator by pressing F5 and wait for the program to stop at breakpoint. 3. When the emulator stops at breakpoint, press F10 once. Then change the R5 value to 5 (was 1) on the Resister window. Press enter to validate the new value. 4. Press F5 to run the emulaltor. And observe the Fault LED on the DCU board. 5. Reset the emulator by pressing ALT+R.	1. The breakpoint is inserted. 2. The buttons change to to indicate emulator running. 3. The value changes to 5 and becomes red. 4. The buttons change to to indicate emulator is running. The red Fault LED blinks once to indicate an IRAM Error.	

APPENDIX A: TEST REPORT SHEET

(Page 1 of 3)

Item Identification

Item Tested:	DCU Application
Item P/N:	
Item Version:	

Tester Identification

Tester:	
Time & Date Started:	
Time & Date Finished:	
Physical Location of Test Execution:	

Equipment Used

Description	Part Number	Rev	Serial Number	Calibration Due Date
Stop Watch				
DCU Test Fixture				
Pulse Generator Test Fixture				
PTU cable DB9-PTU PLUG				
Variable Power Supply (Voltage source min. 10A, 0-50V)				
Multimeter				
2-traces Scope				
Parallel cable				
Banana cables				
Measuring Tape				
Antistatic Mat				
Antistatic Wrist				
PC Running Windows 2000\XP				
PC Running Windows DOS				
DCU				
Complete Door System: Door Control Unit (DCU) Door Operator				

Test Report Sheet (Page 2 of 3)

Software Description	P/N	Version	Date
PTU Application			
PTU Simulator			
CNC Simulator			
Boot Loader Application			
DCU Application			
Neuron Gateway Application			
Signal Generator software			

Comments

Approvals

Tester: _____ Title: _____ Date: _____

Witness: _____ Title: _____ Date: _____
_____ Title: _____ Date: _____
_____ Title: _____ Date: _____
_____ Title: _____ Date: _____

Make copies of this page and the previous one if required.

Test Report Sheet (Page 3 of 3)

Summary of Anomalies

SPR #	Description